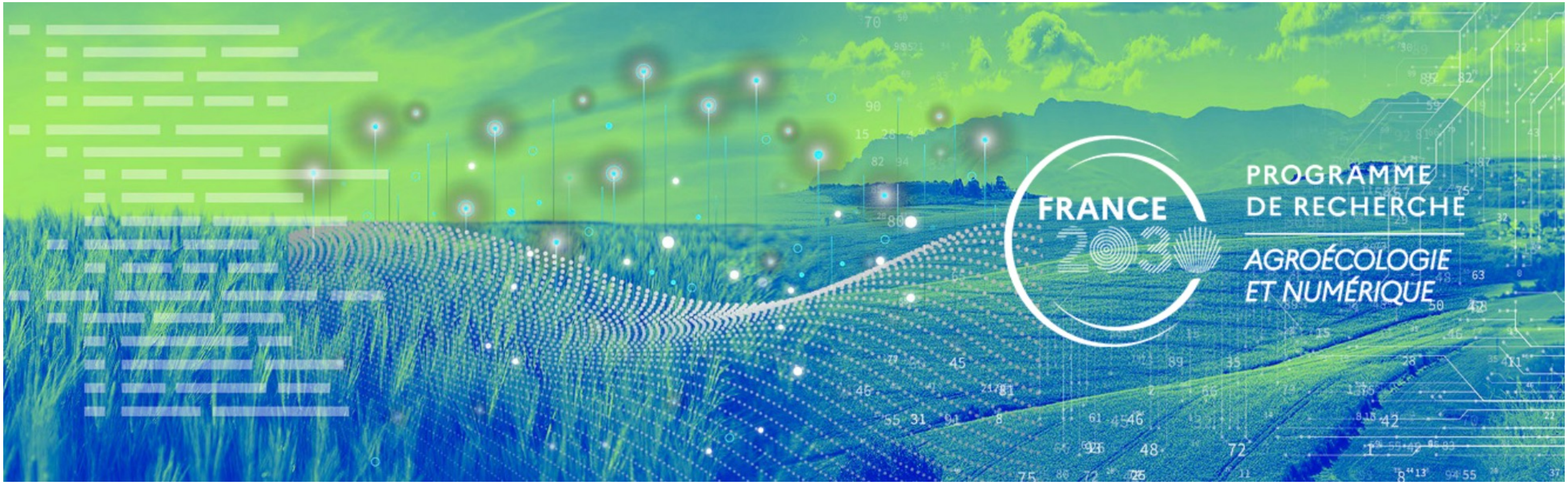


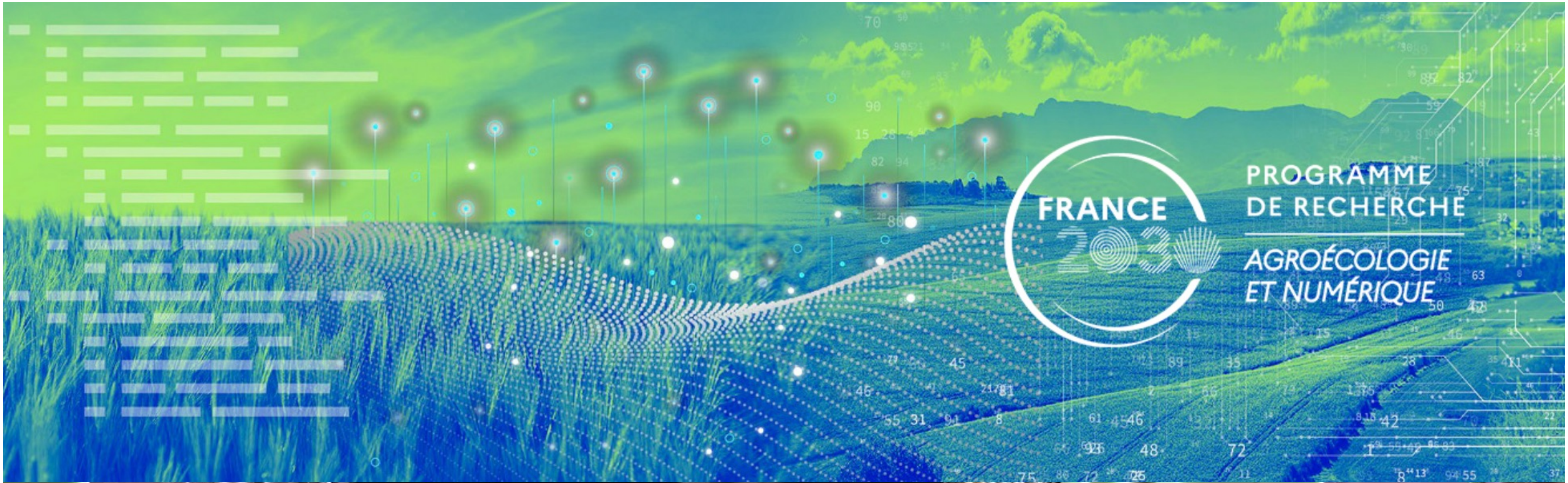
Building the LoRa IoT platform
Part 1: sensor device with PCB v4.1

Prof. Congduc Pham
http://www.univ-pau.fr/~cpham
Université de Pau, France

Congduc.Pham@univ-pau.fr

Wireless Sensors Made Simple
for agroecology & sustainable agriculture

Powered by technologies developed in Intel-IrriS



AGRIFUTUR SENSOR ECOSYSTEM

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Going beyond irrigation

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Wireless Sensors Made Simple
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Powered by technologies developed in Intel-IrriS

- Use the same approach of cost-efficiency and out-of-the-box deployment, expanding the sensor ecosystem
- Complementarity & Adaptability: high-end, low-end, innovative, …
- Target Agroecology, Nature-Based Solutions & Sustainable Agriculture
- Better Qualify & Quantify the impact of these new practices
- More data = more correlation in agroecological system

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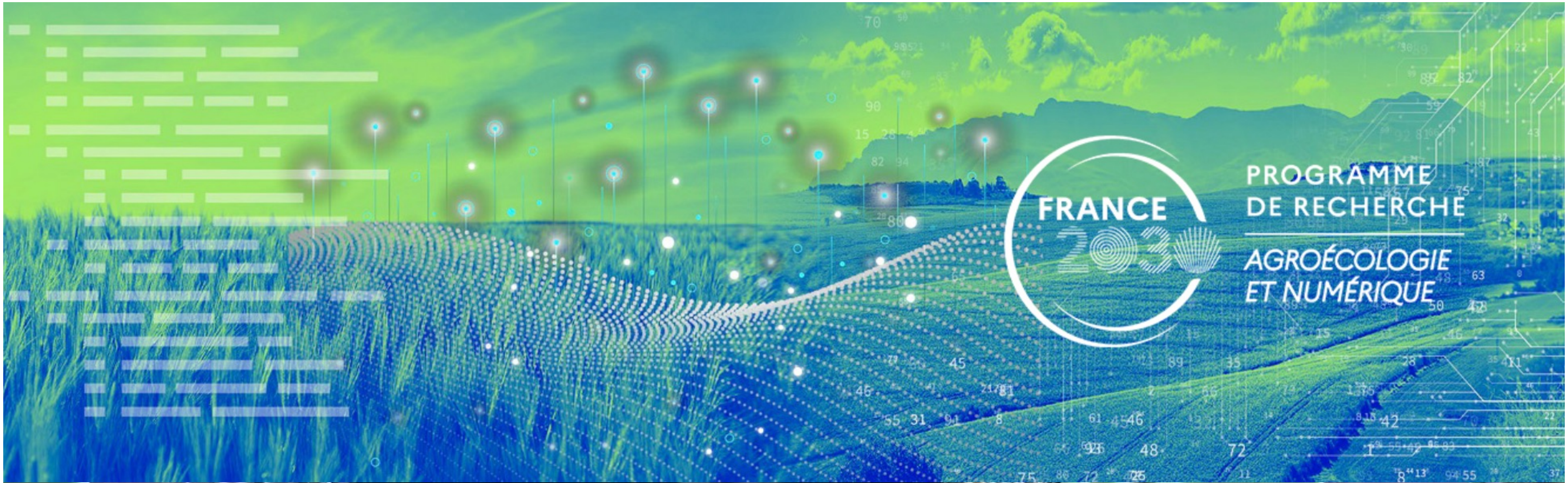
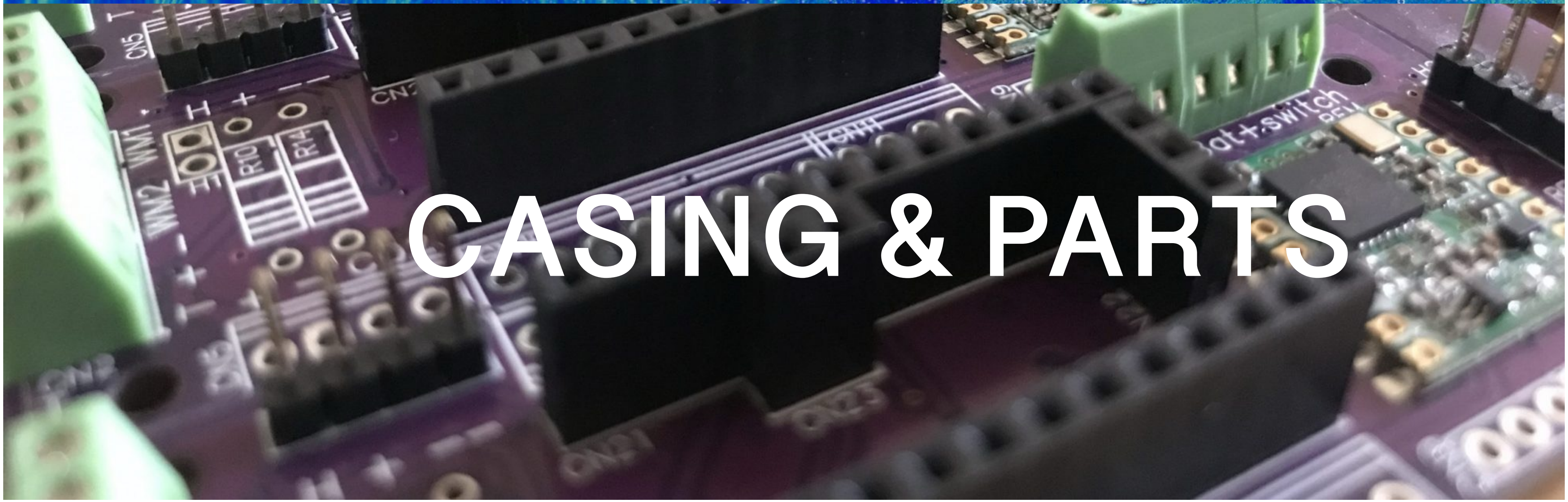


CASING & PARTS



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List of components

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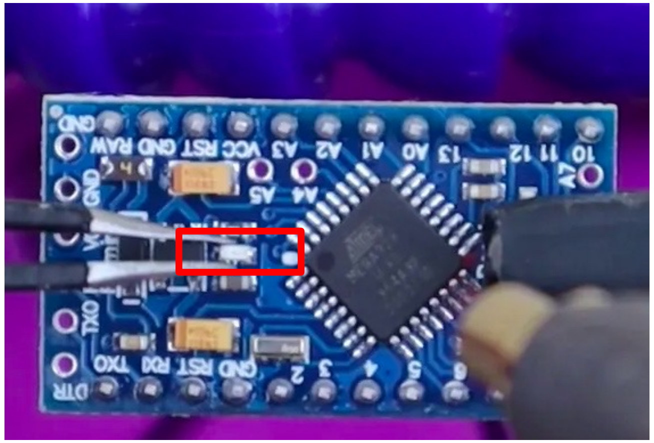
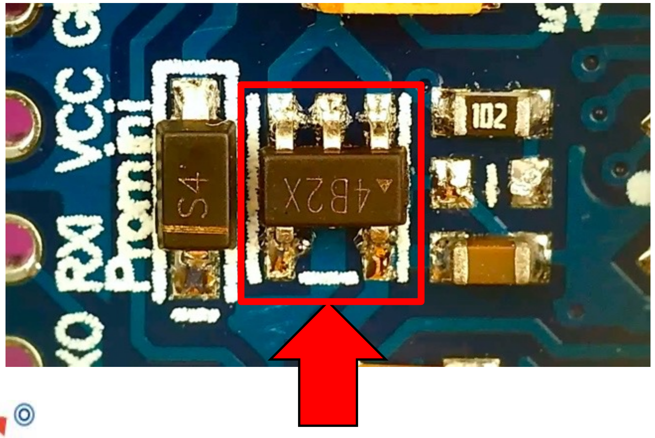
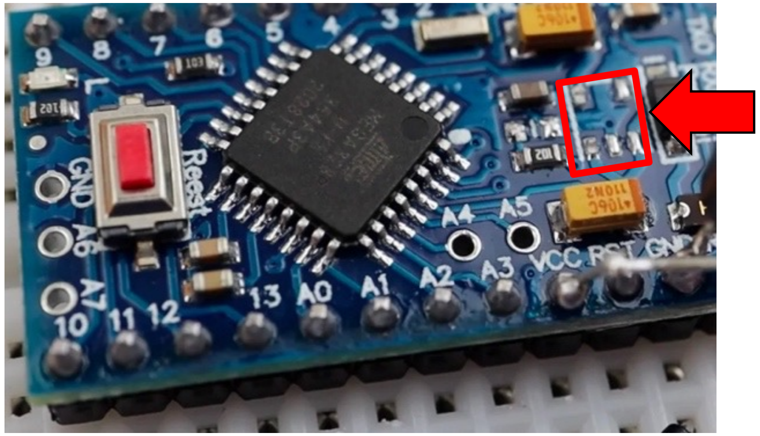
Reduce power consumption





Do not forget to remove the power LED by just clipping it off with some wire cutters



Do not forget to remove the voltage regulator with a small plier

Only inject up to 3.6V through the VCC pin

https://www.the-diy-life.com/making-an-ultra-low-power-arduino-pro/

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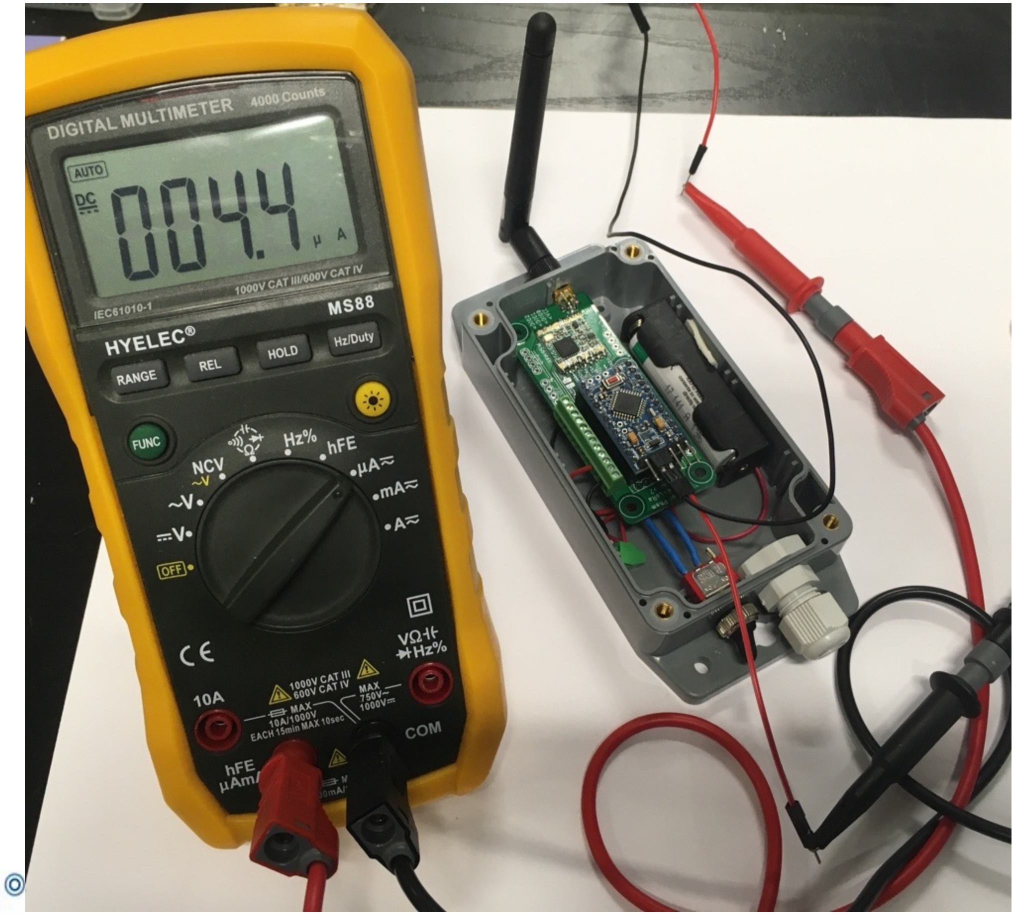
Power consumption in deep sleep



Measured below 5uA in deep sleep, between 2 active periods with transmissions

Expected autonomy with 1 transmission / hour is over 2 years with 2 AA batteries

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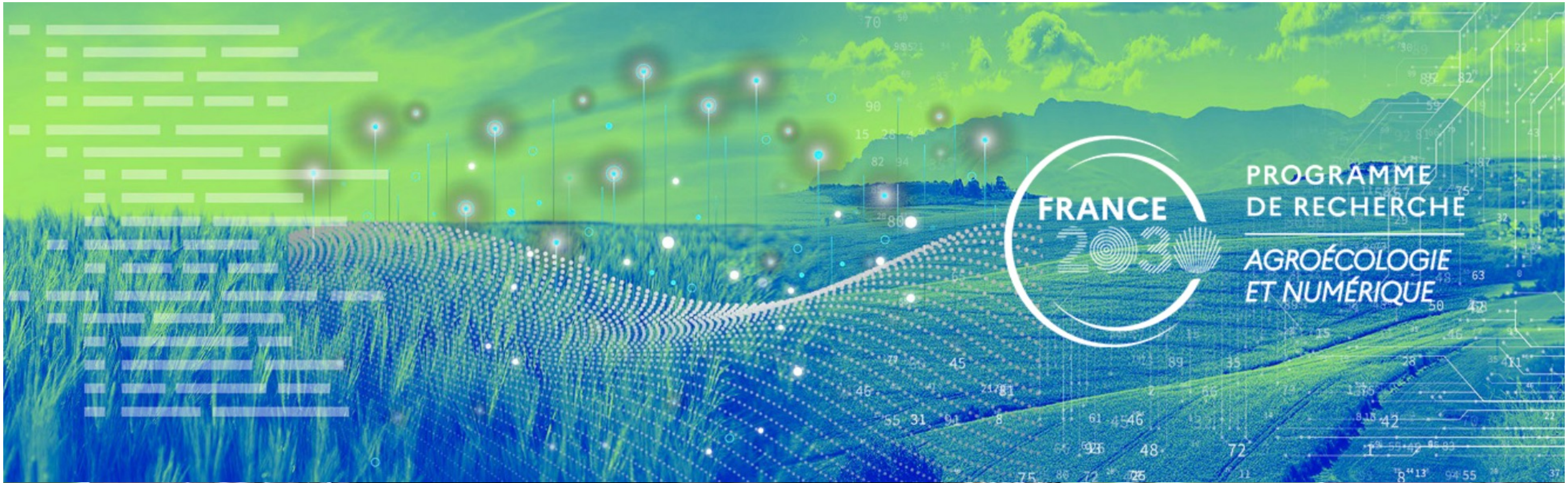
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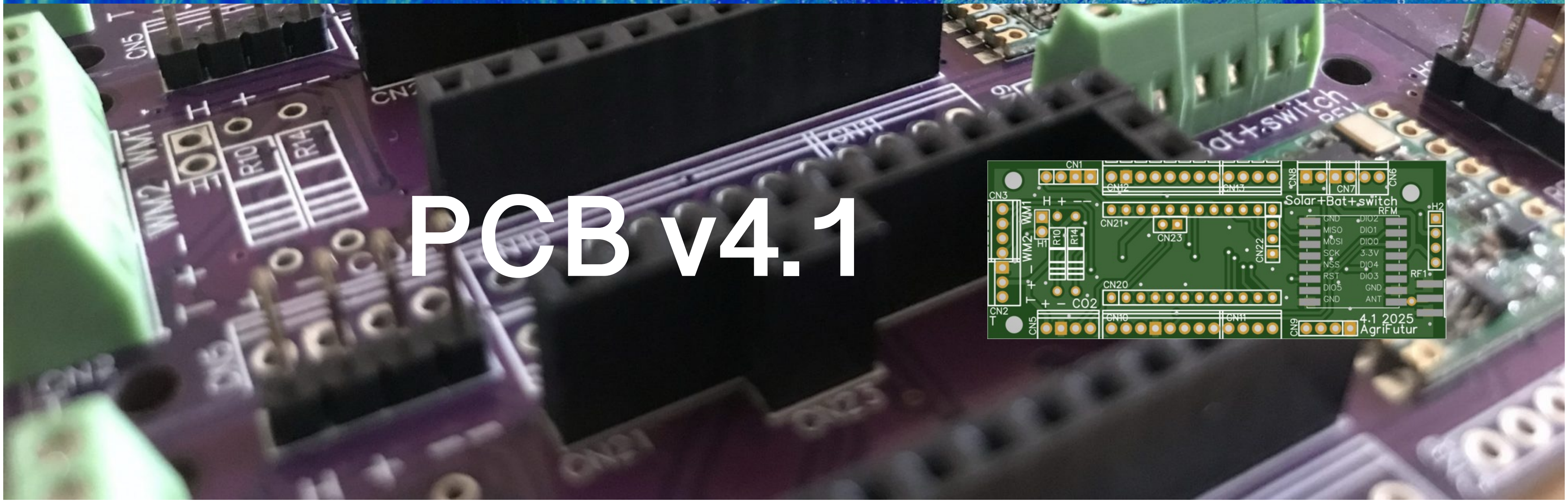
PCB v4.1


Solar+Bat+switch
RFM
4.1 2025
AgriFutur

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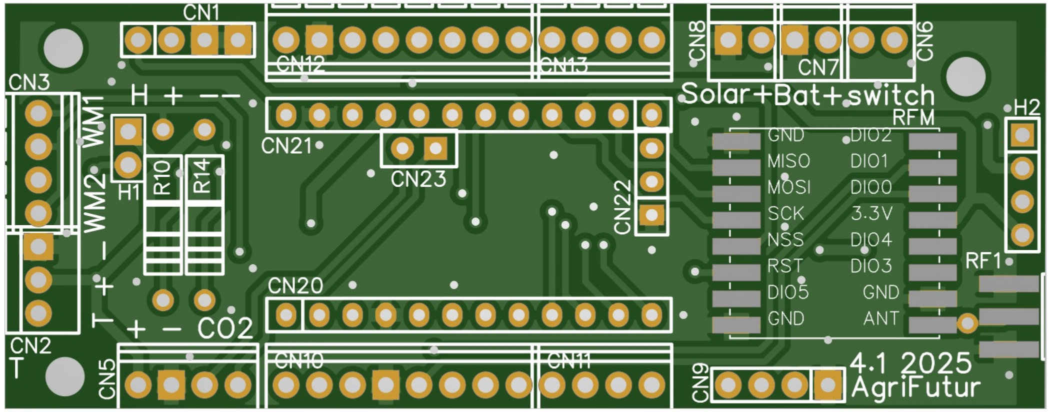
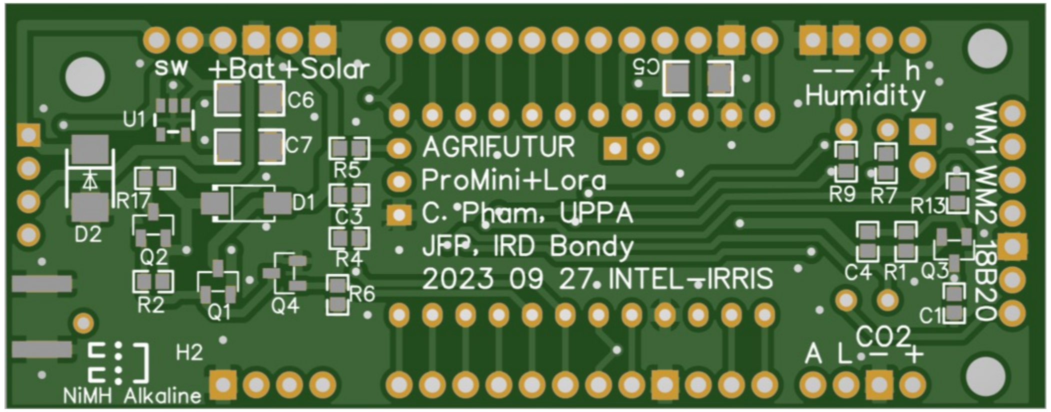
The PCB v4.1 – raw version

- You can order the raw version of IRD PCB, which means only the PCB, without any electronic components soldered by the manufacturer
- It is the so-called DIY approach
- The raw version is not intended to have solar charging capabilities





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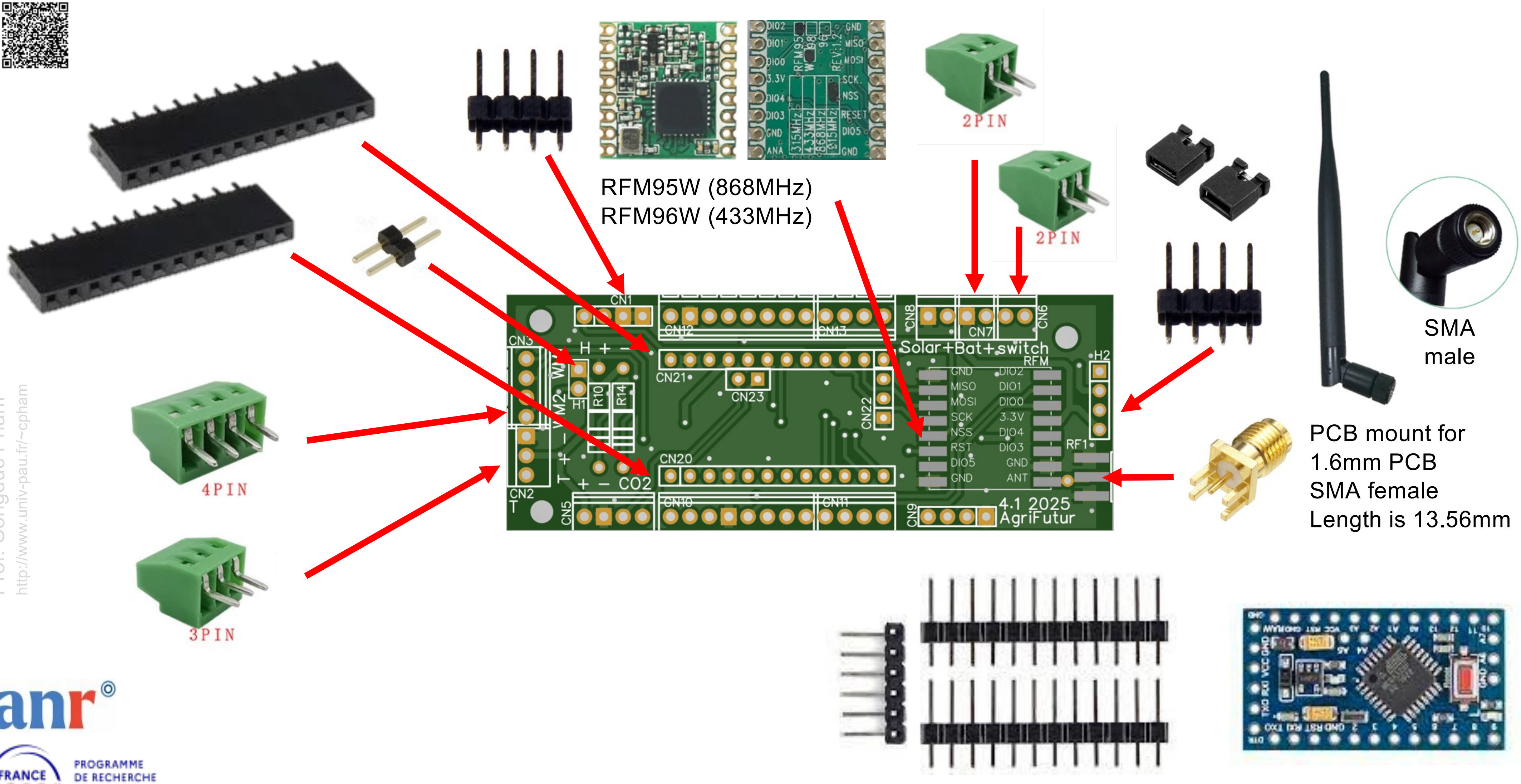
Electronic parts



Arduino ProMini 3.3V 8MHz

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Tutorial video

- This dedicated video from INTEL-IRRIS will show how to solder all the components: Video n°1: https://youtu.be/3jdQ0Uo0phQ
- It features the "old" version of the INTEL-IRRIS PCB but you can easily adapt for the PCB v4.1

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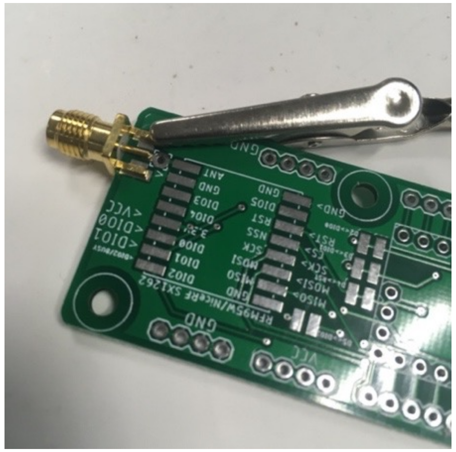
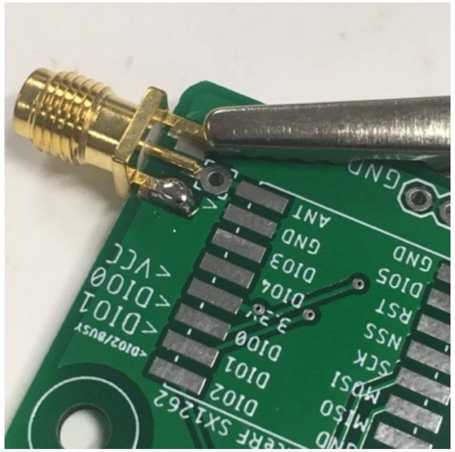
Updates on Video n°1

- Clip the SMA connector to the PCB to correctly align the connector before soldering one of the GND pad, then remove clip and finish soldering
- Or, gently press SMA connector pads so that insertion on PCB is harder and SMA connector will not move during soldering
- Make sure that the GND pads and the central pin are not touching each other, in doubt, do a continuity check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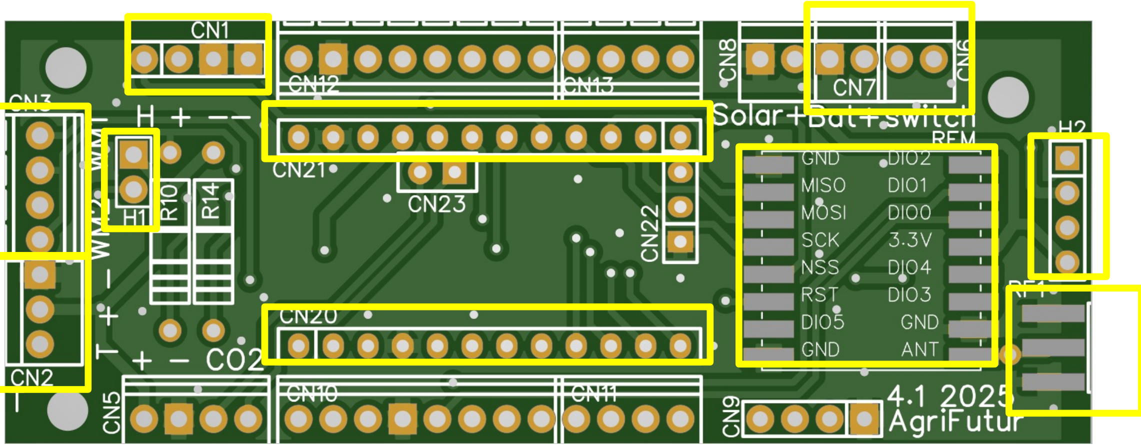
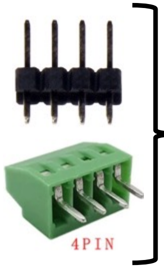
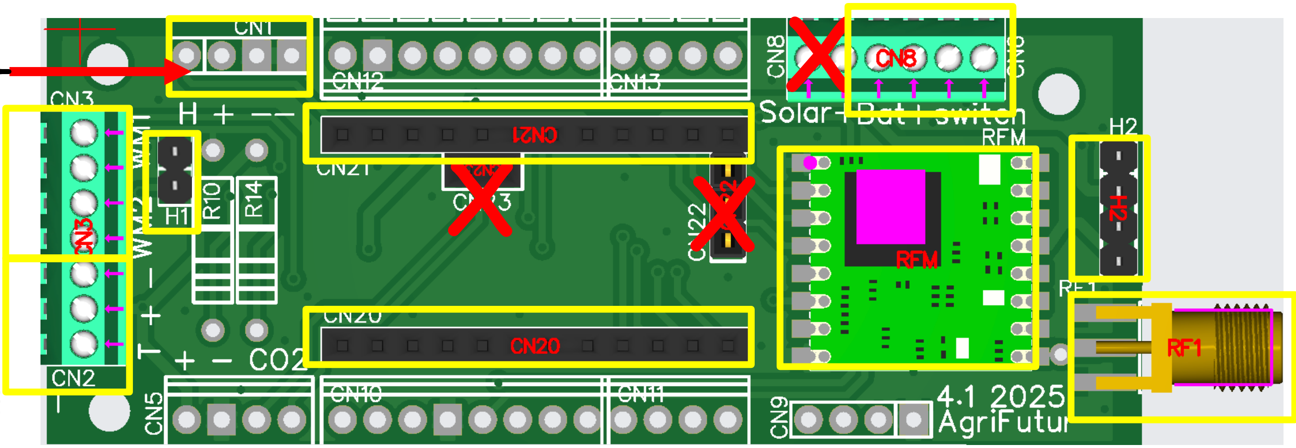
Soldering parts on PCB v4.1

- Solder the various components on the top face







The raw version is not intended to have solar charging capabilities. For solar charging, the full version (see later) is the dedicated PCBA

Not needed for the raw PCB which is not designed to have the solar circuit manually soldered

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Wiring switch & bat on PCB v4.1

- Power wires (from 2xAA alkaline battery holder)

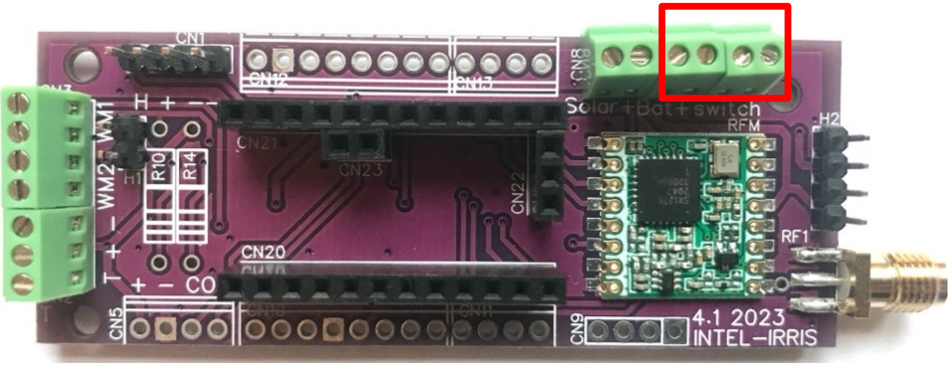
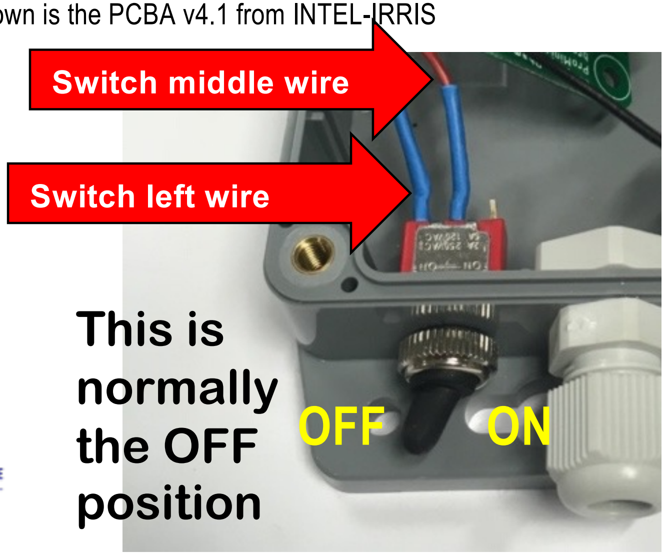


Image shown is the PCBA v4.1 from INTEL-IRRIS



This is normally the OFF position

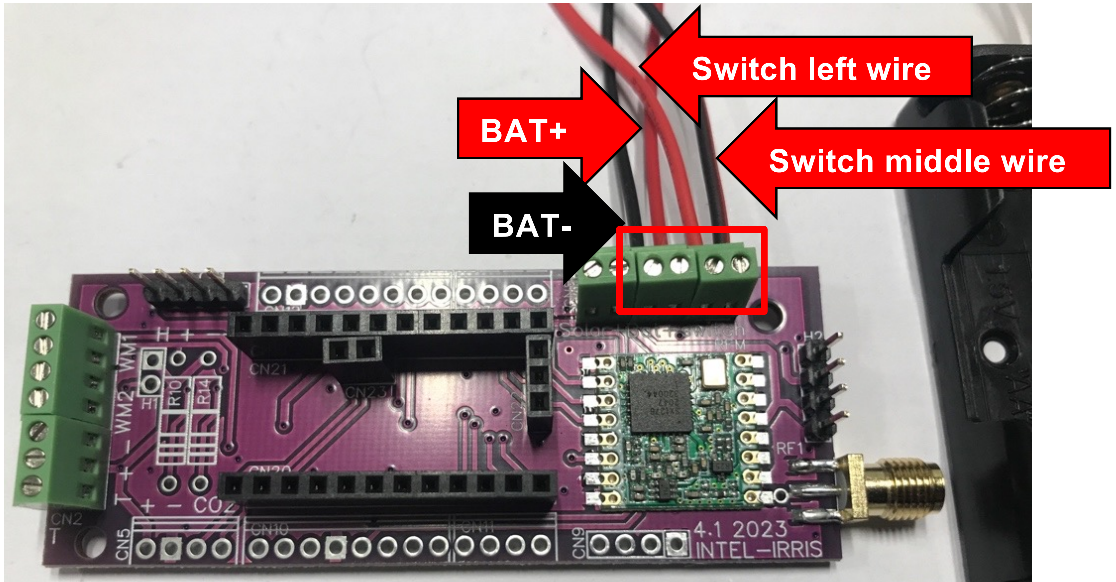


Image shown is the PCBA v4.1 from INTEL-IRRIS

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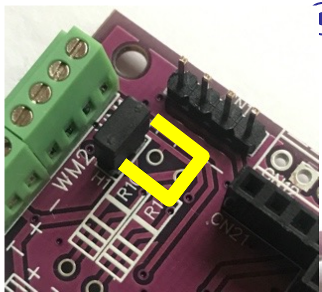
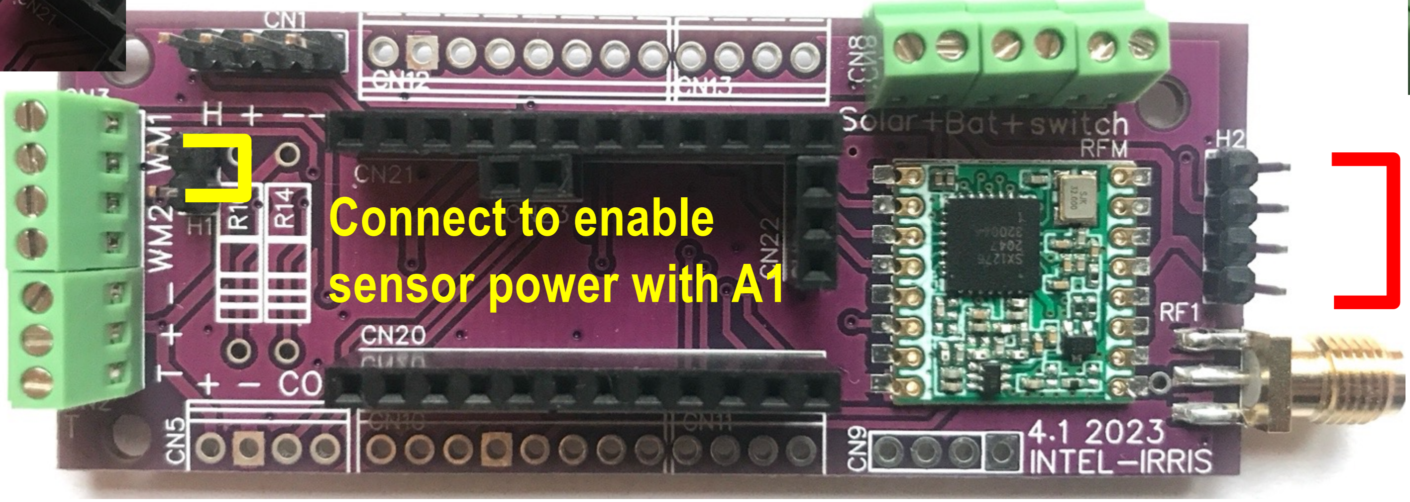
H1 & H2 header on PCB v4.1

- The raw version is not intended to have the solar circuit soldered manually on the back side
  - 2 alkaline AA batteries will be used similar to the simple PCB v2
  - It is necessary to short the 2 external pins of H2 connector and the 2 pins of H1 connector





Connect for alkaline batteries

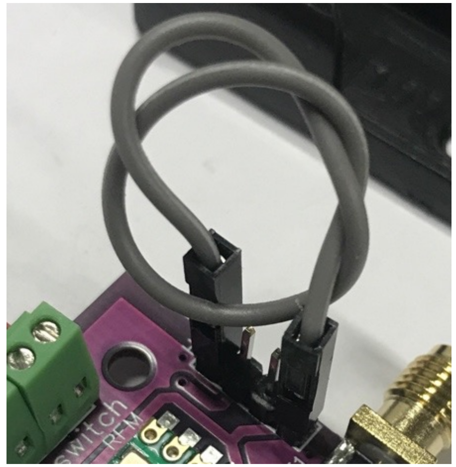


Image shown is the PCBA v4.1 from INTEL-IRRIS

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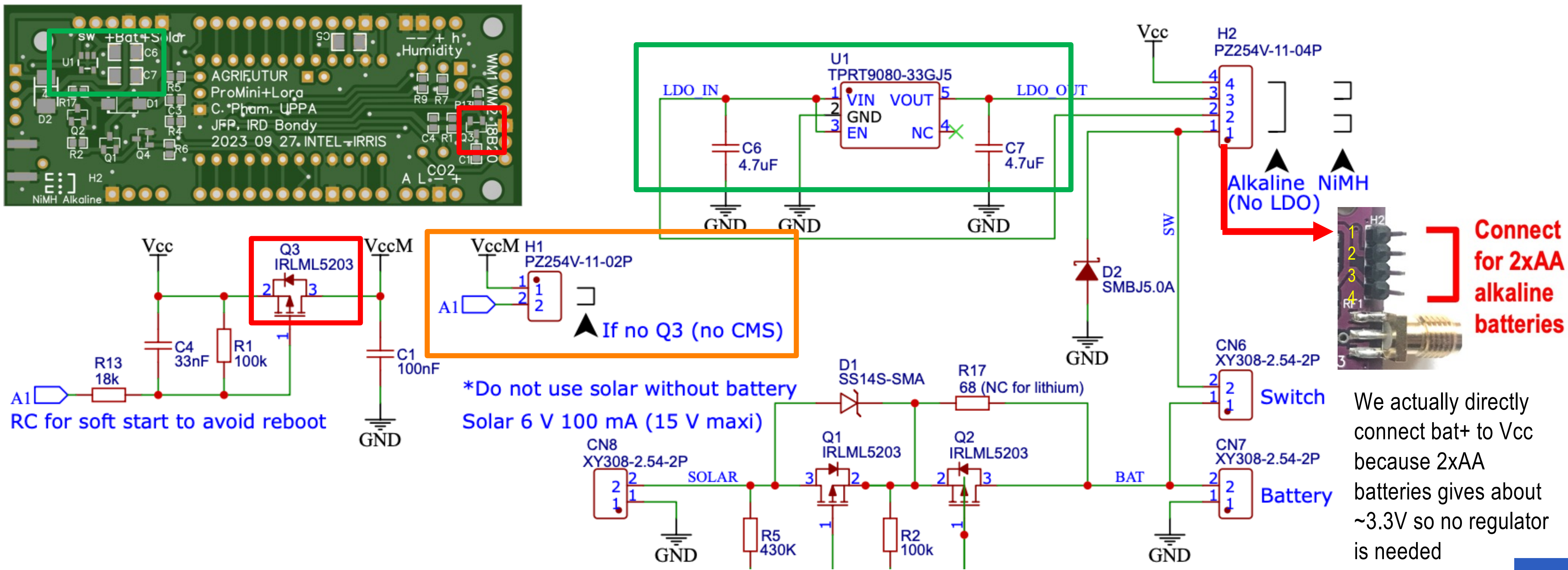
Explanation of previous slide

- As can be seen, the raw PCB does not have Q3 to connect Vcc to VccM, the power line for all physical sensors controlled by A1 pin
- Therefore H1 needs to be shorted to directly connect A1 to VccM



We actually directly connect bat+ to Vcc because 2xAA batteries gives about ~3.3V so no regulator is needed

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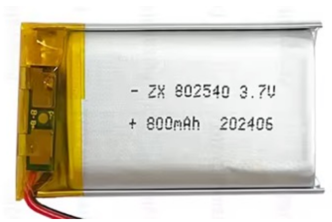
Alternatives for batteries

- Simple 2xAA 1.5V batteries delivering ~3.2V when new can be too low for some settings where the connected sensors needs a stable voltable close to 3.3V, especially when the highest quality for alkaline/lithium AA 1.5V batteries are not easily found
- There are other alternatives as illustrated below



Li-ion or LiPo = 3.7V

1xAA 14500 3.6V

1 x 18650 = 3.7V

3xAA ~ 4.8V

- The 3xAA is the simplest and the cheapest alternative

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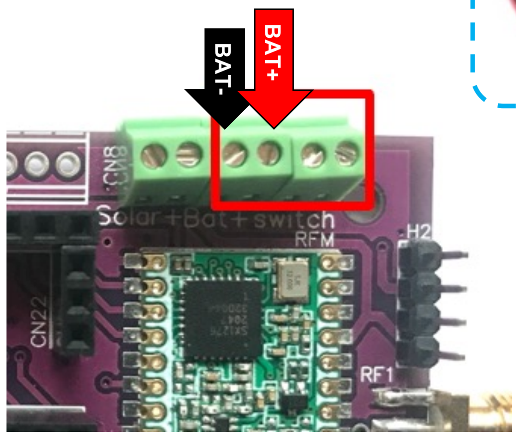
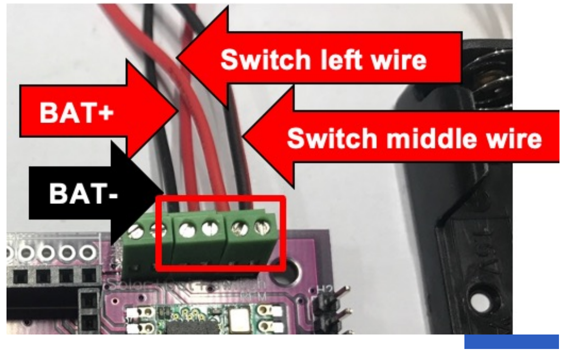
Using 3xAA alkaline batteries

- The 3xAA battery holder is too big for the selected case. If we keep the same case, here is a possible solution
- In addition to the already 2xAA battery holder, add a 1xAA battery holder
- Connect 2xAA bat- (black) to BAT-
- Connect 2xAA bat+ (red) to 1xAA bat- (black)
- Connect 1xAA bat+ (red) to BAT+
- Both battery holders can be put in the case , see next slide





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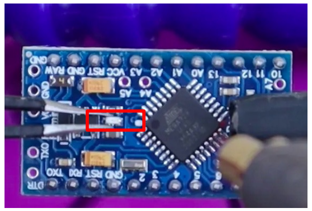
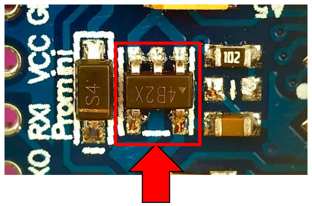
Using 3xAA, keep Pro Mini regulator

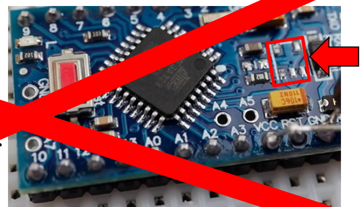
- With 3xAA providing ~ 4.8V, we **MUST KEEP** the Arduino Pro Mini on-board voltage regulator!





Do not forget to remove the power LED by just clipping it off with some wire cutters

Do not forget to remove the voltage regulator with a small plier



Only inject up to 3.6V through the VCC pin

Connect for 3xAA alkaline batteries

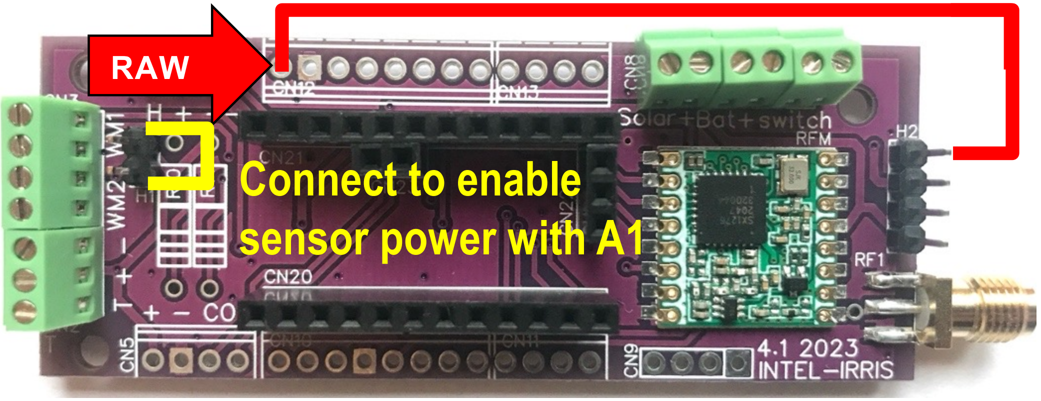
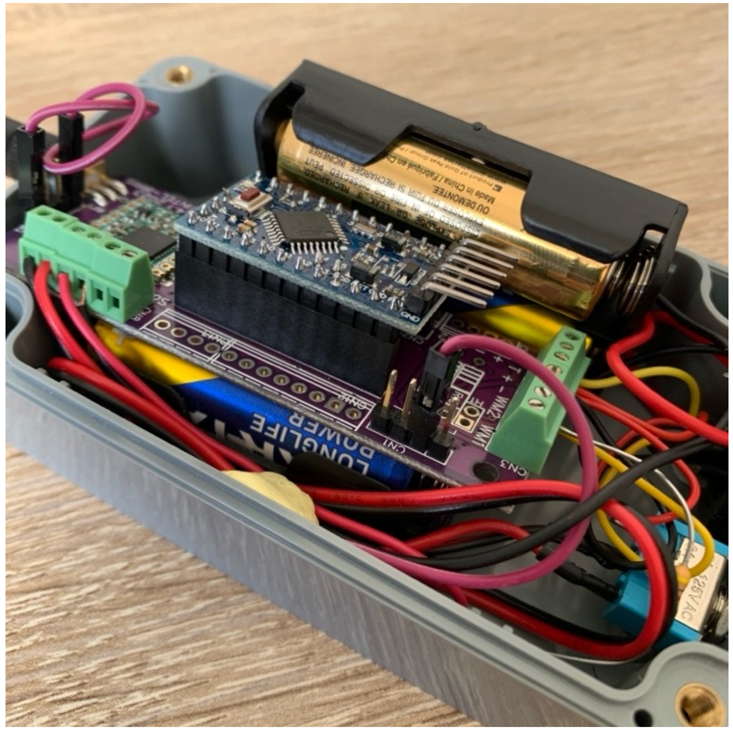


Image shown is the PCBA v4.1 from INTEL-IRRIS



Both battery holders can be put in the case

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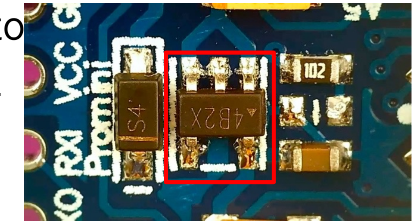
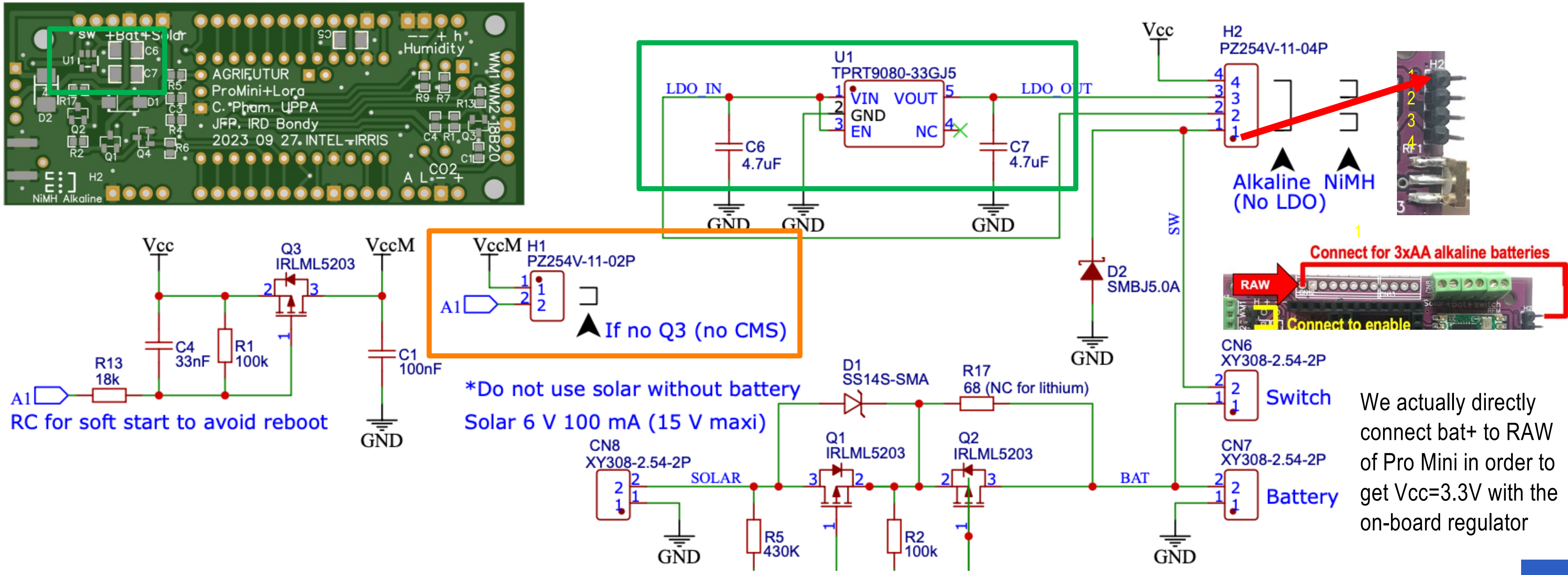
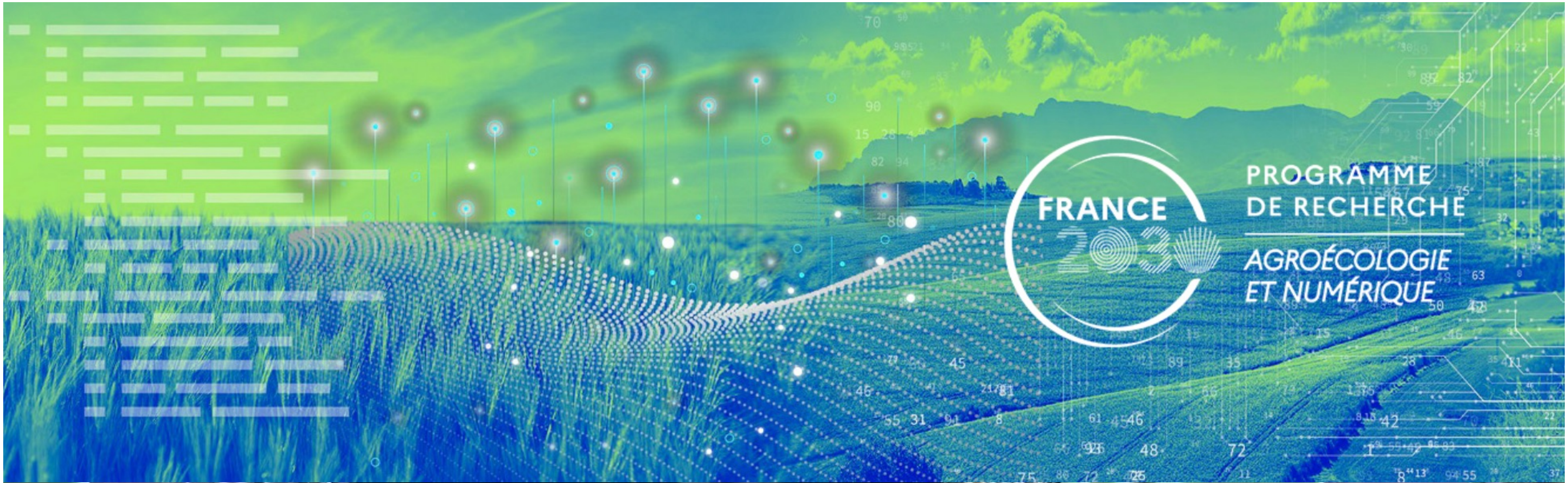
Explanation of previous slide

- With 3xAA batteries providing 4.8V, we need a voltage regulator to get down to 3.3V. As the raw PCB does not have a regulator (U1) we will <u>use the regulator of the Pro Mini that we keep for this 3xAA batteries version</u>
- Therefore we connect bat+ to RAW of Pro Mini to use its regulator which will deliver 3.3V to Vcc – consumption will be about 45uA in deep sleep
- H1 still needs to be shorted to directly connect A1 to VccM





Prof. Congduc Pham



PCBA v4.1

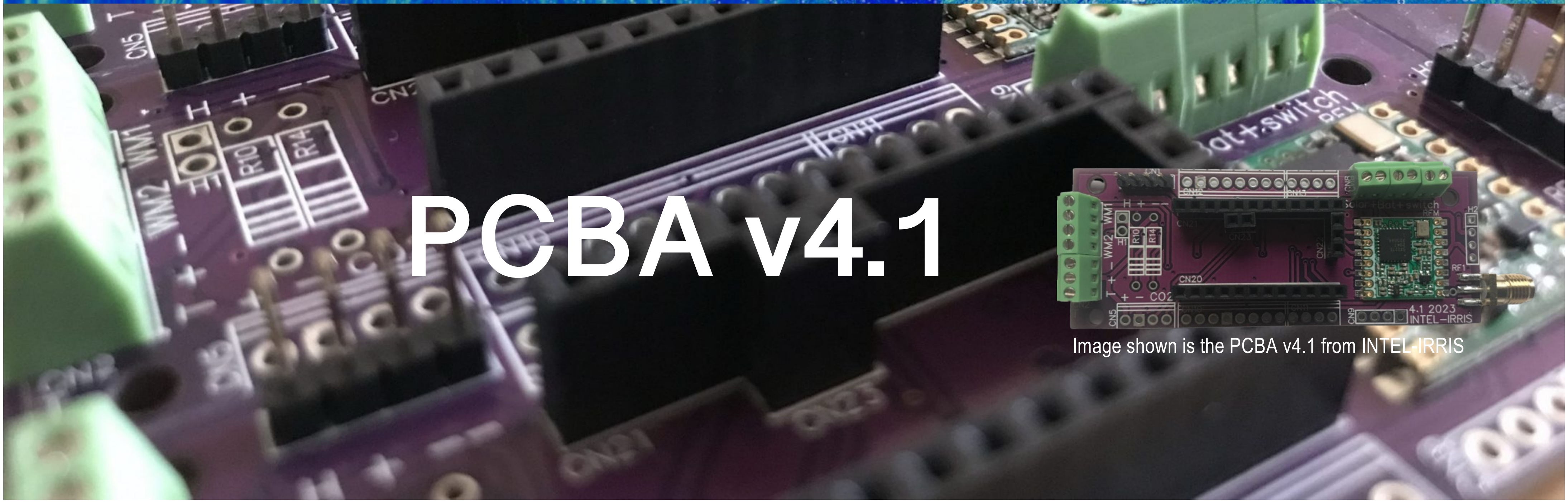


Image shown is the PCBA v4.1 from INTEL-IRRIS

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Wiring with PCBA v4.1

- The PCBA is already fully assembled, including the resistors for the temperature and watermark sensors (on the back side)

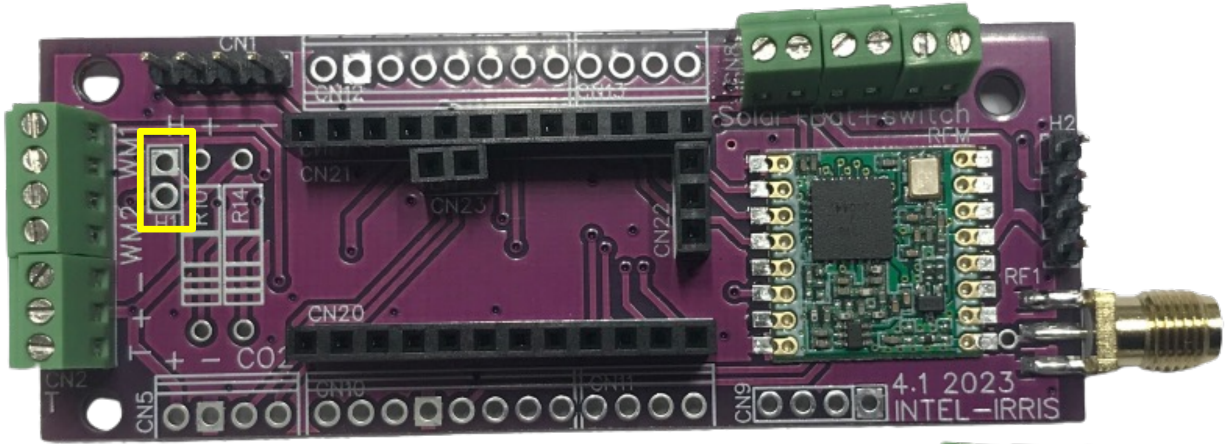

Image shown is the PCBA v4.1 from INTEL-IRRIS

The full version does not need to have header for H1

The solar circuit is on the back side

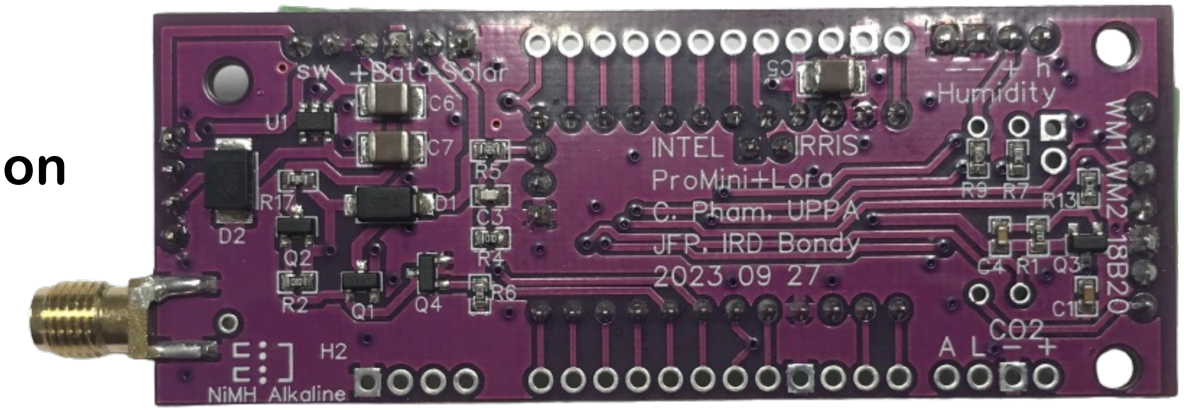

Image shown is the PCBA v4.1 from INTEL-IRRIS

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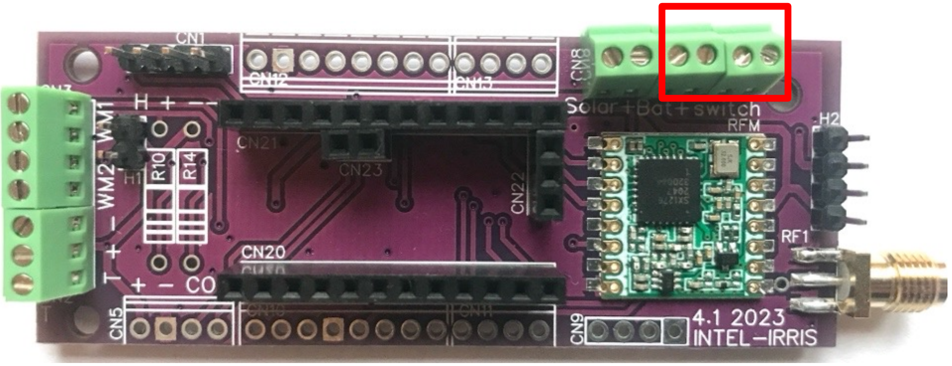
Wiring with PCBA v4.1

- Power wires (from NiMh batteries or AA alkaline batteries)



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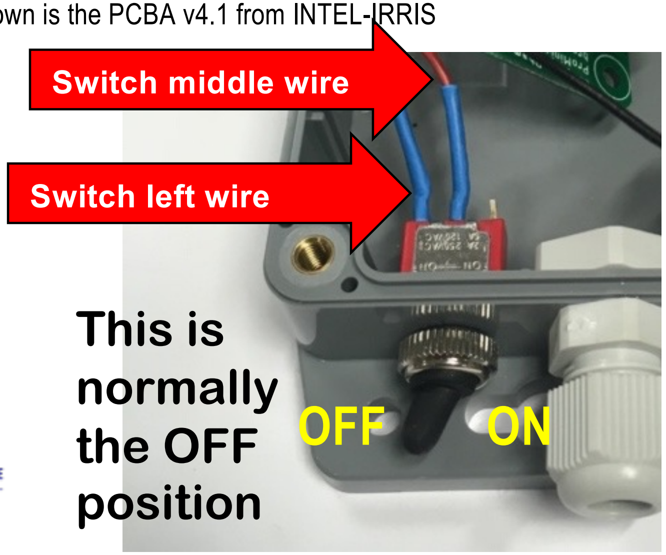

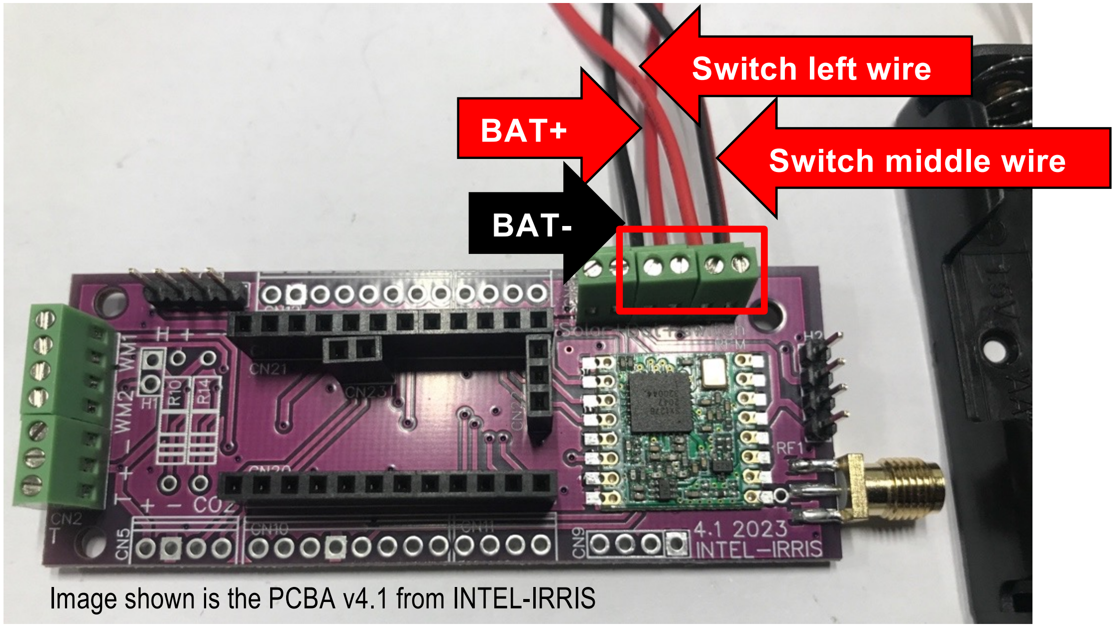

Image shown is the PCBA v4.1 from INTEL-IRRIS

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Wiring with PCBA v4.1

- The full version (PCBA) comes with the solar charging circuit
- **BUT** you can still power it with 2 alkaline AA batteries without using a solar panel to avoid external fragile parts
- You can select between solar charging with NiMh batteries or no solar charging with 2 alkaline AA batteries

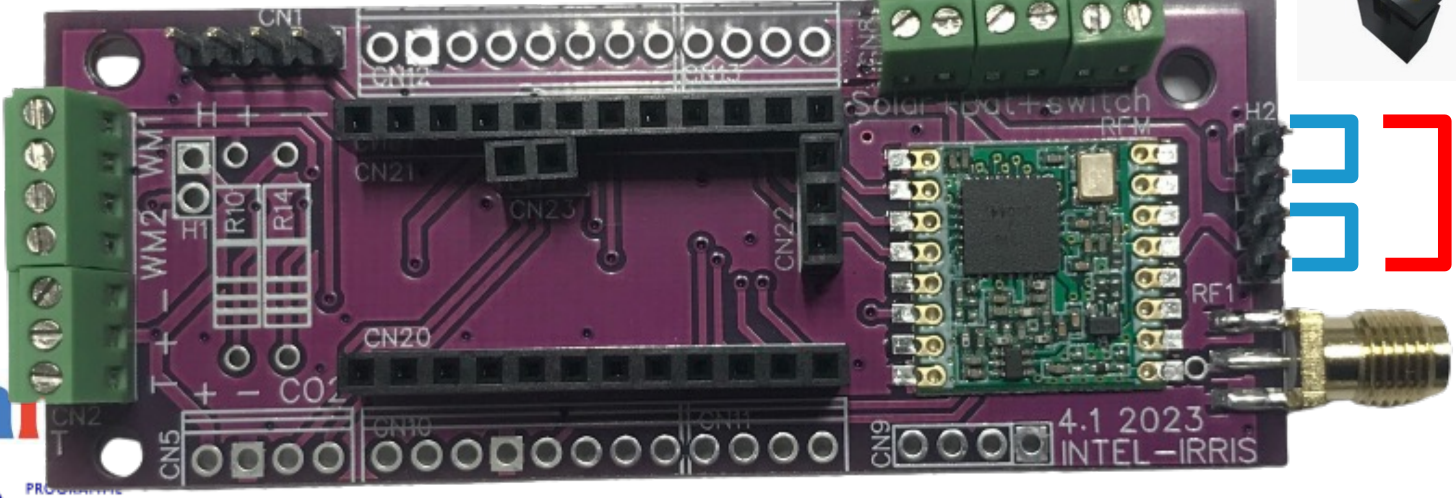
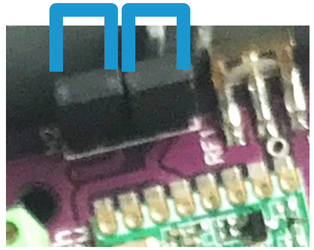
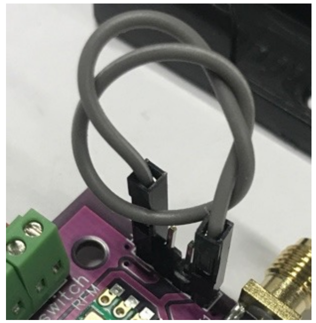


Image shown is the PCBA v4.1 from INTEL-IRRIS

Connect for SOLAR with NiMh

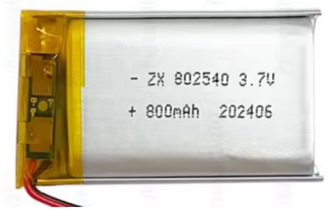


Connect for alkaline batteries



Alternatives for no rechargeable batteries

- Simple 2xAA 1.5V batteries delivering ~3.2V when new can be too low for some settings where the connected sensors needs a stable voltable close to 3.3V, especially when the highest quality for alkaline/lithium AA 1.5V batteries are not easily found
- There are other alternatives as illustrated below



Li-ion or LiPo = 3.7V

1xAA 14500 3.6V

1 x 18650 = 3.7V

3xAA ~ 4.8V

- The 3xAA is the simplest and the cheapest alternative

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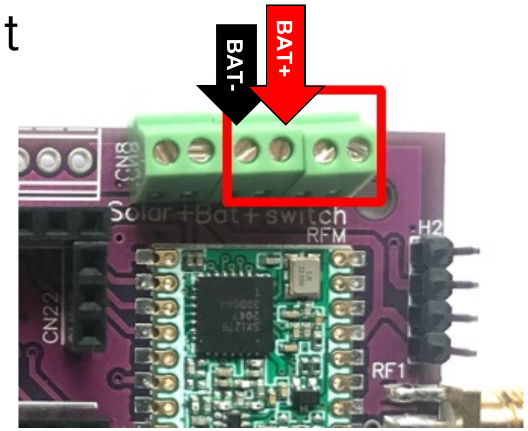
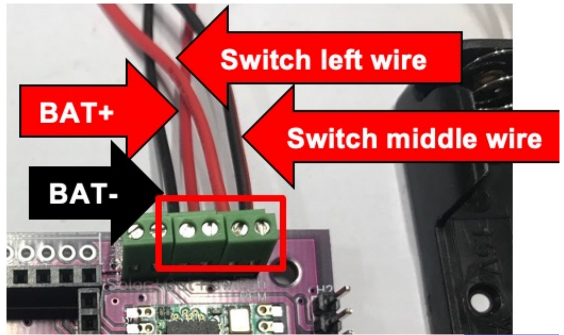
Using 3xAA alkaline batteries

- The 3xAA battery holder is too big for the selected case. If we keep the same case, here is a possible solution
- In addition to the already 2xAA battery holder, add a 1xAA battery holder
- Connect 2xAA bat- (black) to BAT-
- Connect 2xAA bat+ (red) to 1xAA bat- (black)
- Connect 1xAA bat+ (red) to BAT+
- Both battery holder can be put in the case , see next slide





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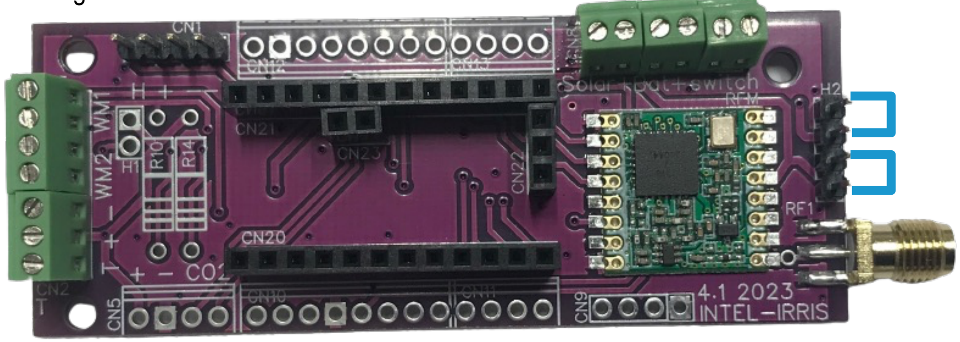
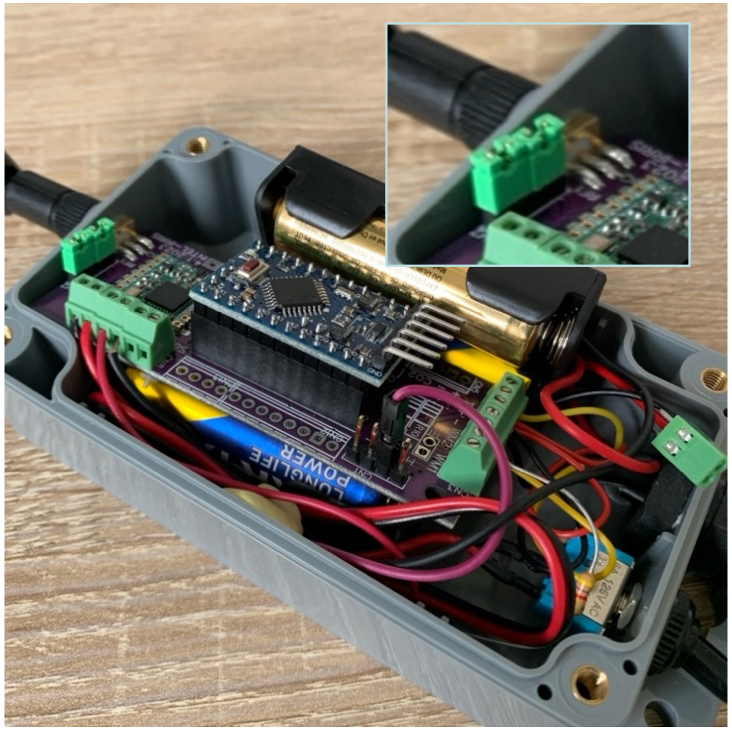
Using 3xAA, use PCBA voltage regulator

- With 3xAA providing ~ 4.8V, we can use the PCBAv4.1 on-board voltage regulator that is more efficient than the one of the Pro Mini
- The regulator on the Pro Mini should then be removed with the PCBA version
- Consumption will be about 55uA in deep sleep

Image shown is the PCBA v4.1 from INTEL-IRRIS

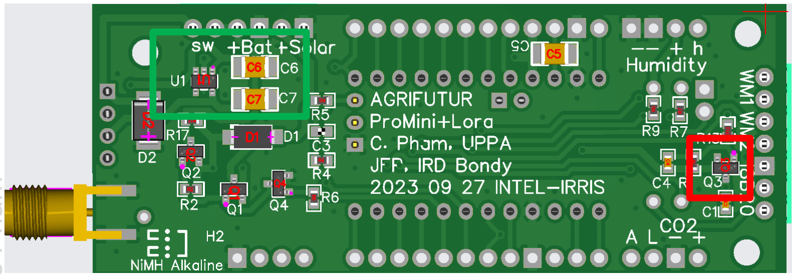


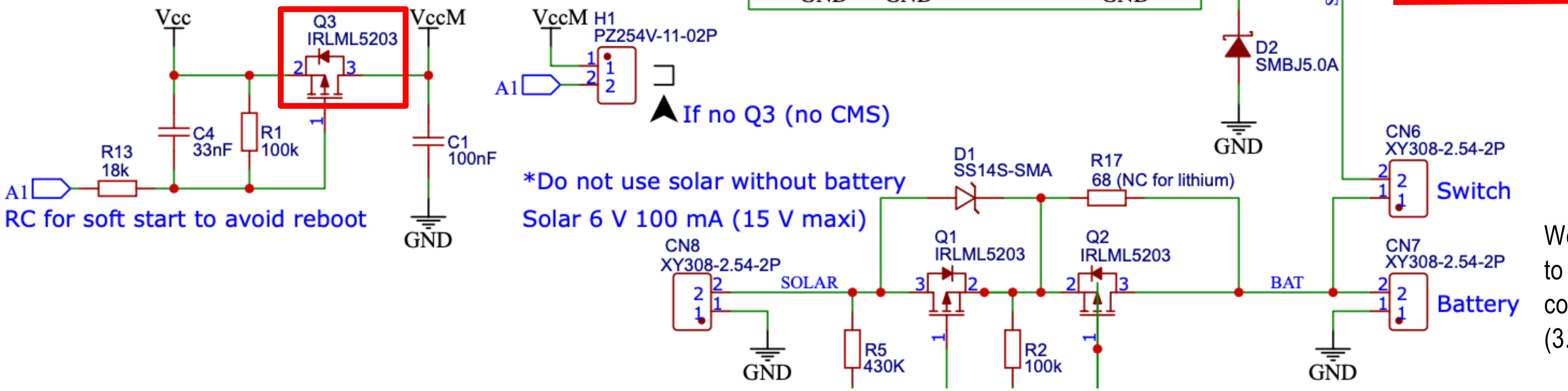


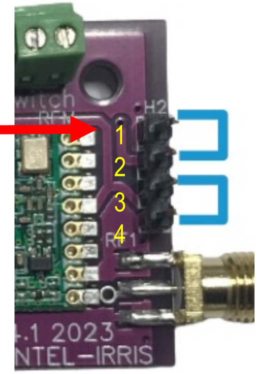
Both battery holders can be put in the case

Explanation of previous slide

- With PCBAv4.1, all components are present: Q3 allows to inject Vcc into VccM when A1 is HIGH; regulator U1 can be used to get down from 4.8V to 3.3V







We first connect bat+ to LDO IN, then we connect LDO OUT (3.3V) to Vcc

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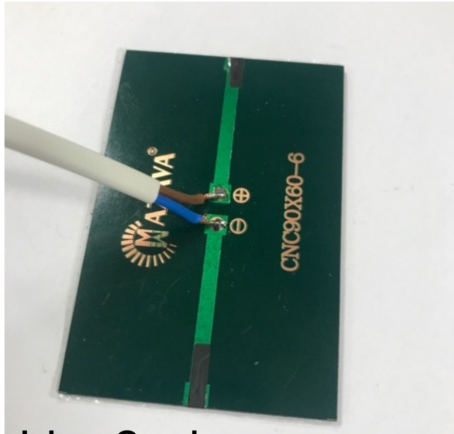
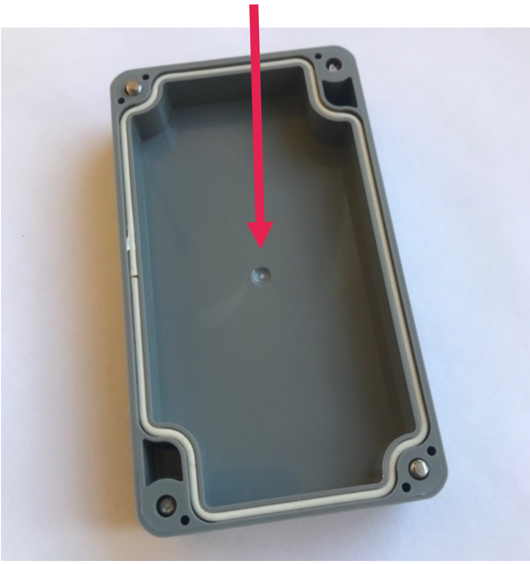
PCBA: Installing solar panel (1)

- Mini solar panel
  6 V 0.6 W 100 mA 60x90mm

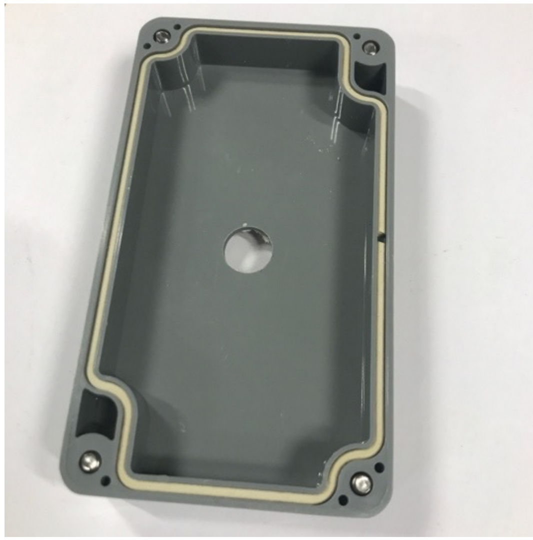


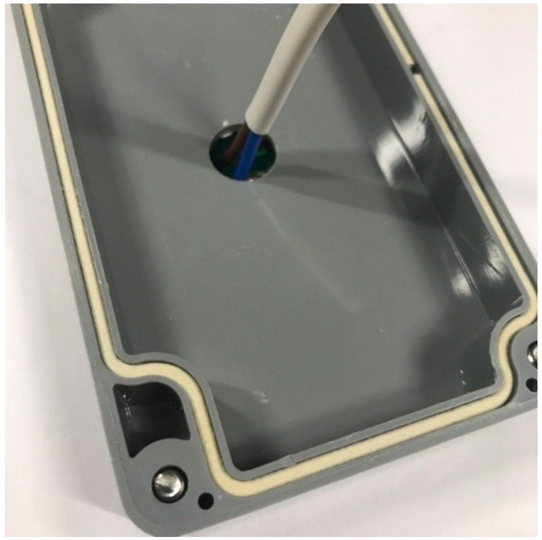
Solder 2 wires on + and -

The case cover has a mark indicating the center



Drill a hole with a 10mm to 12mm drill bit for metal





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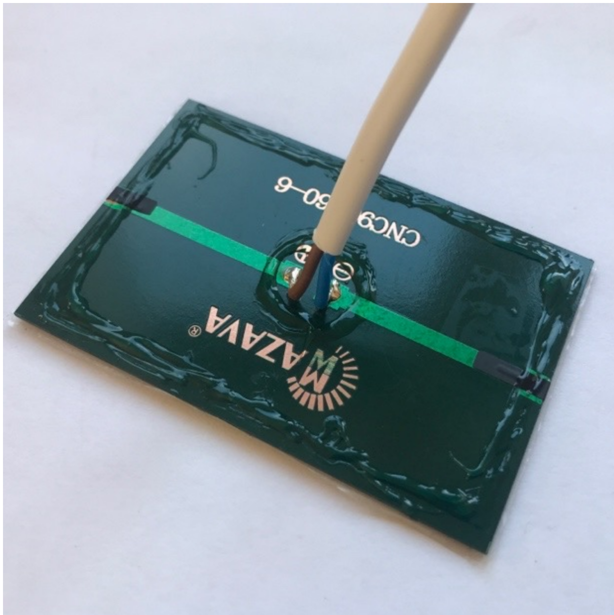
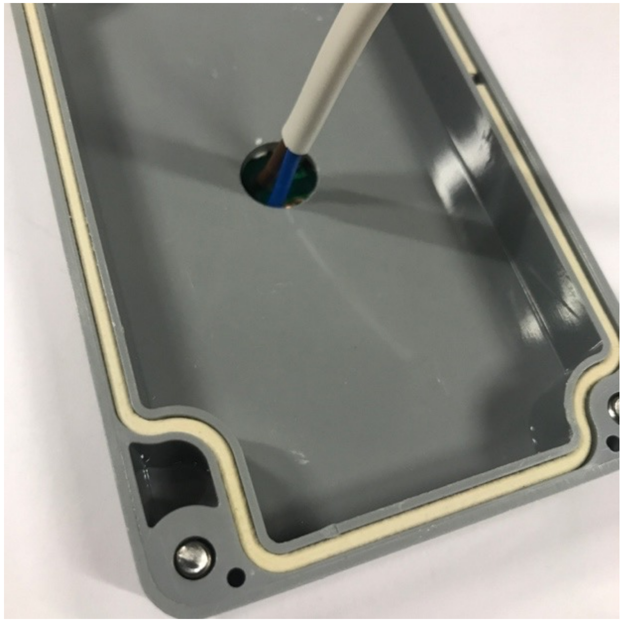
PCBA: Installing solar panel (2)

- Use silicon or glue to assemble the solar panel on the case cover
- Make sure there is no discontinuity of glue, also add glue around the centered hole surrounding the wires





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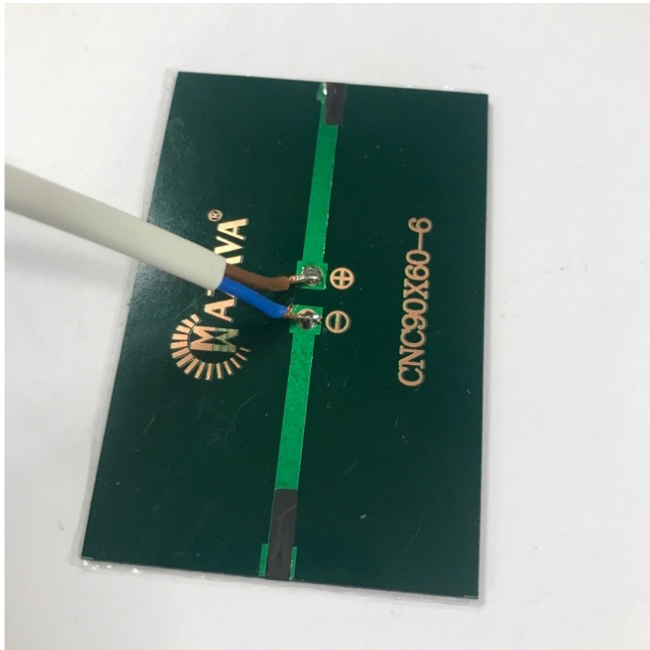
Wiring with PCB1 v4.1

- Solar panel wires



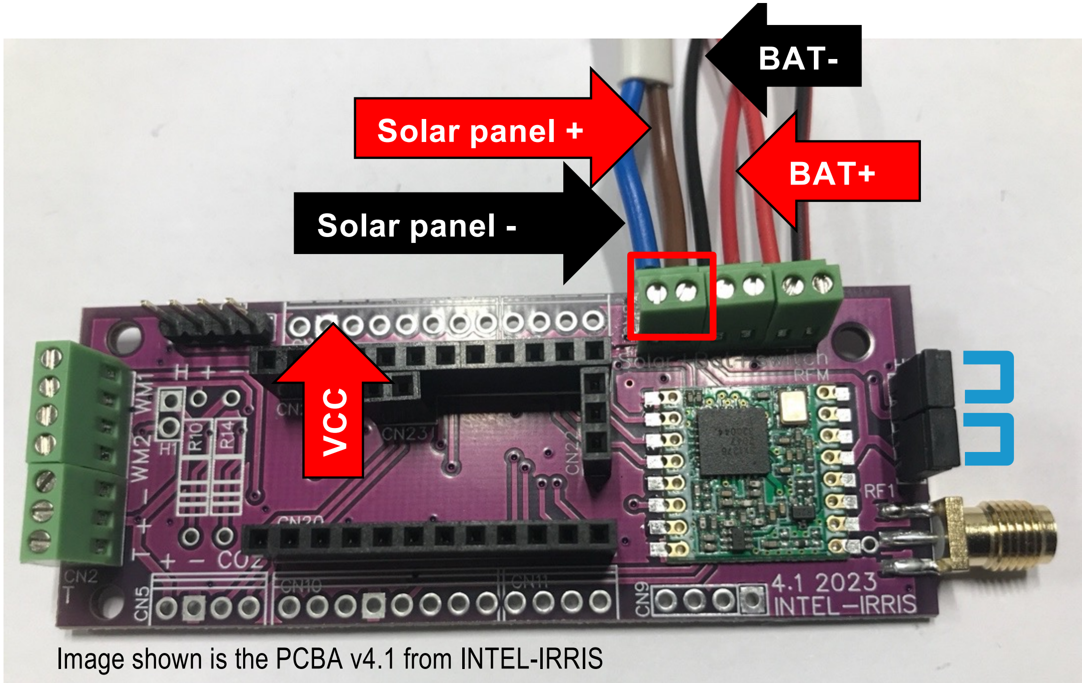
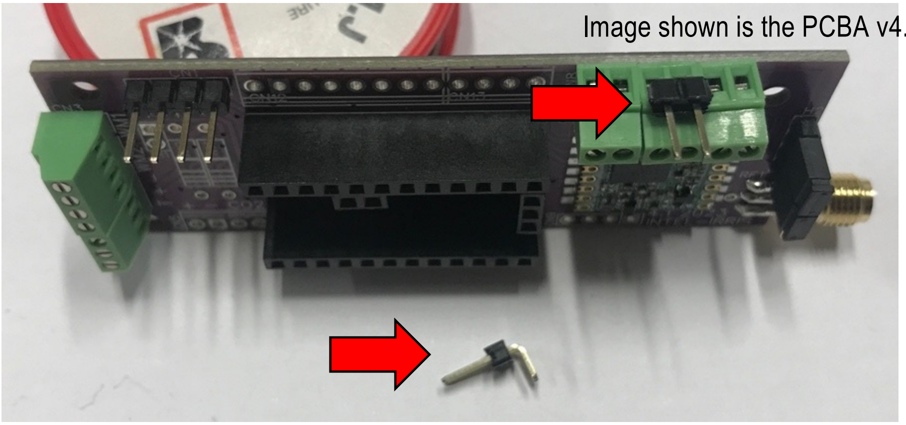


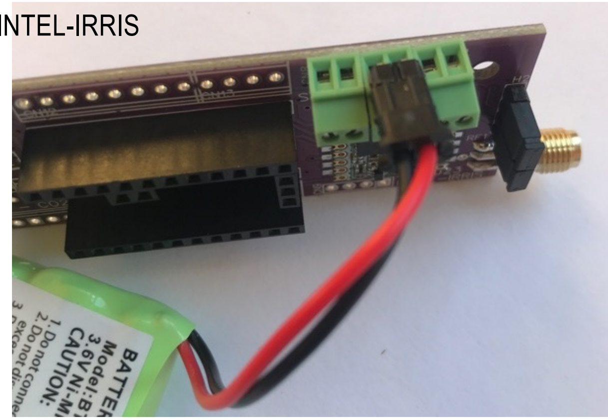
Image shown is the PCBA v4.1 from INTEL-IRRIS

Wiring with PCB1 v4.1

- Connect NiMh batterie packs with a 2-pin female connector
- You can screw in a 2-pin male 90° header

Image shown is the PCBA v4.1 from INTEL-IRRIS





You can buy NiMh battery packs with a 2 pin connector that can be used for fast connection

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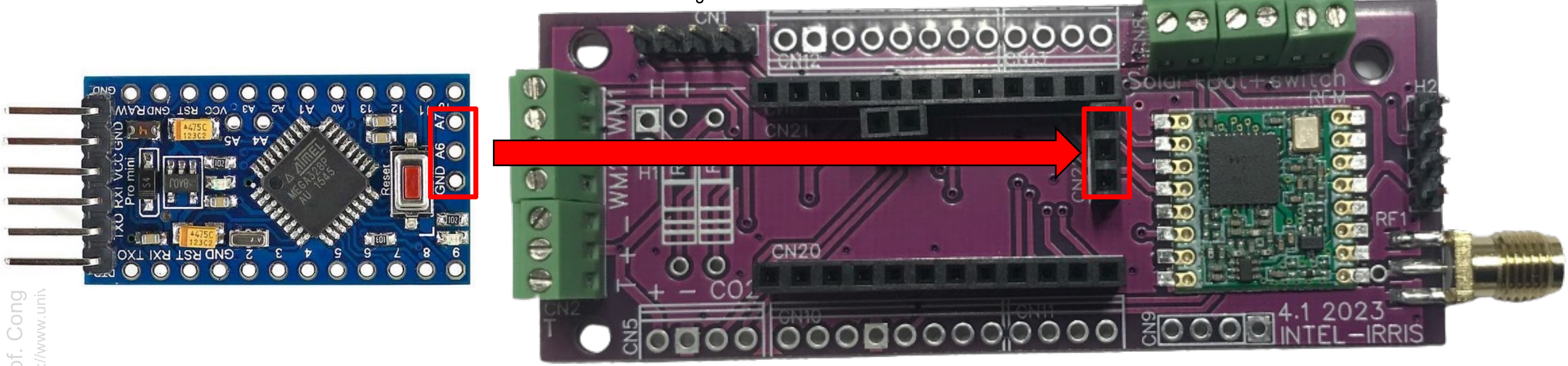
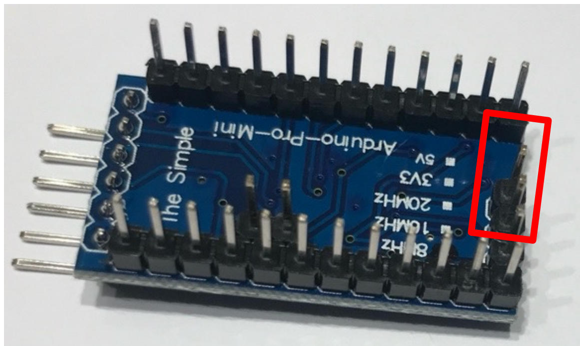
IMPORTANT: Arduino ProMini

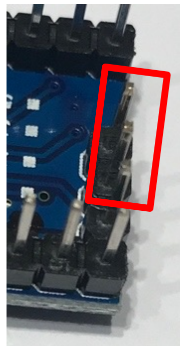
- For solar charging, using the PCBA v4.1, the Arduino ProMini MUST have pin A7 connected!

Image shown is the PCBA v4.1 from INTEL-IRRIS



- So, it is MANDATORY to solder a 3-pin header for GND, A6 & A7 on the Arduino ProMini board





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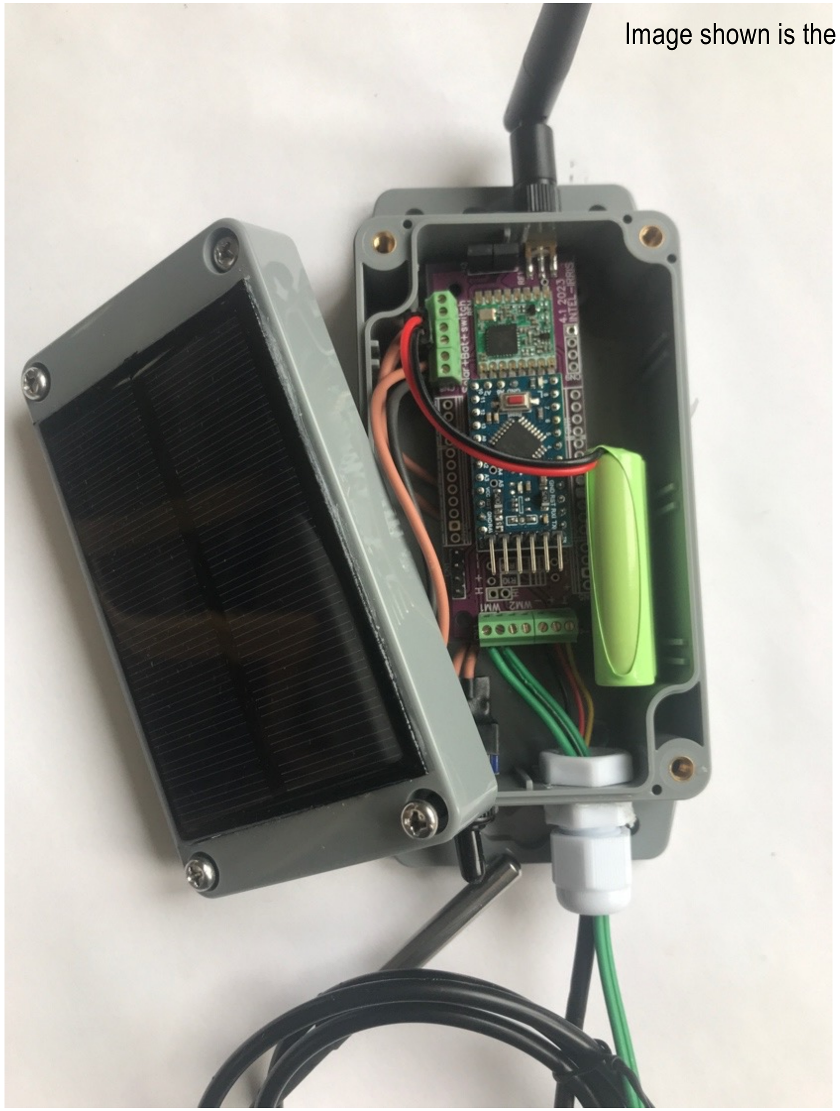
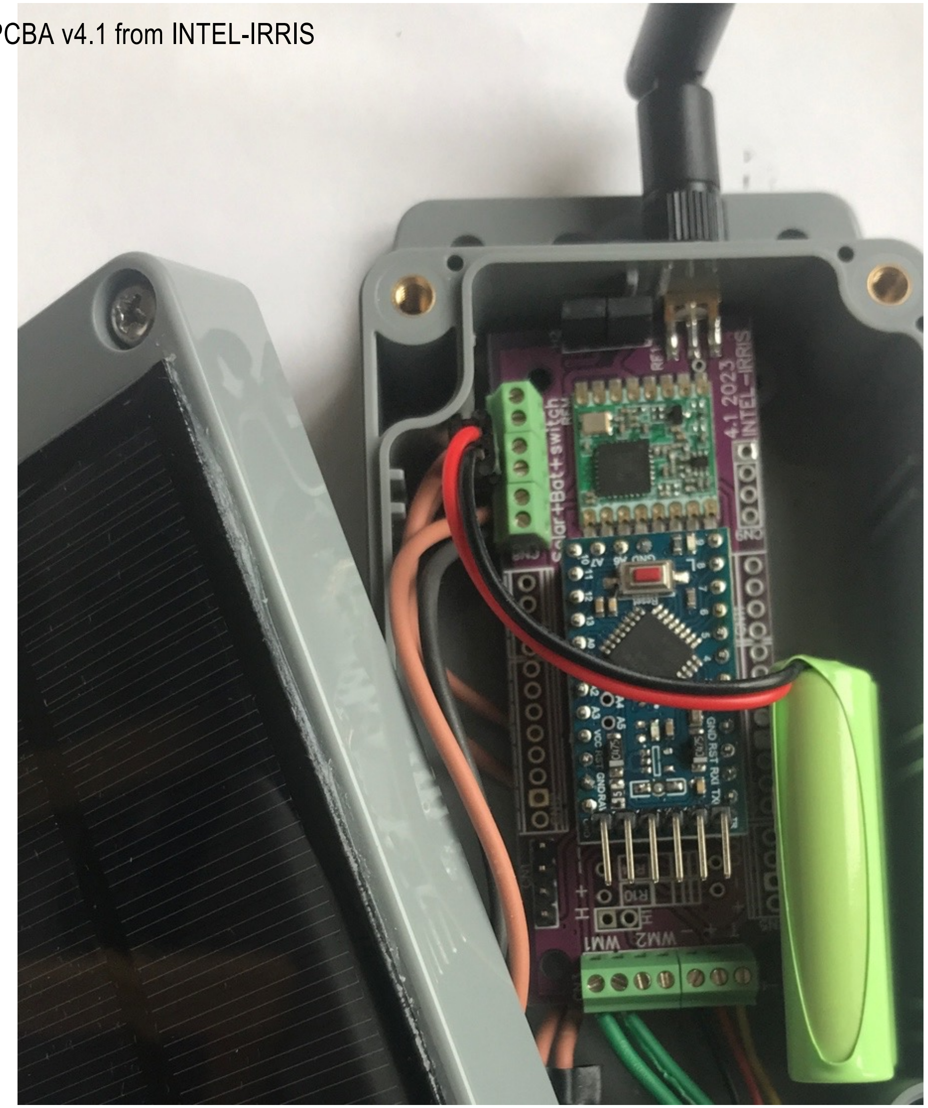
Soil device with PCBA v4.1, final result

Image shown is the PCBA v4.1 from INTEL-IRRIS





Prof. Congduc Pham
http://www.univ-pau.fr/~cpham

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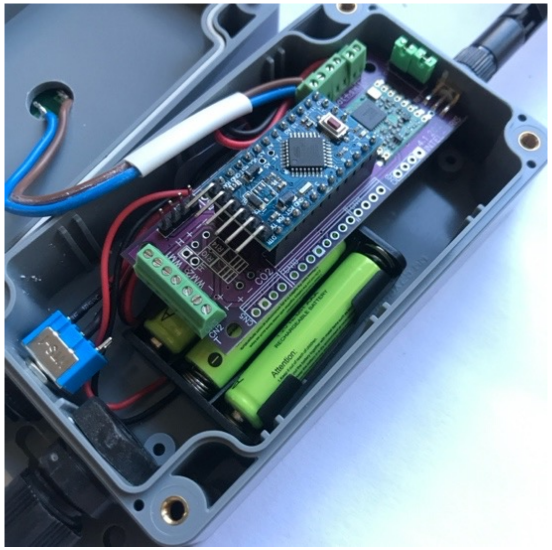
Using 3xAAA batteries

- You can alternatively use 3xAAA NiMh rechargeable batteries with a battery holder
- This would allow you to slip the 3xAAA battery holder under the PCBA board



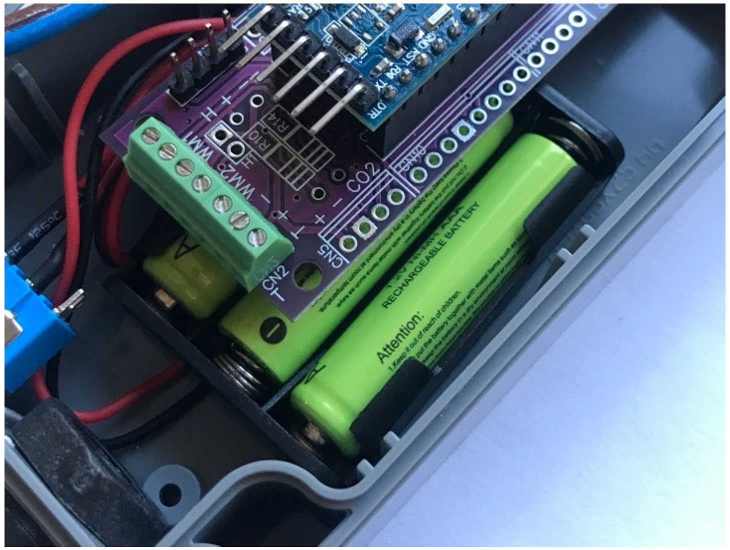


Image shown is the PCBA v4.1 from INTEL-IRRIS

Externally charging NiMH batteries (1)

- Normally, the solar panel should charge the NiMH batteries during the day, with sufficient power for regular operation of the device
- However, if the device has been stored for a long period in the dark, the battery charge level may have dropped too much
- In this case, it will need to be charged by an external source of power
- If you are using 3xAAA NiMh rechargeable batteries with a battery holder, it is straightforward to charge the individual AAA batteries with an NiMh battery charger
- If you are using a 3xAAA battery pack, you can directly use the PCBA to charge the pack as described in the next slide

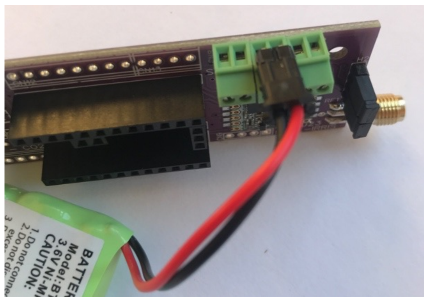
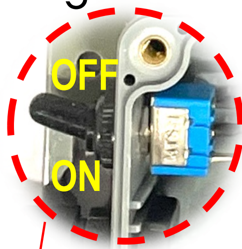


Image shown is the PCBA v4.1 from INTEL-IRRIS

Externally charging NiMH batteries (2)

- With a solar panel and the 3xAAA NiMh battery pack connected to the PCBA, it is possible to directly charge the battery pack using the on-board regulator for slow charging mode
- First, the Arduino ProMini microcontroller must be OFF
- Then, disconnect the 2 wires of the solar panel
- And connect instead an external DC source at 5V

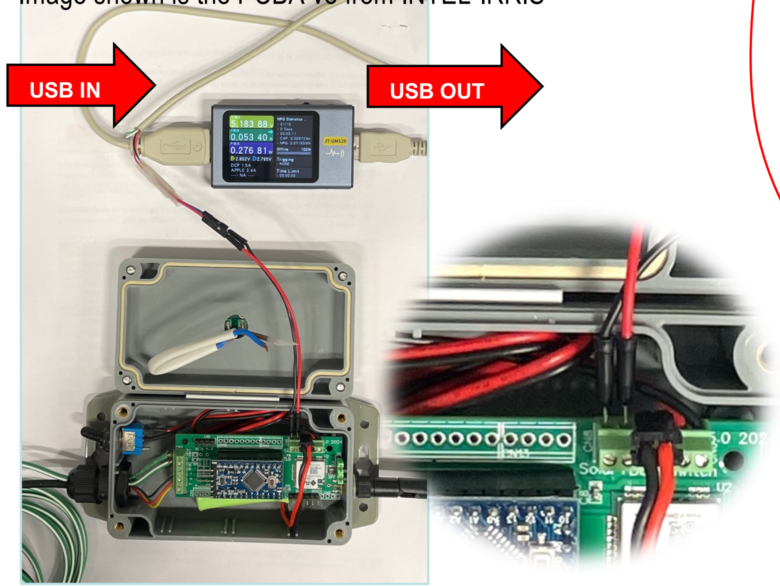


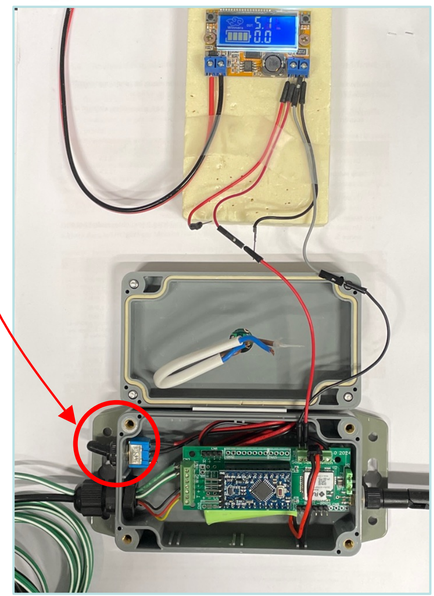
Make sure that switch is on OFF position

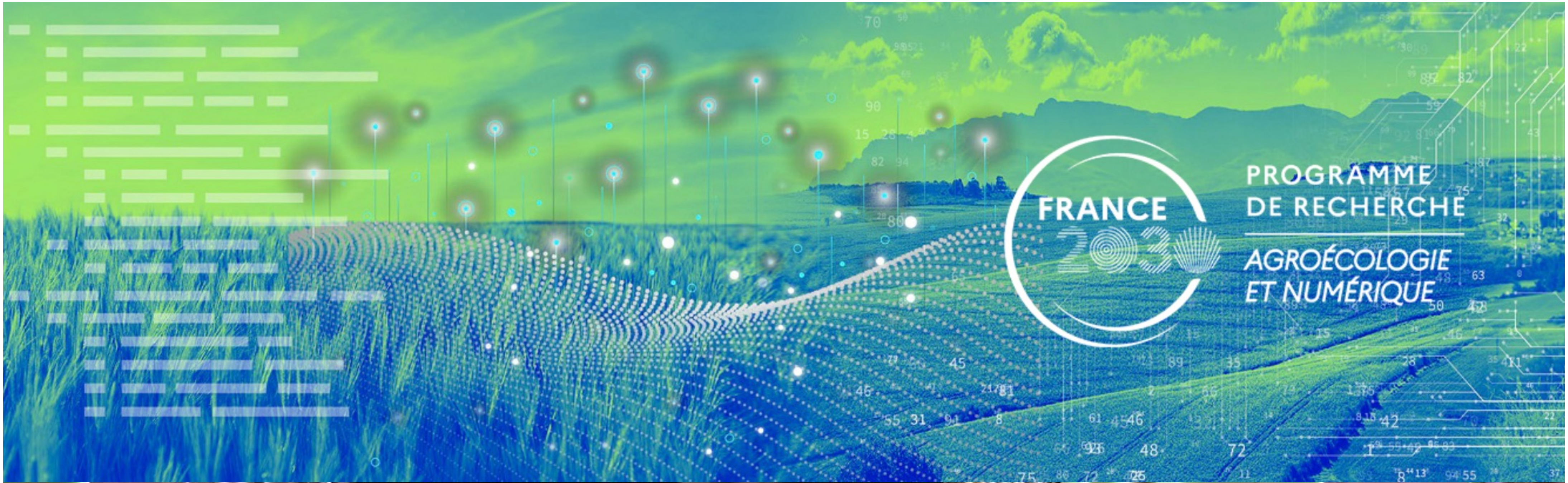
You can simply take power from a USB source – with a modified USB cable – here we show with a USB multimeter in-between

Or, you can use an adjustable DC-DC step-down buck converter

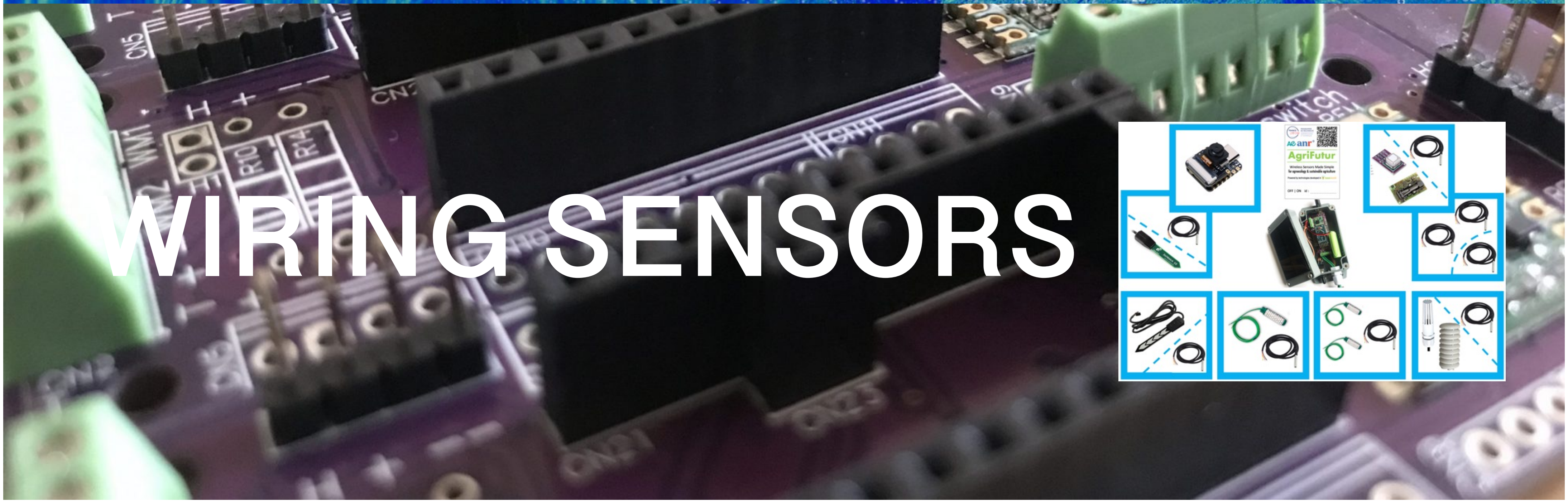
Image shown is the PCBA v5 from INTEL-IRRIS







WIRING SENSORS



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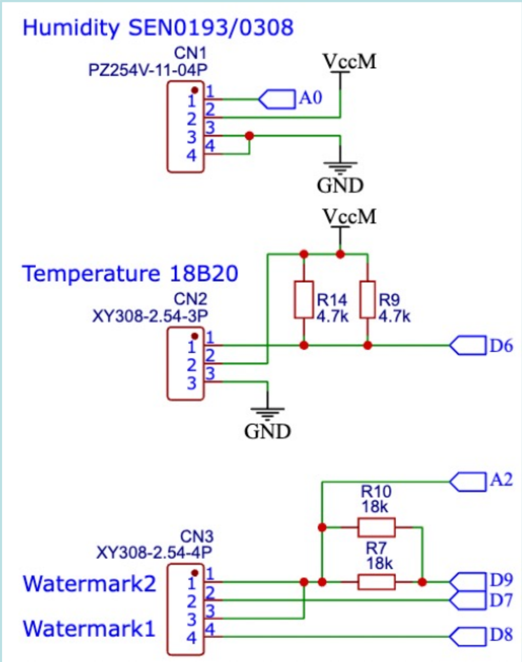
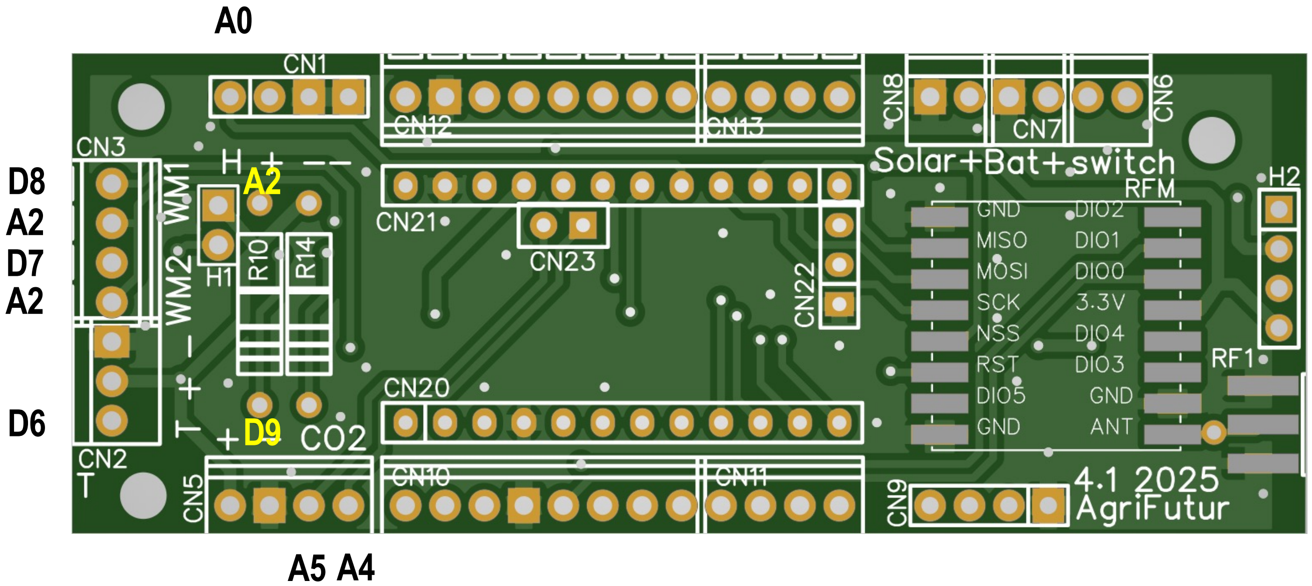
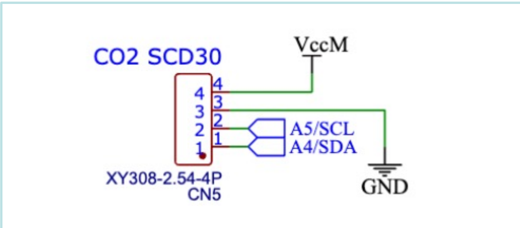
PCB/PCBA v4.1 pin description

- The PCB has the following pre-defined wiring to Arduino ProMini pin
  https://github.com/CongducPham/PEPR_AgriFutur/blob/main/PCBs/IRD_PCB_4_1/Schematic_IISS_4_1_2023-09-27.pdf







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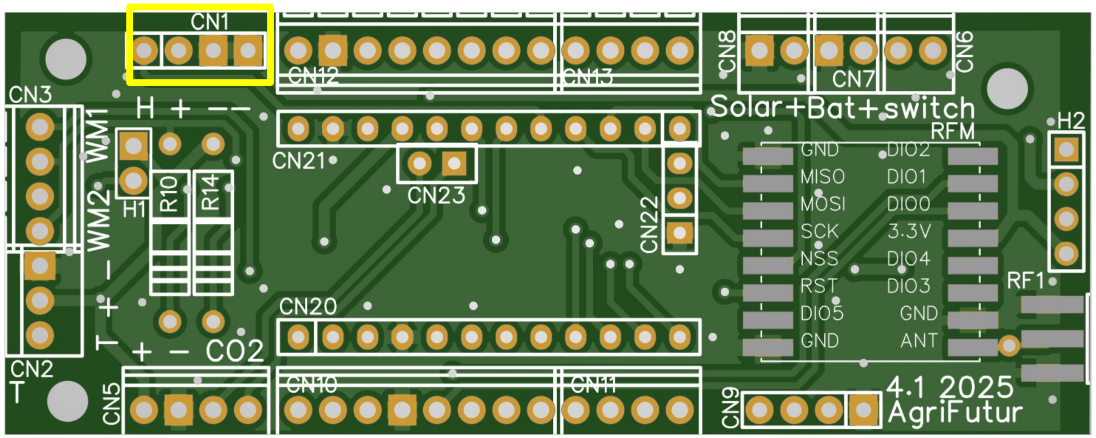
Wiring sensors with PCB/PCBA v4.1

- SEN0308 capacitive (or other capacitive)



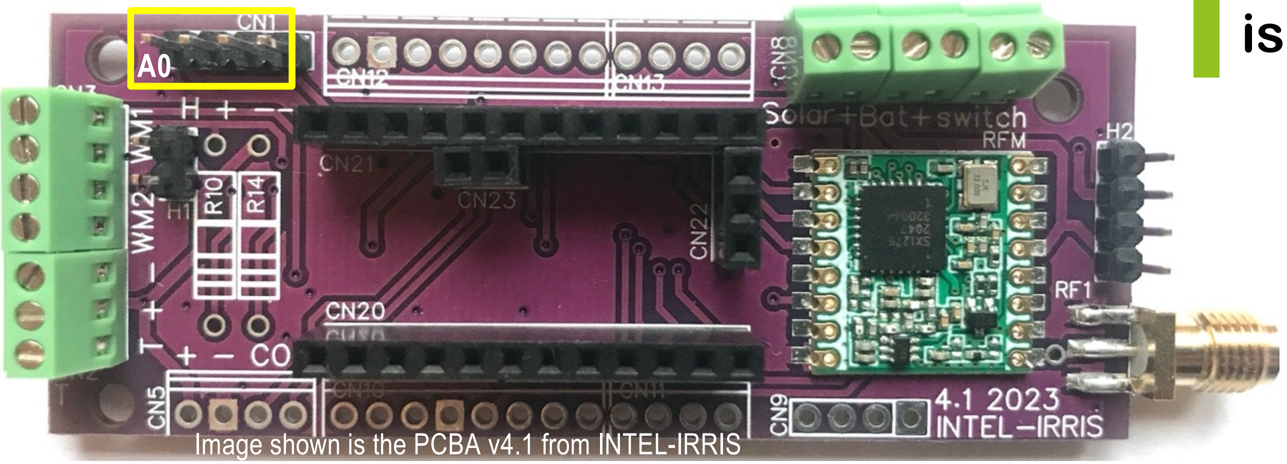
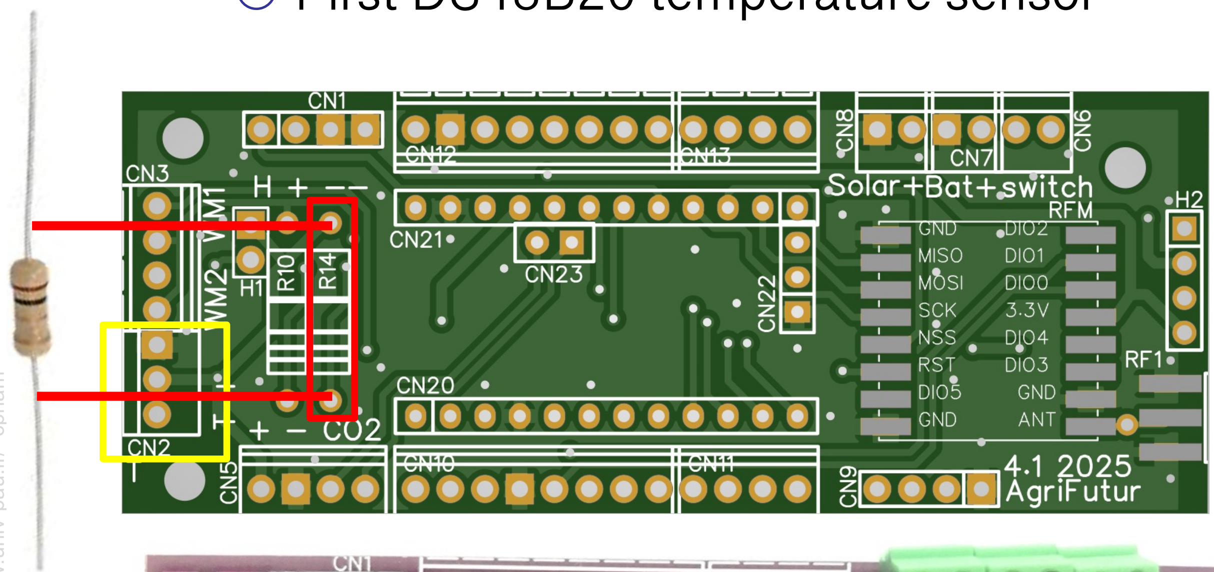


Image shown is the PCBA v4.1 from INTEL-IRRIS

Just connect the sensor in the dedicated header
– – are the 2 black wires
+ is red and H is yellow
H is the analog output and is linked to A0

Wiring sensors on PCB/PCBA v4.1

- First DS18B20 temperature sensor



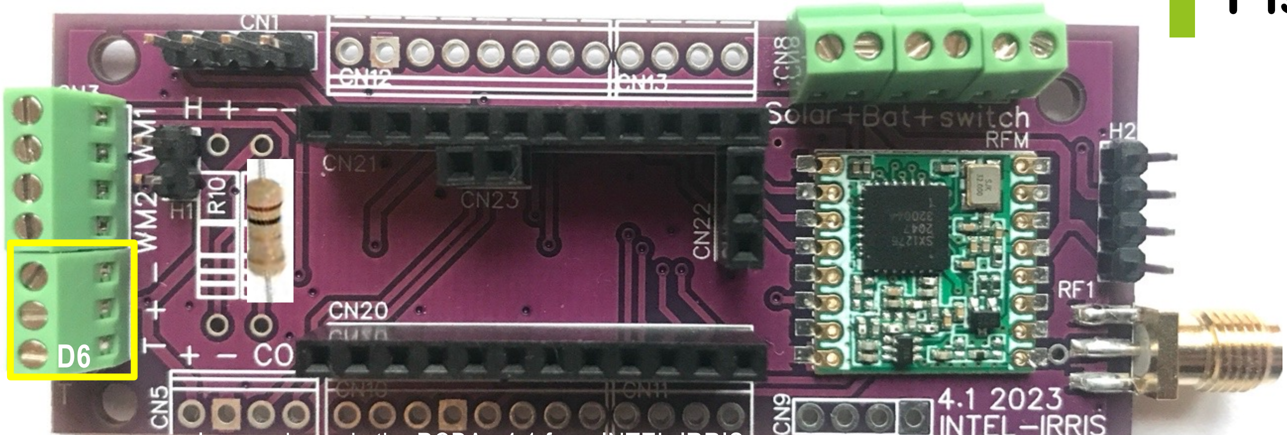
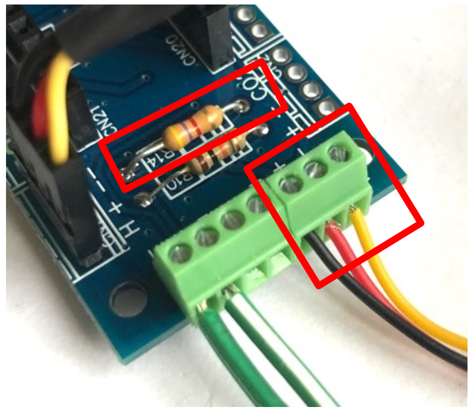


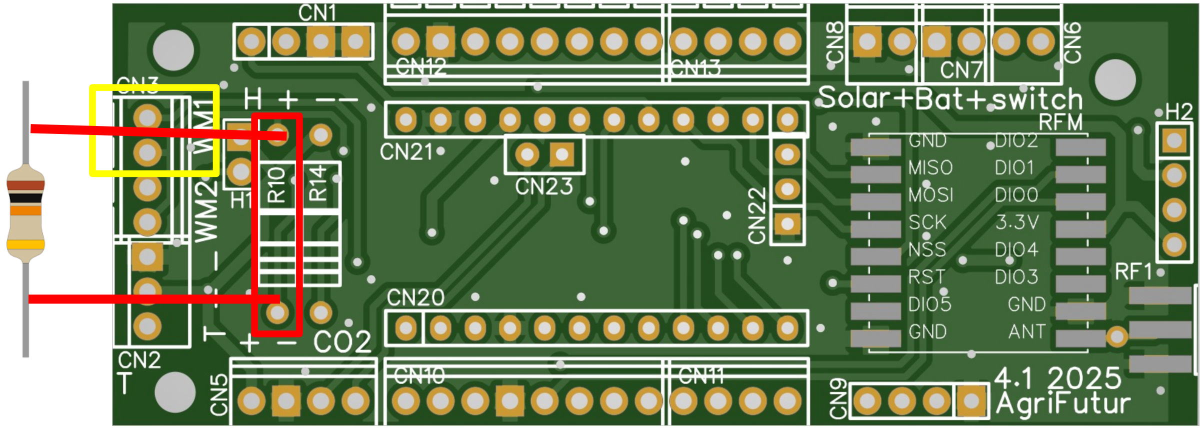
Image shown is the PCBA v4.1 from INTEL-IRRIS

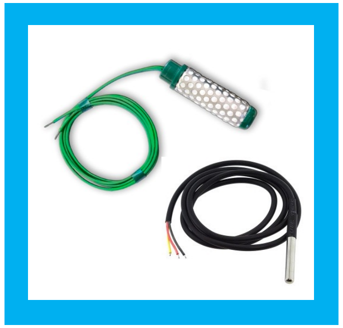
Solder a 4.7kOhms resistor (raw PCB only) then wire in the dedicated terminal block T+ – : Yellow, Red, Black wires T is linked to D6



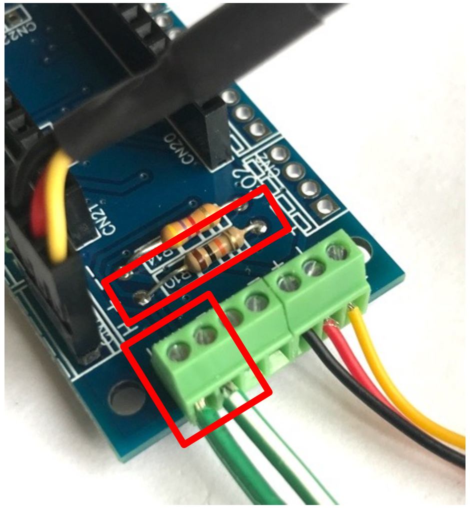
Wiring sensors on PCB/PCBA v4.1

- First Watermark





Solder a 10kOhms resistor (raw PCB only) then wire in the dedicated WM1 terminal block



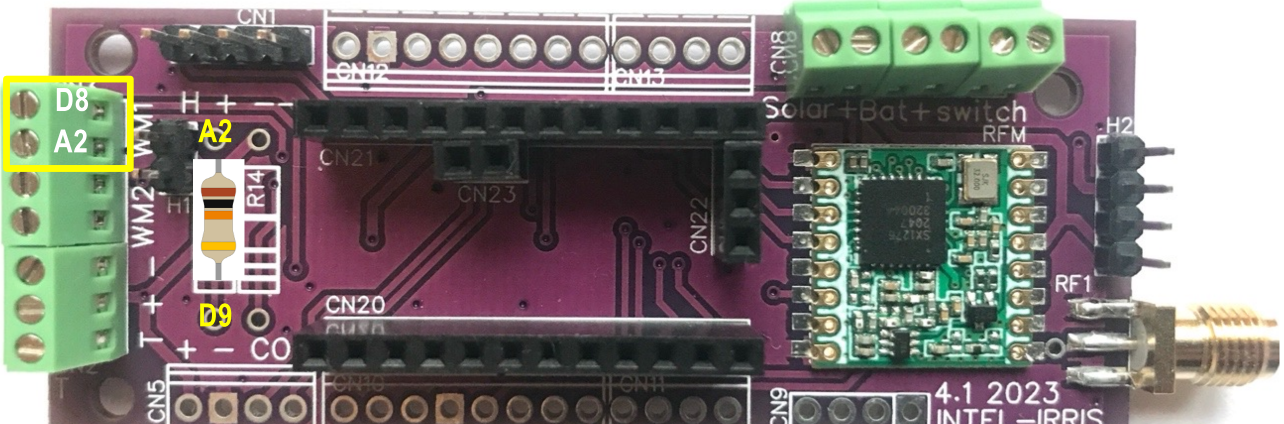
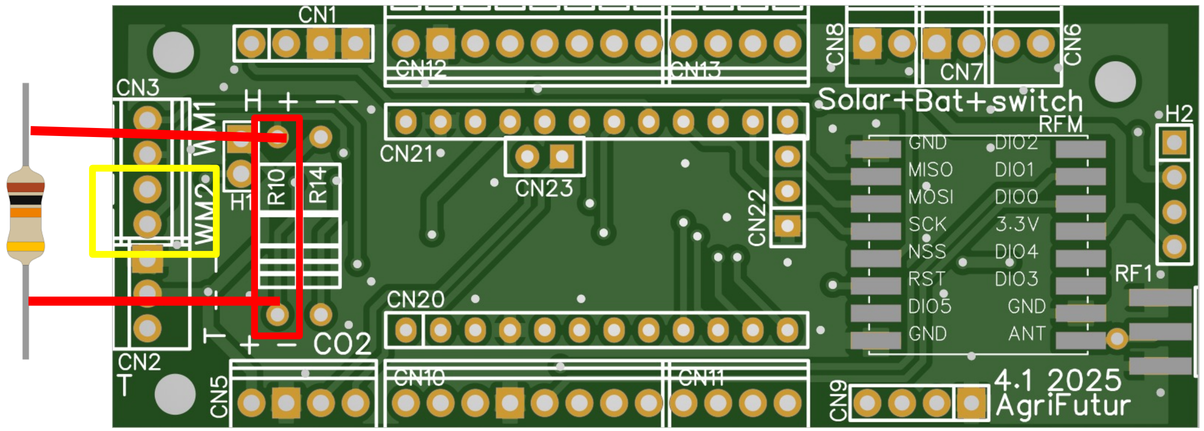


Image shown is the PCBA v4.1 from INTEL-IRRIS

Wiring sensors on PCB/PCBA v4.1

- Second Watermark



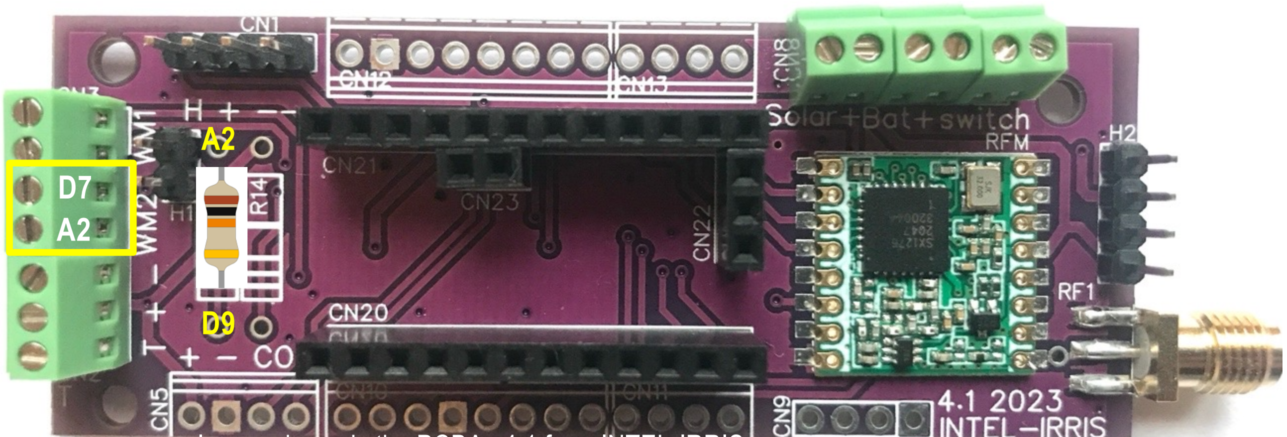
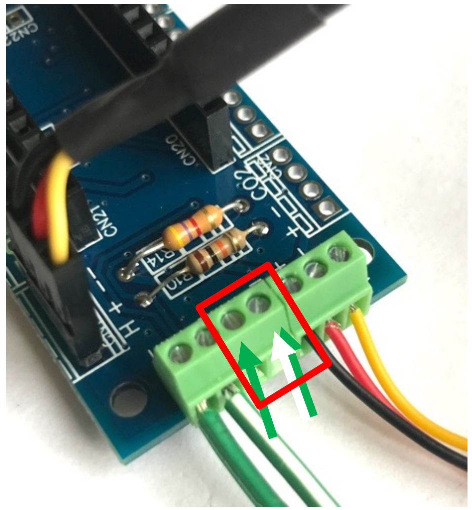


Image shown is the PCBA v4.1 from INTEL-IRRIS

No additional resistor needed just wire in the dedicated WM2 terminal block



Wiring sensors on PCB/PCBA v4.1

- 2nd & 3rd DS18B20 temperature sensor

For 2nd DS18B20: connect yellow to D7 and connect a 4.7kOms resistor between yellow and + taken on the CN1 connector. You can use a Dupont wire or a jumper.

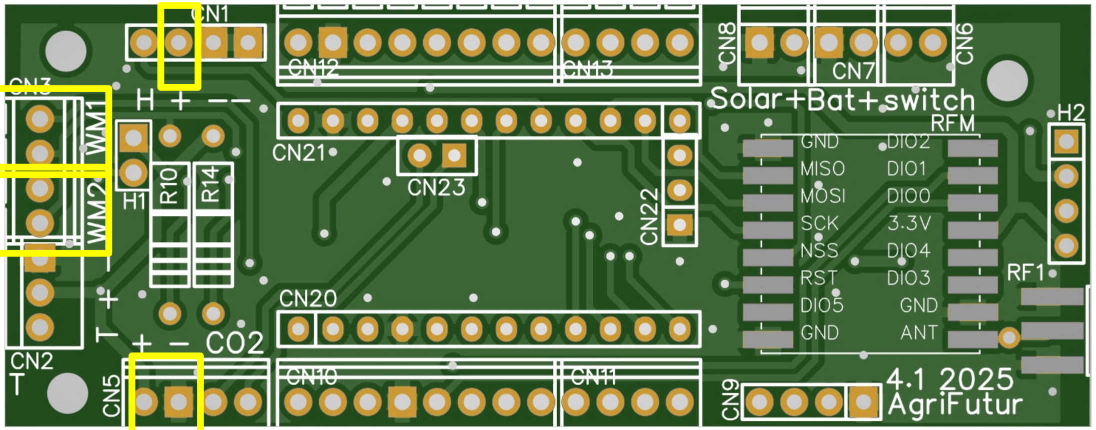
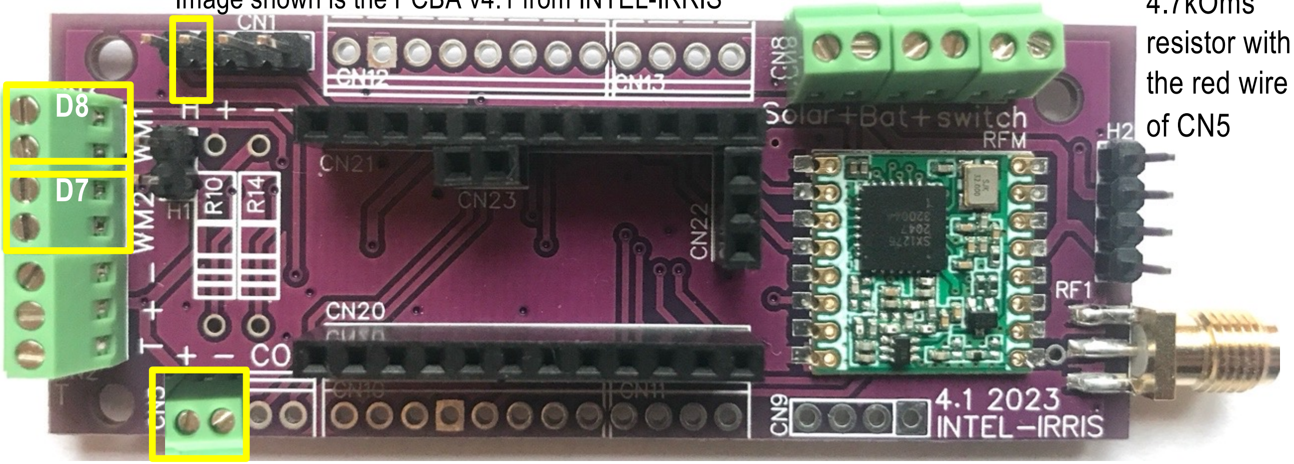
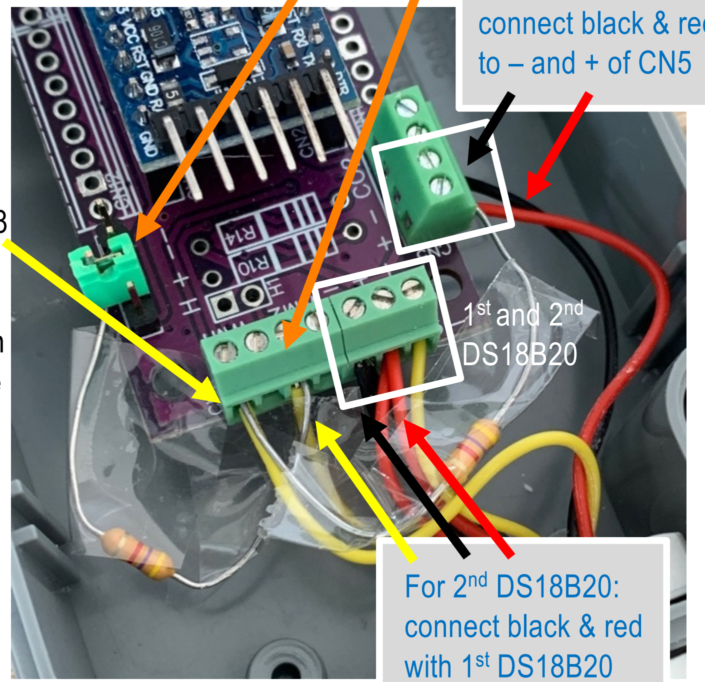


Image shown is the PCBA v4.1 from INTEL-IRRIS



For 3rd DS18B20: connect yellow to D8 and the 4.7kOms resistor with the red wire of CN5

For 3rd DS18B20: connect black & red to – and + of CN5



For 2nd DS18B20: connect black & red with 1st DS18B20 black & red

Wiring sensors on PCB/PCBA v4.1

- SHT2x for ambient temp/hum

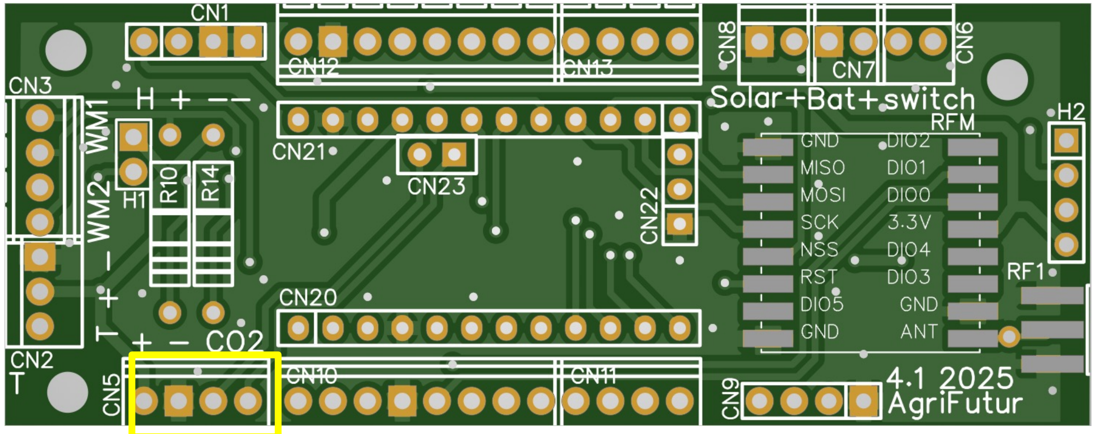
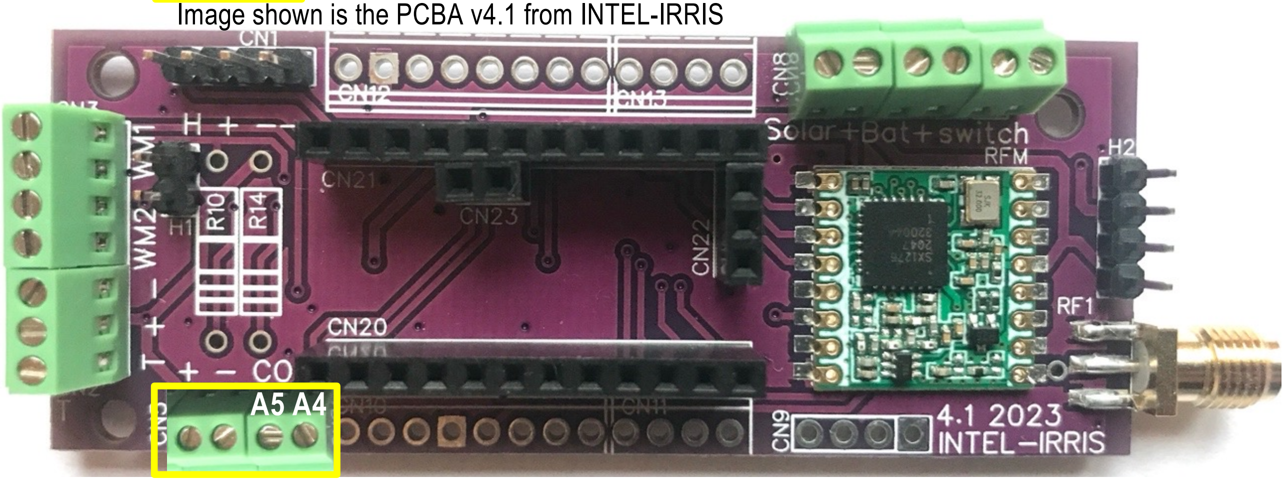
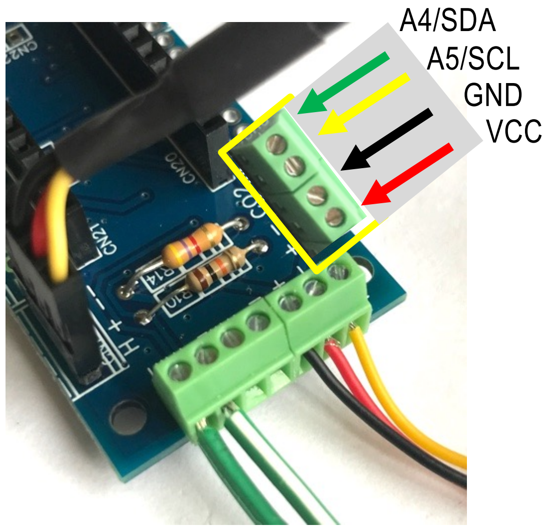


Image shown is the PCBA v4.1 from INTEL-IRRIS





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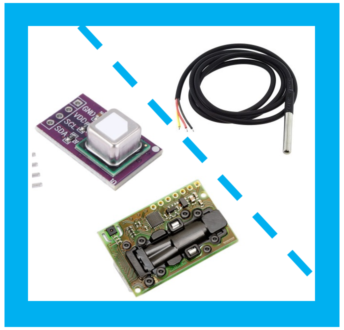
Wiring sensors on PCB/PCBA v4.1

- Sensirion SCD30/40 for CO_2



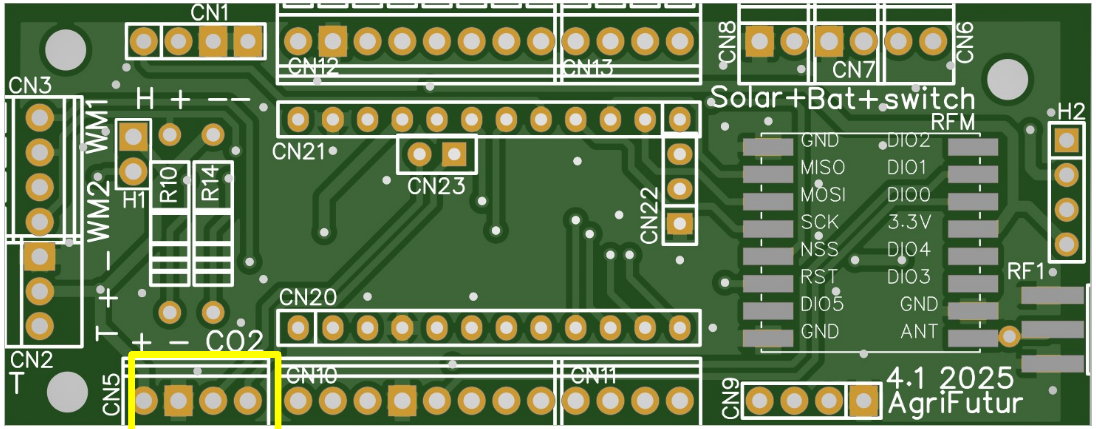
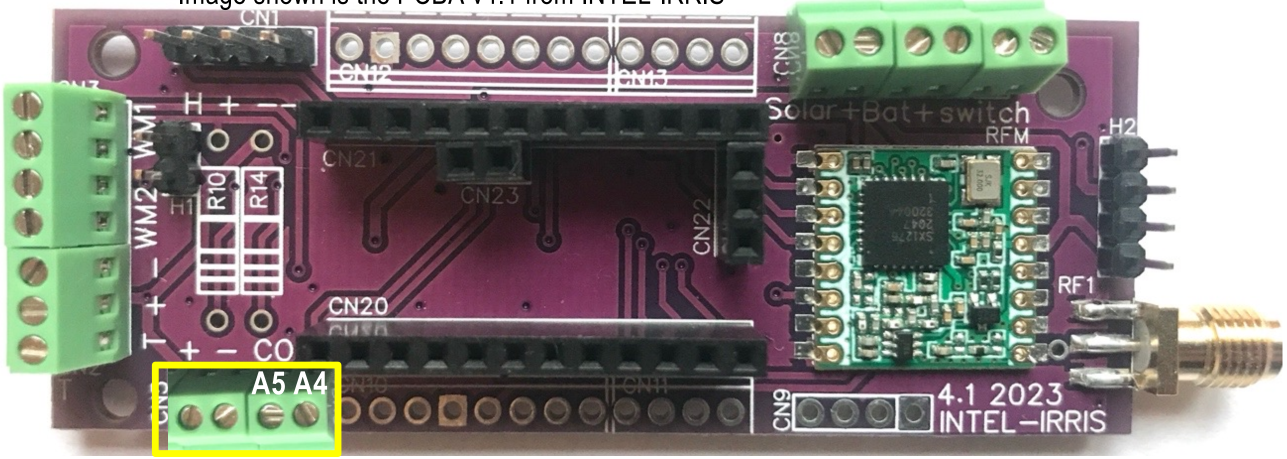
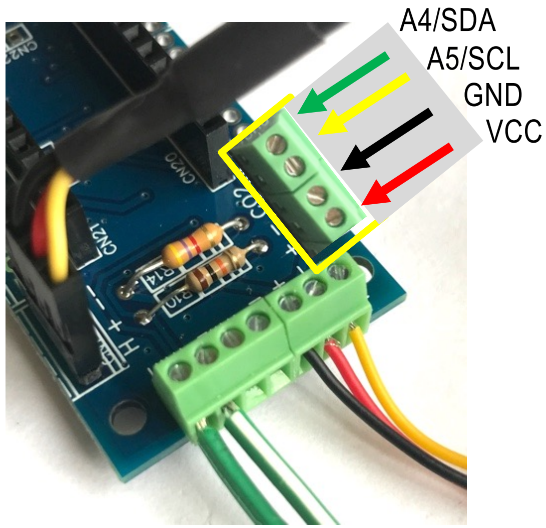


Image shown is the PCBA v4.1 from INTEL-IRRIS

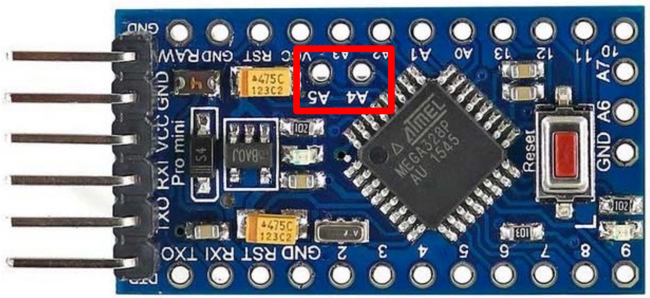


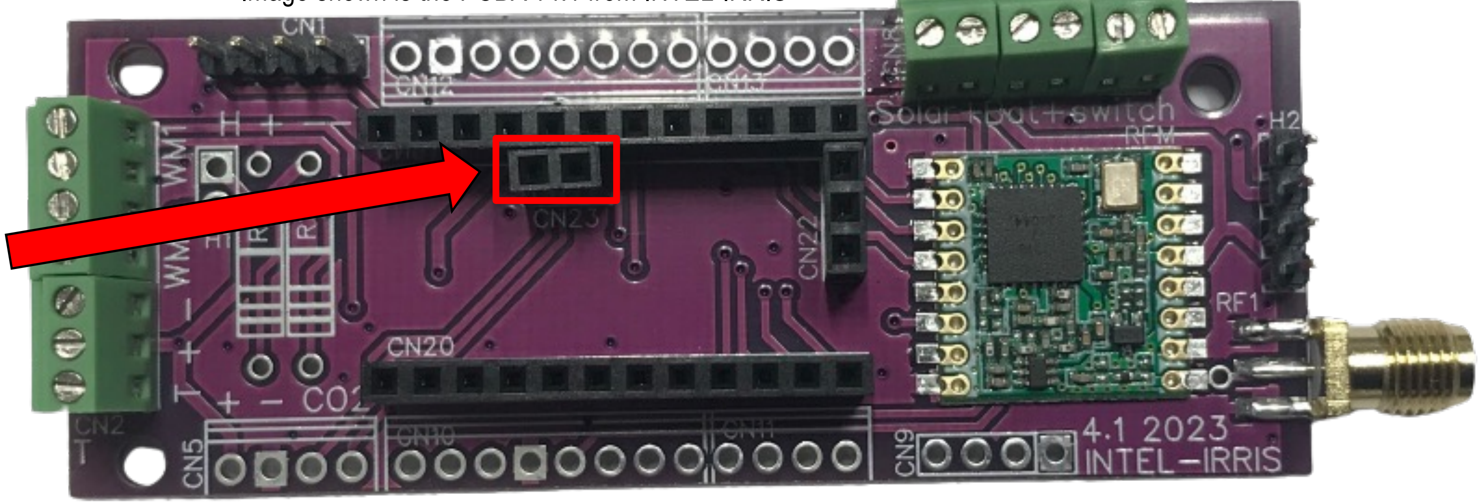


IMPORTANT: Arduino ProMini

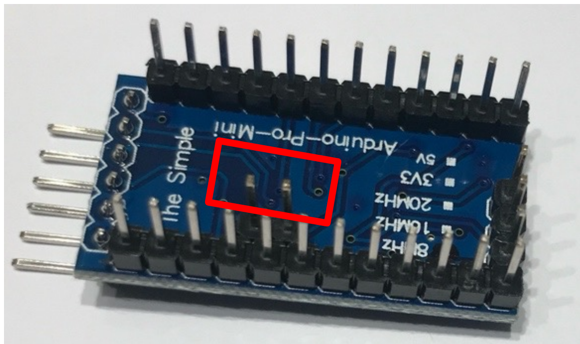
- To use I2C, required for Sensirion SHT and SCD sensors, the Arduino ProMini MUST have pin A4 & A5 connected!

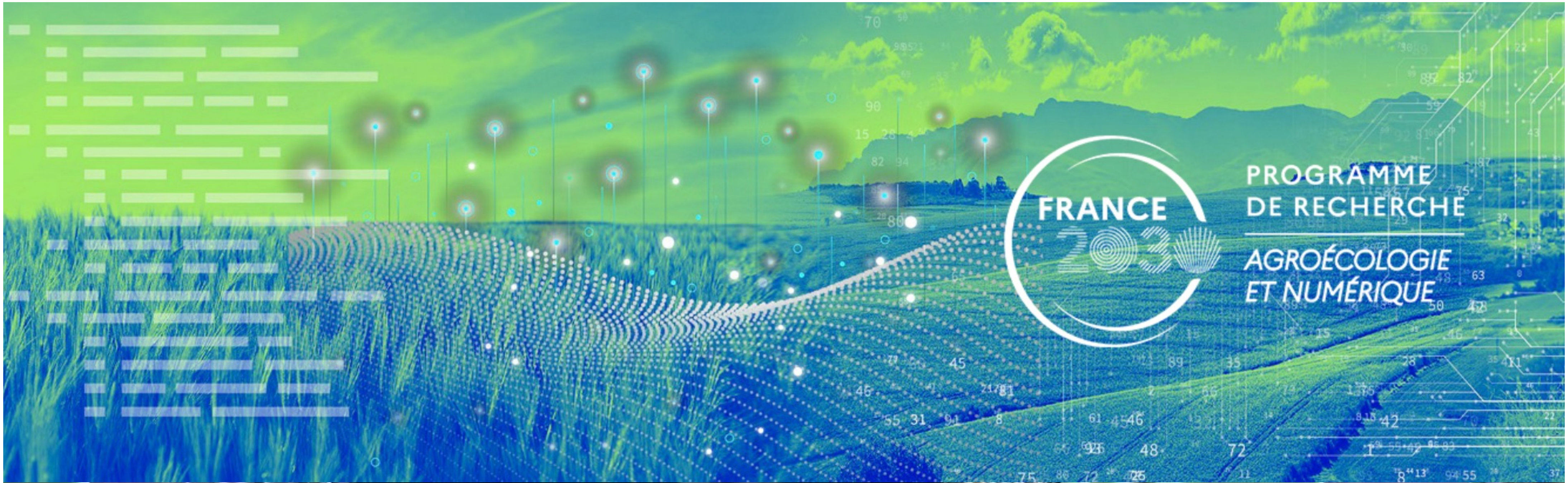
Image shown is the PCBA v4.1 from INTEL-IRRIS

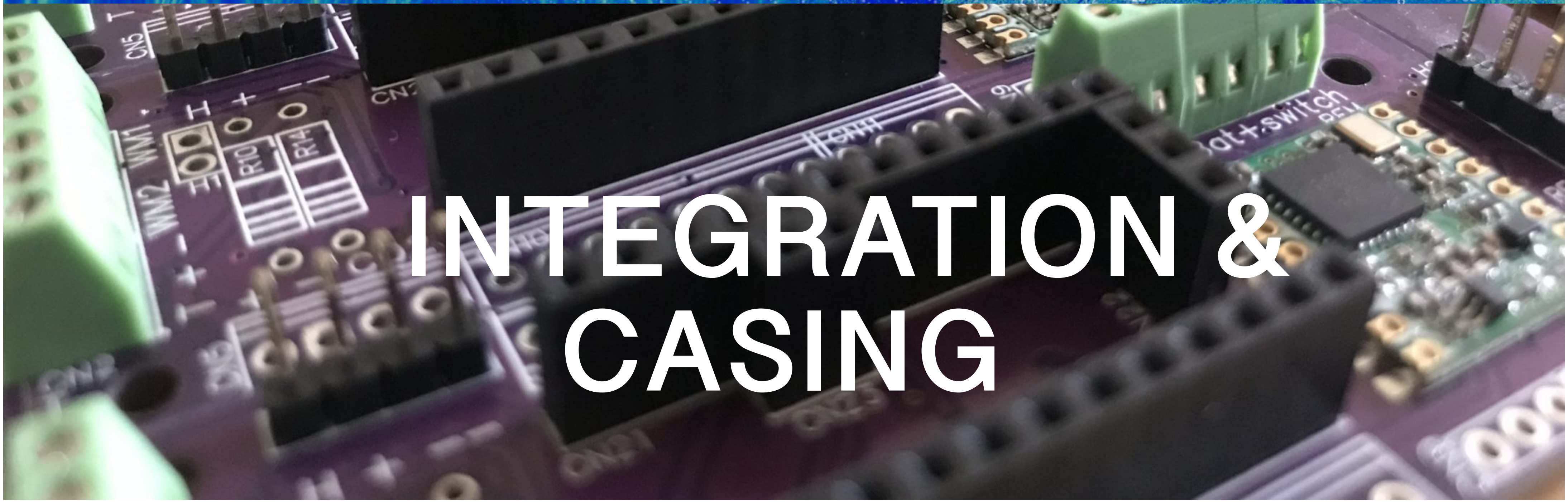




- So, it is MANDATORY to solder a 2-pin header for A4 & A5 on the Arduino ProMini board







INTEGRATION & CASING

Get an enclosure for outdoor usage

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Here, it is an IP65 box which dimension is 115 x 65 x 40mm
https://www.gotronic.fr/art-boitier-abs-etanche-g304m-17977.htm

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Drilling machine and drilling bits

At least a simple cordless drilling machine is necessary
If you have a (small) bench drilling machine it is of course better
Then you need an assortment of drilling bits for metal, not for wood nor concrete! Here, you will mainly need 7mm and 13mm bits
It is also interesting to have step drill bits

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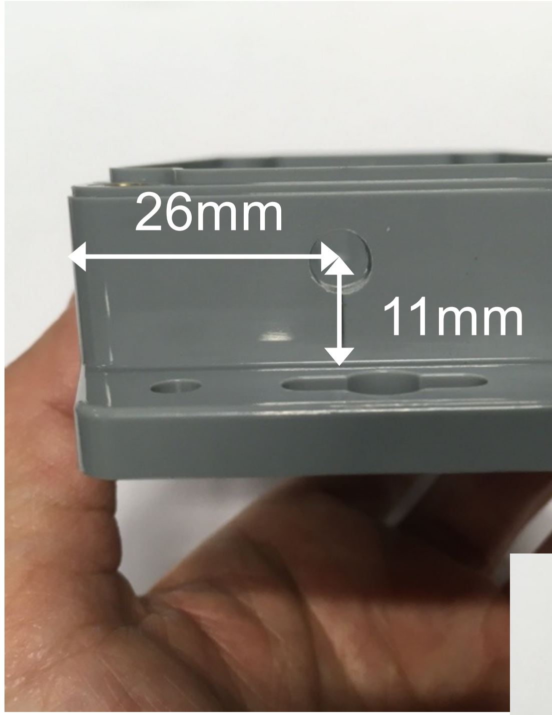
Drill a hole for the SMA connector



If 3xAAA battery holder is used, having 11mm is mandatory!

26mm for the right edge:
- measure from the flat side as the corner is round
11mm from the outside bottom
use a 7mm drill bit for metal, not for wood nor concrete!

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Placing the PCB board

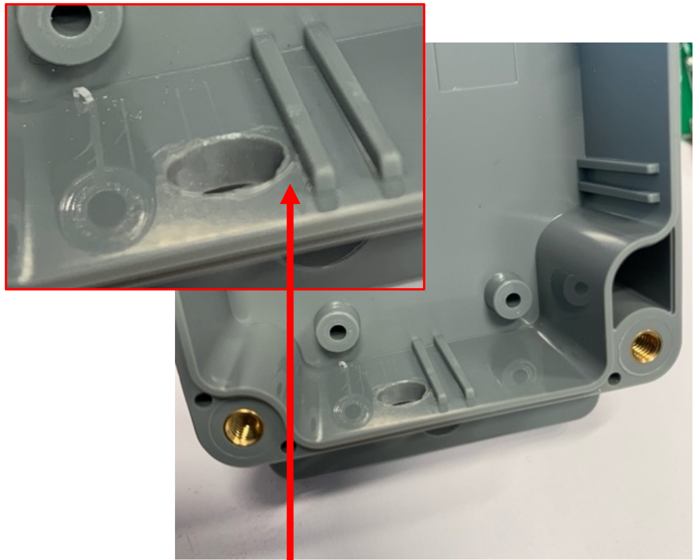

Image shown is the PCBA v4.1 from INTEL-IRRIS

With this particular enclosure case, it is better to have the hole about 1mm next to the plastic reinforcement part

The board can be placed with the SMA connector going through the hole. The 2-AA battery holder is put next to the PCB

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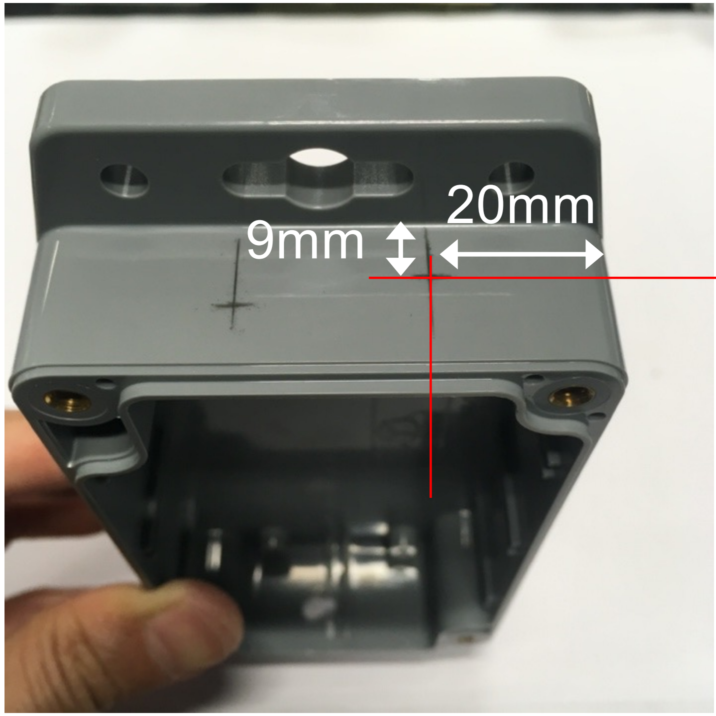
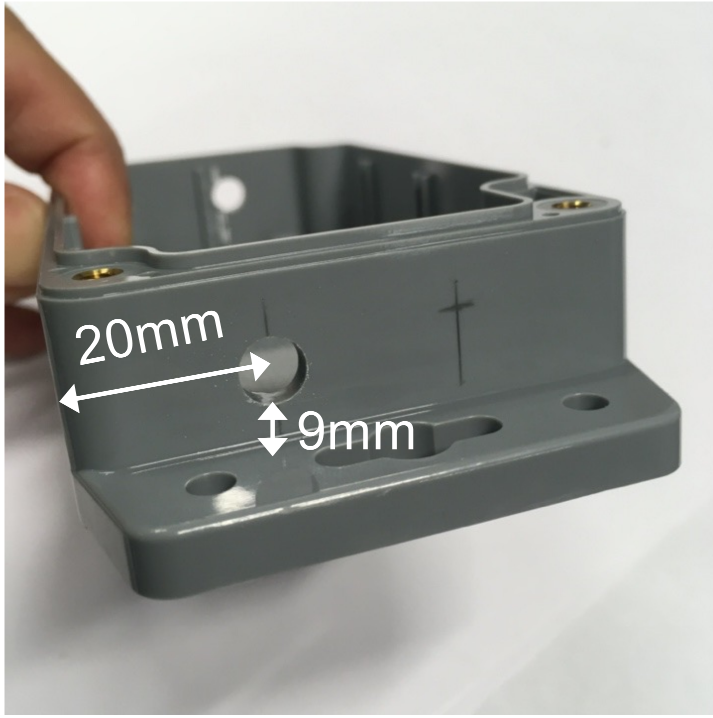
Drill a hole for the external power switch





20mm for the right edge:

- measure from the flat side as the corner is round

9mm from the outside bottom

use a 7mm drill bit for metal

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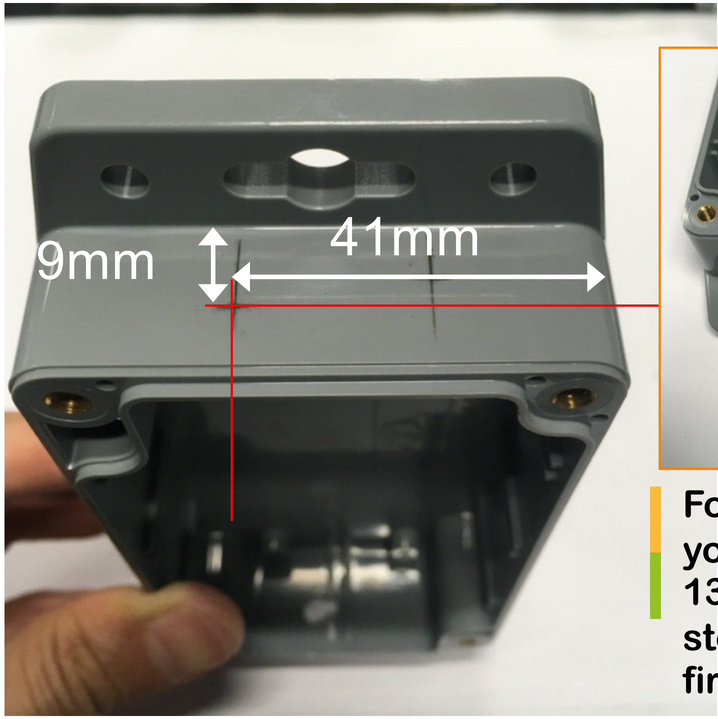
Drill a hole for the cable gland



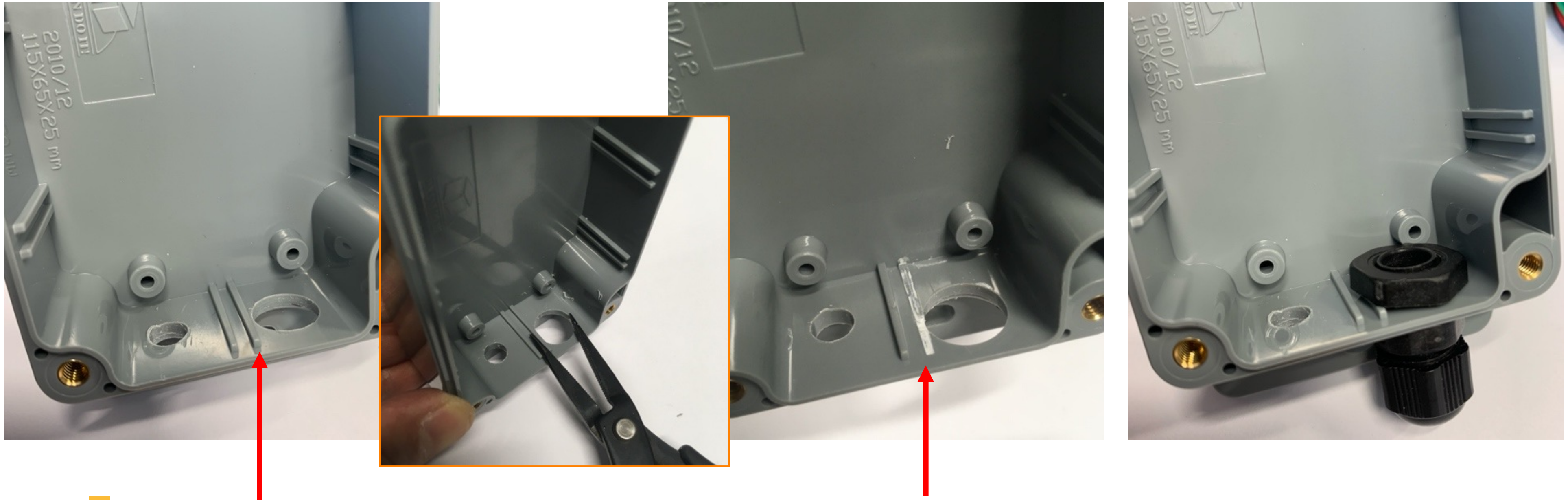
For a PG7 cable gland, you may need 12mm or 13mm hole. If 13mm, a step drill bit can be used first to get 12mm.

41mm for the right edge:
- measure from the flat side as the corner is round

9mm from the outside bottom

use a 13mm or 12mm drill bit for metal. If 13mm, it is recommended to use a step drill bit to first get a 12mm hole before

Remove unwanted plastic part



It may be needed to remove the plastic reinforcement part for this particular enclosure: use a flat cutter for instance to remove and smooth the inside part (a small plier can be used first to remove most of the part)

But that may be not necessary if you can securely screw in the cable gland (right-most photo)

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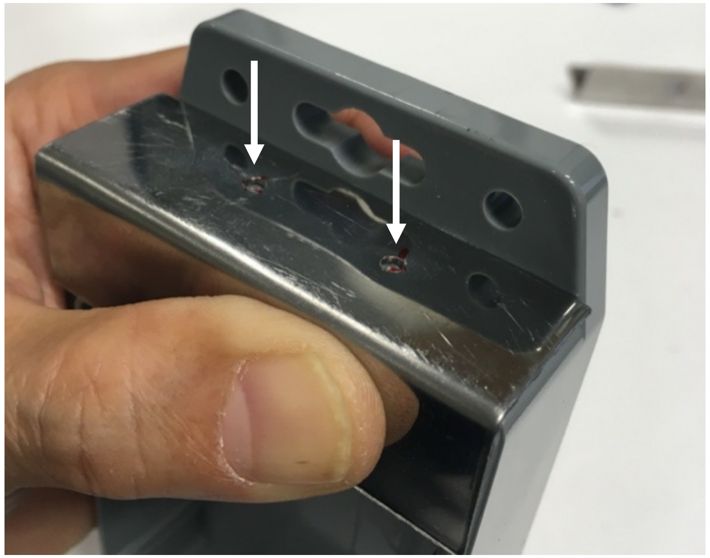
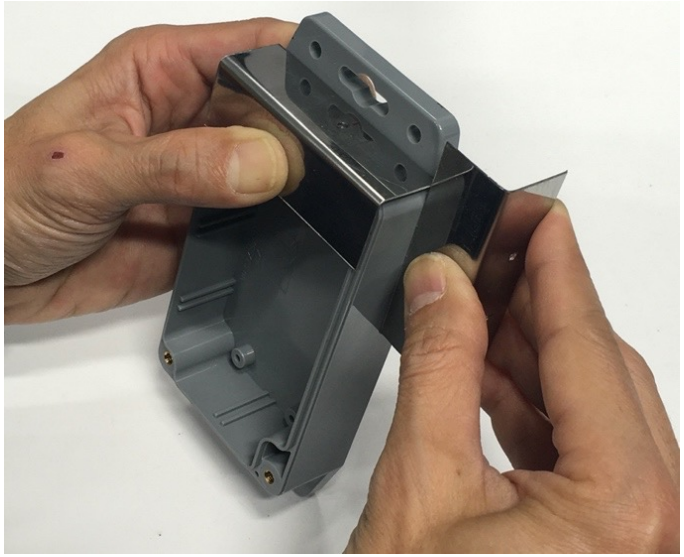
Going for larger production





If you need to prepare many enclosures, it may be faster to first make an assembly jig (here a piece of metal) for marking the holes

But be sure to precisely align the edge of the jig with the edge of the enclosure

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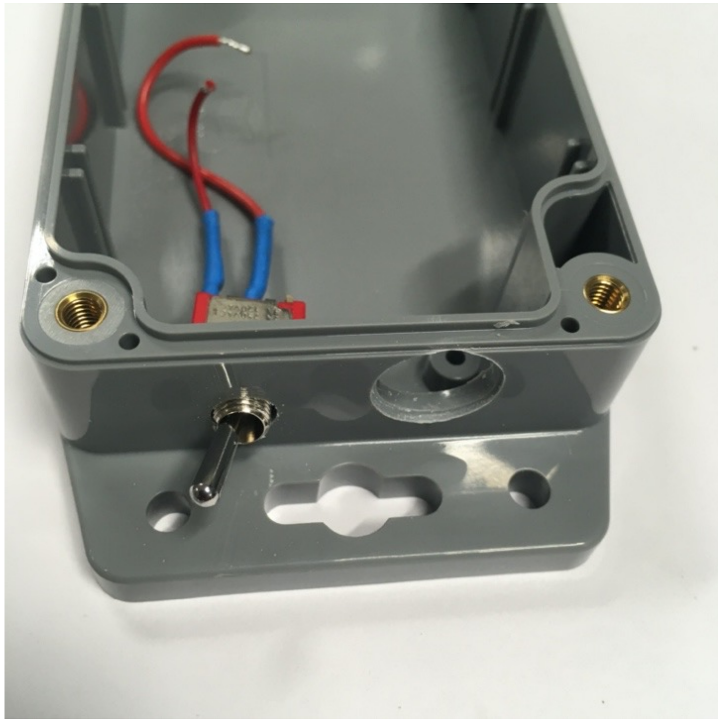
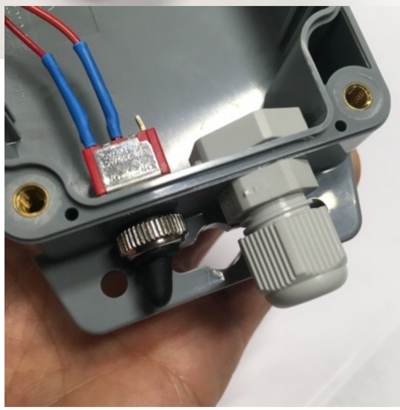
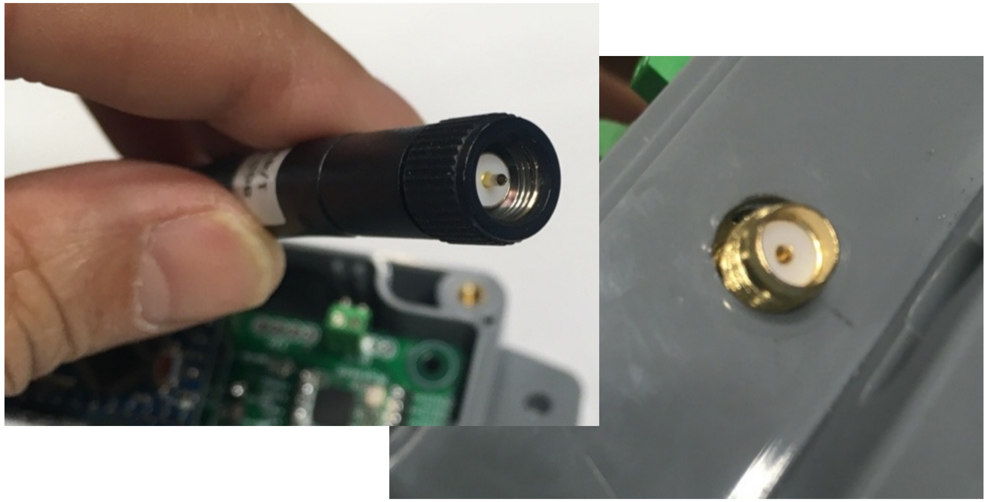
Placing power switch and cable gland



- test that everything is OK
- the switch has a water-proof rubber cap which should be carefully tighten



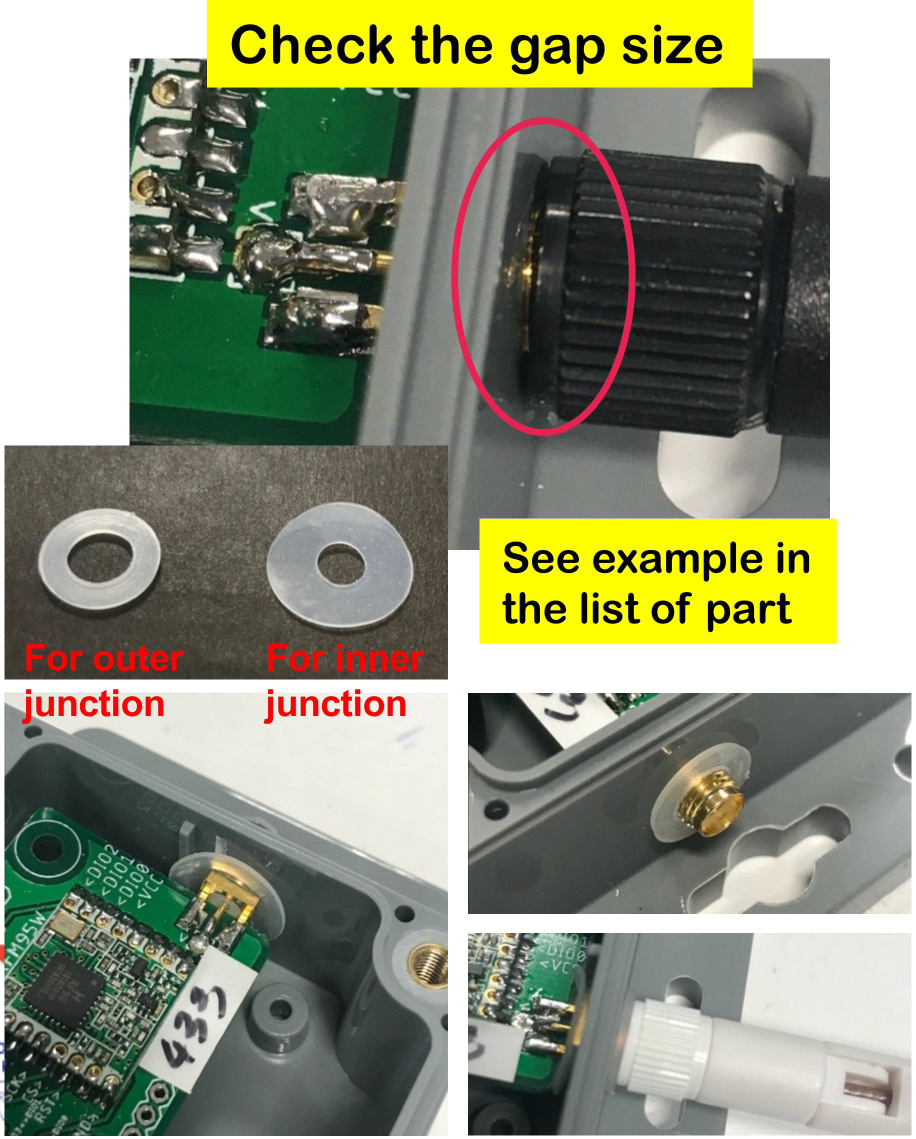
Connecting antenna



Be sure to connect the matching antenna
Here, SMA female with SMA male antenna
Need to screw the antenna in <u>all the way</u>

The antenna junction is critical because this is where rain water can come in

Waterproofing the antenna junction



Even when the antenna is screwed in all the way, there might still be a gap

Even with no apparent gap, it is necessary to waterproof the junction

Take flat silicon seals for that purpose, but do not take it too thick or too large!

Too thick: the antenna will not be screwed in all the way!

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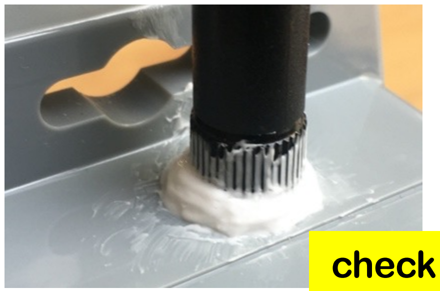
Do not have or can not use flat seal ?

Maybe the gap is too big? Use silicon joint sealant

First, screw in the antenna <u>all the way</u>

Put small amount of silicon around the antenna junction (use a flat screw driver or other flat tool)

Use a <u>wet</u> toothpick to finish and clean the silicon all around the antenna junction



check especially the back side

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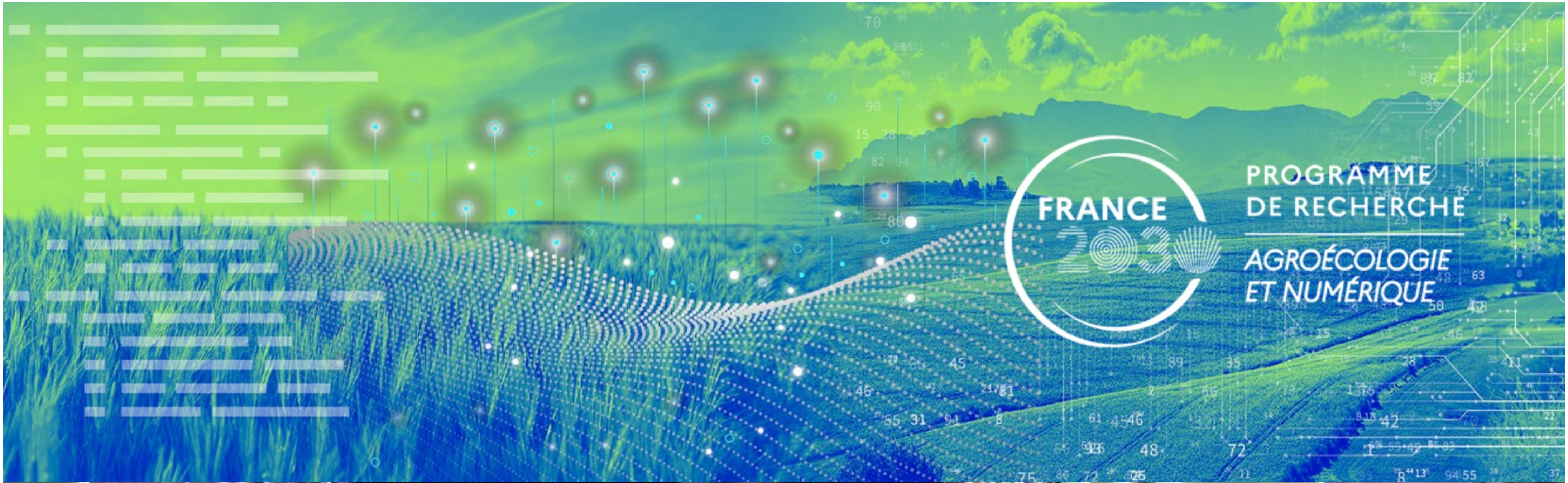

PROGRAM THE MICROCONTROLLER

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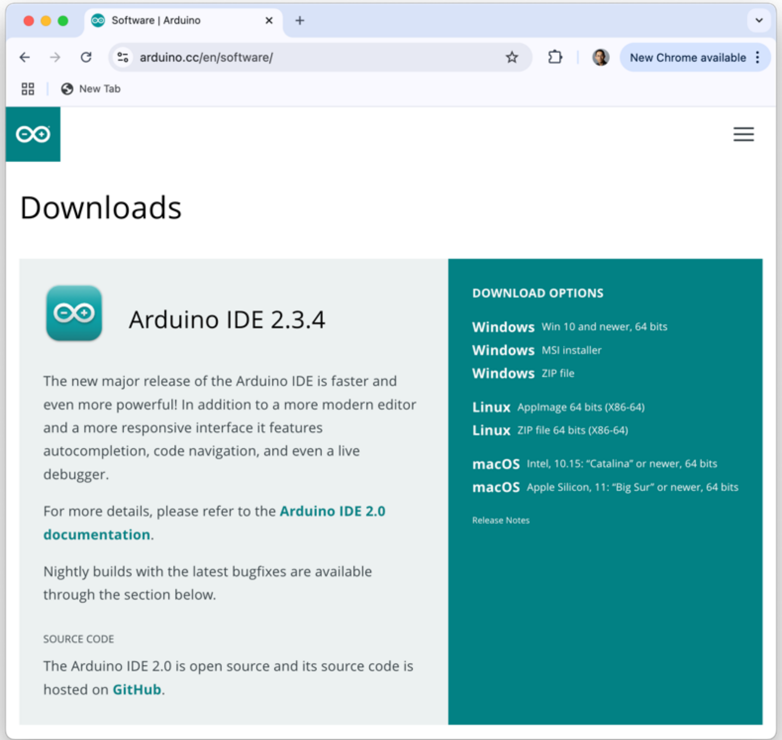
Getting the software: Arduino IDE



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Install latest version of Arduino IDE from https://www.arduino.cc/en/software

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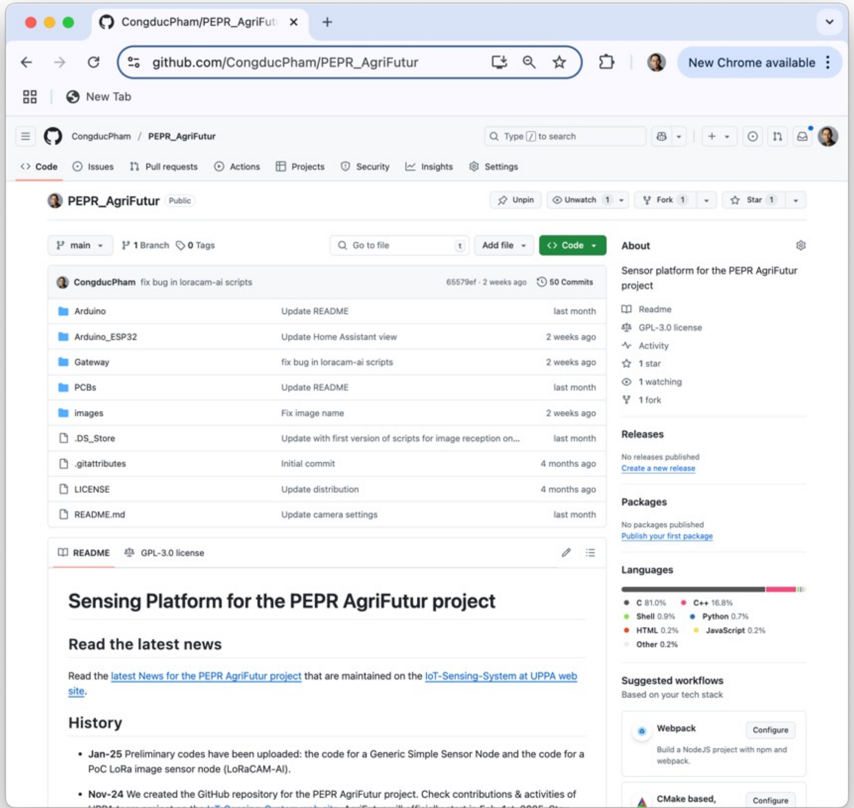
Getting the AgriFutur code



On your computer, create a `sketch` folder

Then download the whole repository as ZIP file

Unzip the file and copy the content of `Arduino` folder into your `sketch` folder

With the `librairies` folder

The entire AgriFutur GitHub repository is hosted here
https://github.com/CongducPham/PEPR_AgriFutur

GitHub

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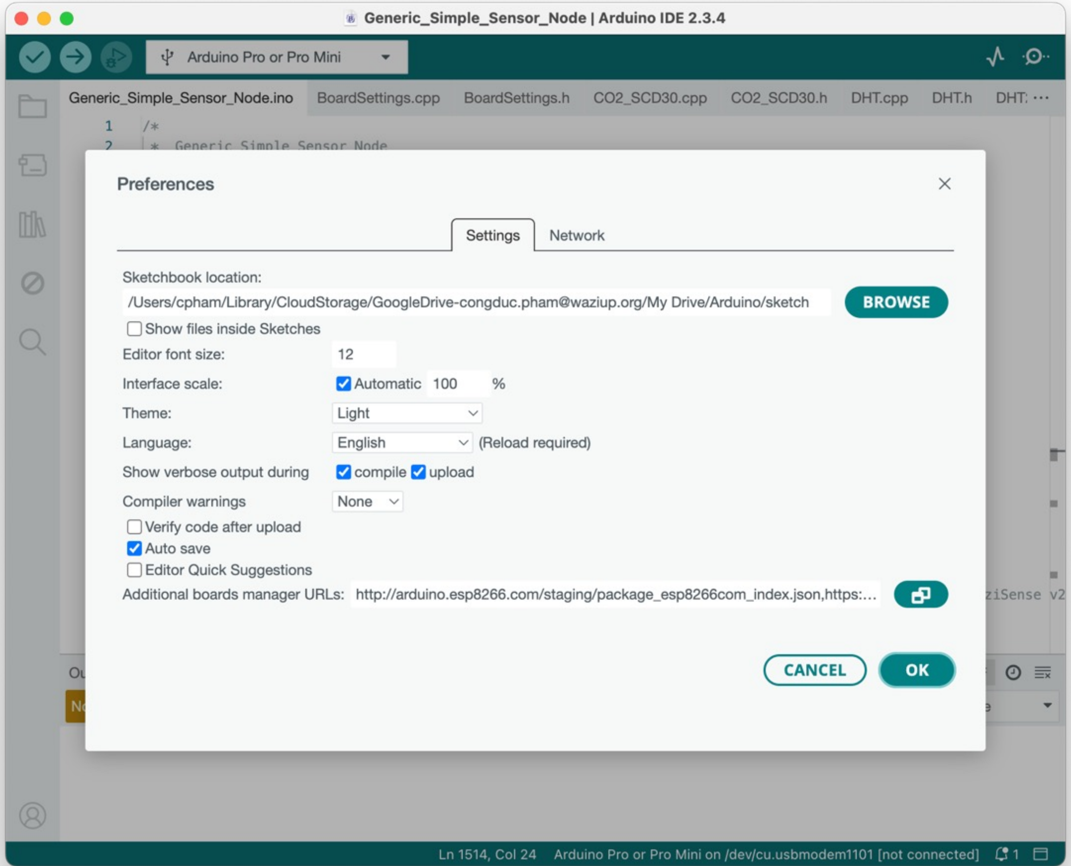
Setting your Arduino IDE



Run Arduino IDE, open Preferences
Indicate your `sketch` folder in Sketchbook location

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Configuration for PCB/PCBA v4.1

- For raw version, uncomment `IRD_PCB` in BoardSettings.h
- For PCBA version, also uncomment `IRD_PCBA`

```
////////////////////////////////////////////////////////////////////
// uncomment for IRD PCB board (both v4.1 and v5)
#define IRD_PCB
// also uncomment for IRD PCB board fully assembled by manufacturer
// with all components, including solar circuit (both v4.1 and v5)
#define IRD_PCBA
```

- ONLY for solar version, uncomment `SOLAR_BAT`

```
////////////////////////////////////////////////////////////////////
// uncomment only if the IRD PCB or PCBA is running on solar panel
// MUST be commented if running on alkaline battery
// code for SOLAR_BAT has been written by Jean-François Printanier from IRD
#define SOLAR_BAT
// do not change if you are not knowing what you are doing
#define NIMH
```

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Compiling the soil sensor code

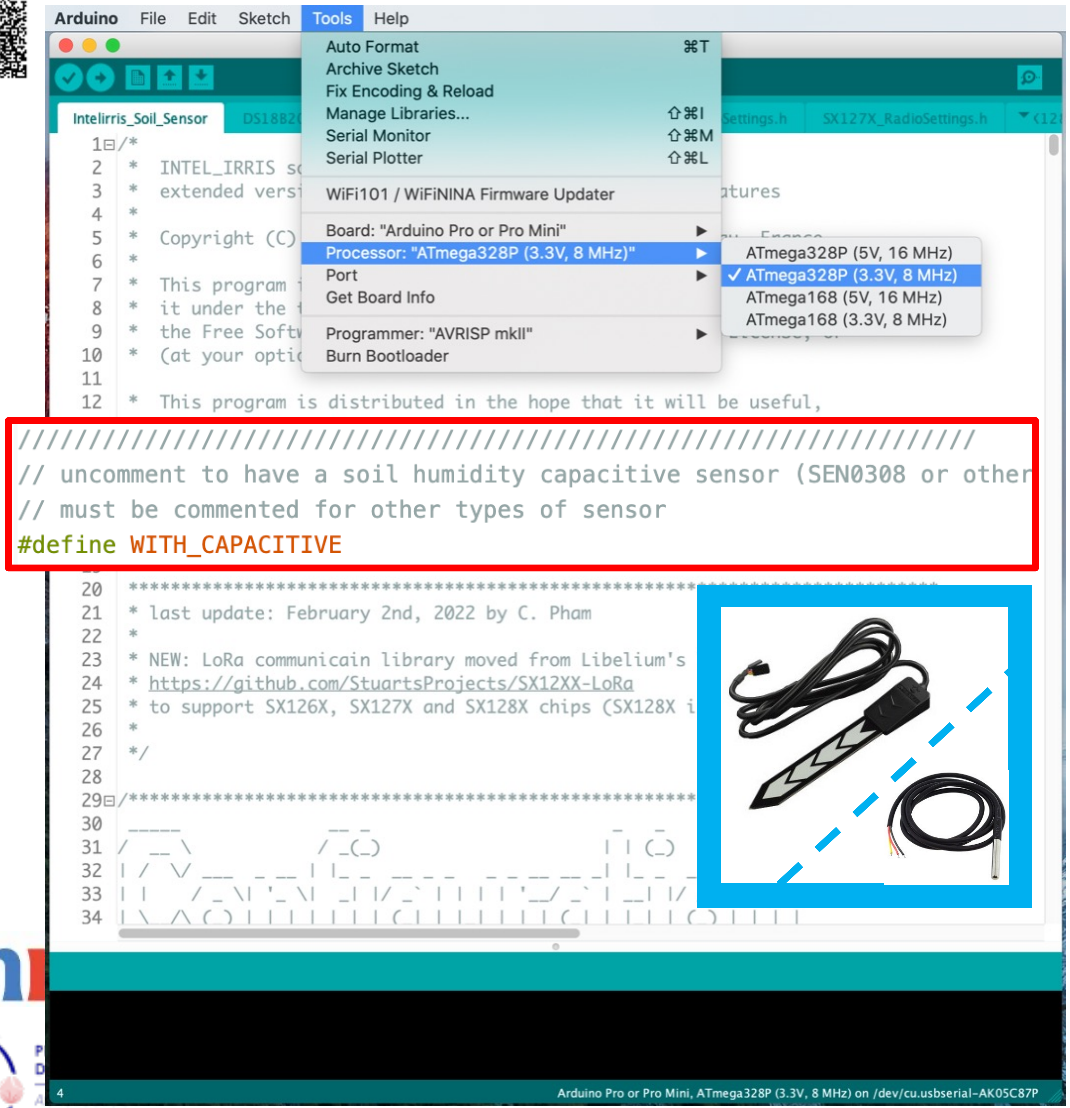
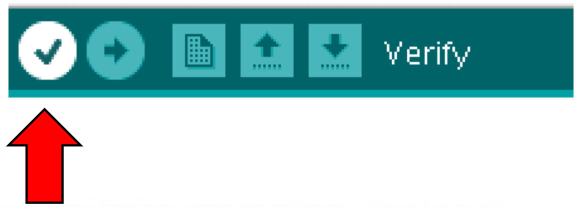


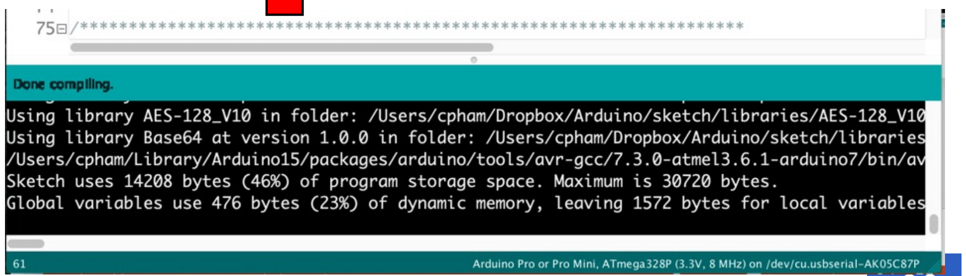
Image shown is the Arduino IDE with INTEL-IRRIS code

Open Generic_Simple_Sensor_Node sketch – default configuration builds a capacitive sensor device

Select the ProMini board, 3.3V and 8MHz version

Then click on the "verify" button





Never transmit without antenna

- NEVER, NEVER transmit without an antenna
- Doing so can damage the radio module
- If your board is already connected to the radio module and you need to flash the board, then connect the antenna
- If you need to update the existing code and your device already run a code that transmit data, connect the antenna
- It is safer when programming the device to remove the Arduino board from the female header and program it disconnected from the radio module
- If you deploy a device, make sure that the antenna is correctly connected before powering on the device and realizing any transmission test

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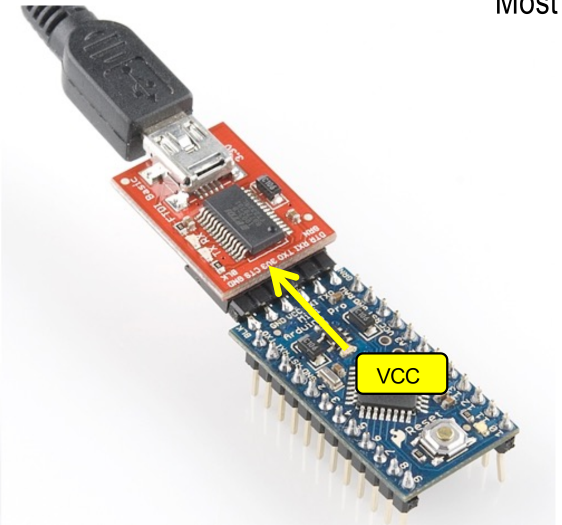
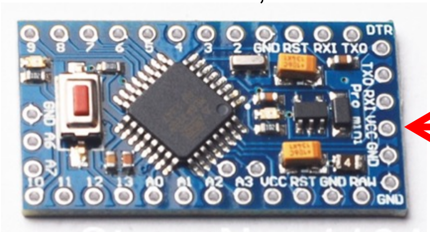
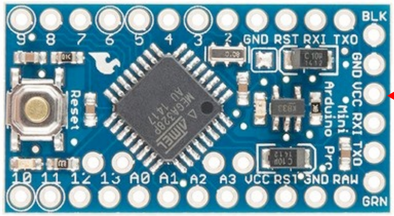
Connecting with an FTDI cable

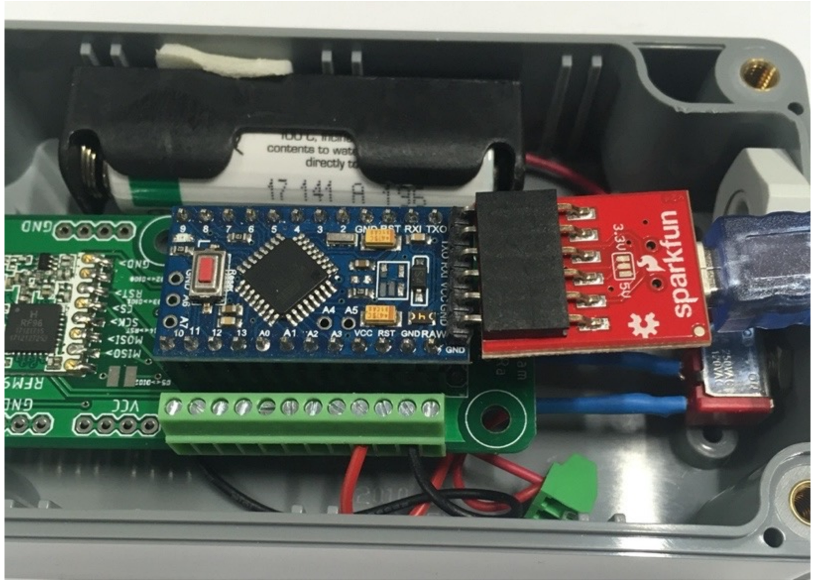


Most Chinese clone version, check the VCC pin





Original Sparkfun version



For the ProMini, you need to have an FTDI breakout cable working at 3.3v

Check the VCC pin position and make it to correspond to the VCC pin of the FTDI breakout.

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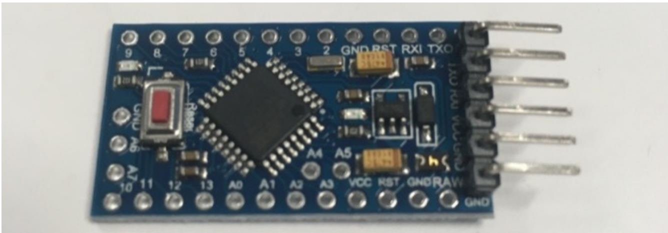
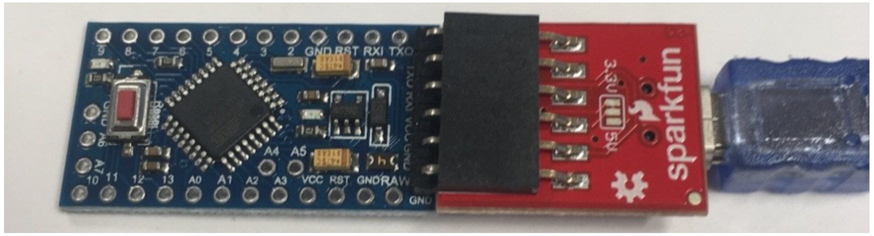
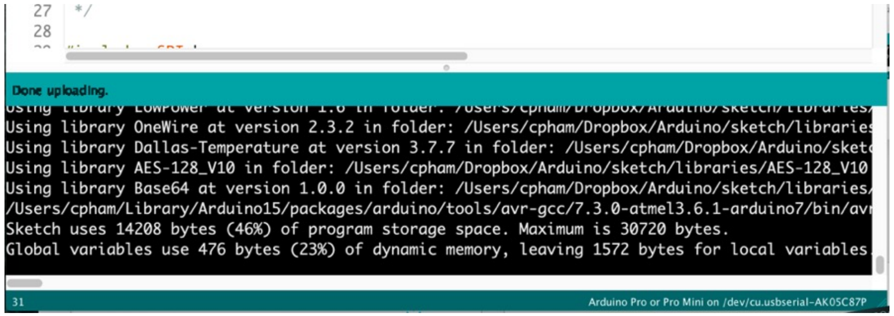
Check the Arduino board

- First, just solder the programming header





- To flash the Arduino board with the AgriFutur code
- See slides from "Getting the software: Arduino IDE" to "Uploading to your board"
- If uploading is successful, then you can continue with soldering the remaining header pins, otherwise, take another board as this one may have hardware issue



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Select serial port for uploading

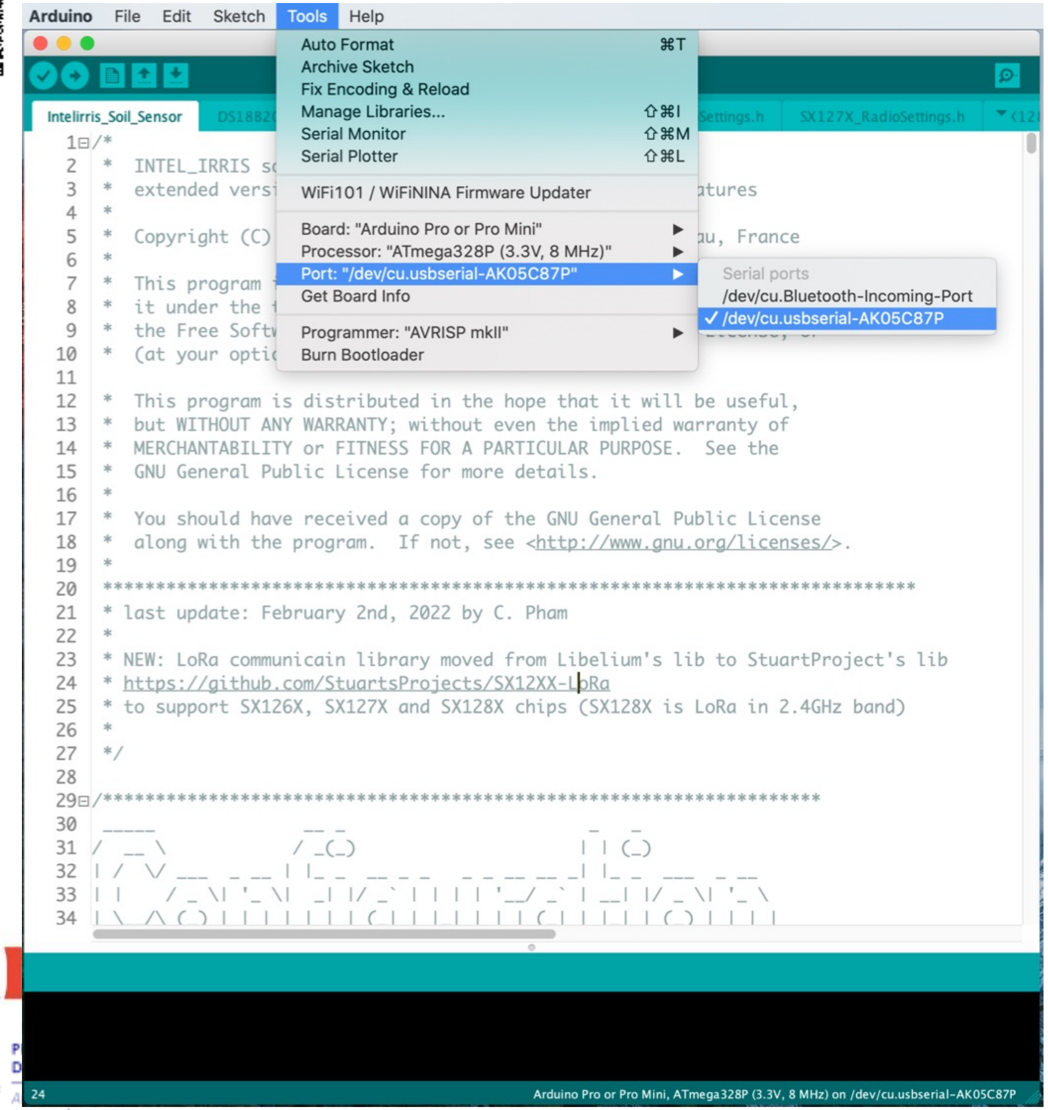


Image shown is the Arduino IDE with INTEL-IRRIS code

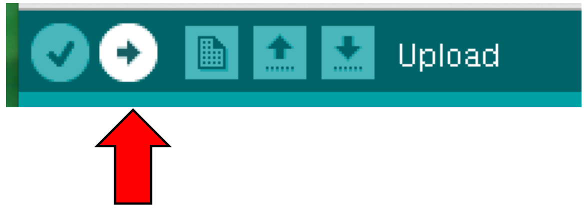
After connecting the cable to your computer/laptop USB port, try to find the serial port

If you don't find it, you may need to install specific drivers

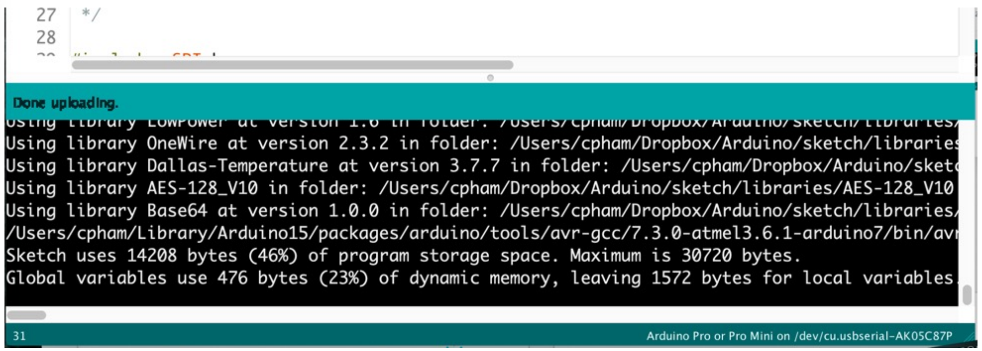
https://learn.sparkfun.com/tutorials/how-to-install-ch340-drivers/all

Uploading to your board

Click on the "upload" button



And wait until upload is completed



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Checking that device is operational

/dev/cu.usbserial-AK05C87P

Send

```
SX1276,868099968Hz,SF12,BW125000,CR4:5,LDRO_On,SyncWord_0x34,IQNormal,Preamble_8
SX1276,SLEEP,Version_12,PacketMode_LoRa,Explicit,CRC_On,AGCauto_On,LNAgain_0,LNAboostHF_On,LNAboostLF_On

Reg   0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F
0x00  00 80 1A 0B 00 52 D9 06 66 FC 09 10 1B 12 00 00
0x10  00 00 00 00 00 00 00 00 00 00 00 00 00 72 C4 64
0x20  00 08 FF FF 00 00 0C 00 00 00 00 00 00 50 14 40
0x30  00 03 05 27 1C 0A 03 0A 42 34 40 1D 00 A1 00 00
0x40  70 00 12 24 2D 00 03 00 04 23 00 09 05 84 32 2B
Get back previous sx1272 config
Using packet sequence number of 171
Forced to use default parameters
Using node addr of 8
Using idle period of 30
Setting Power: 14
node addr: 8
SX127X successfully configured
Sending \!SH1/346.00
use XLPP format for transmission to WaziGate
end-device uses native LoRaWAN packet format

plain payload hex
00 67 08 75
Encrypting
encrypted payload
A0 34 2F 4C
calculate MIC with NwkSKey
transmitted LoRaWAN-like packet:
MHDR[1] | DevAddr[4] | FCtrl[1] | FCnt[2] | FPort[1] | EncryptedPayload | MIC[4]
40 AA 1D 01 26 00 00 00 01 A0 34 2F 4C D1 B8 13 A9
[base64 LoRaWAN HEADER+CIPHER+MIC]:QKodASYAAAABoDQvTNG4E6k=
--> CS1
--> CAD 183
OK1
CRC,752E
LoRa pkt size 17
LoRa pkt seq 171
LoRa Sent in 1506
Switch to power saving mode
```

Autoscroll · Show timestamp · No line ending · 38400 baud · Clear output

- **Open serial monitor**
- **Set baud rate to 38400**
- **See output from board**
- **Check that transmission is OK**

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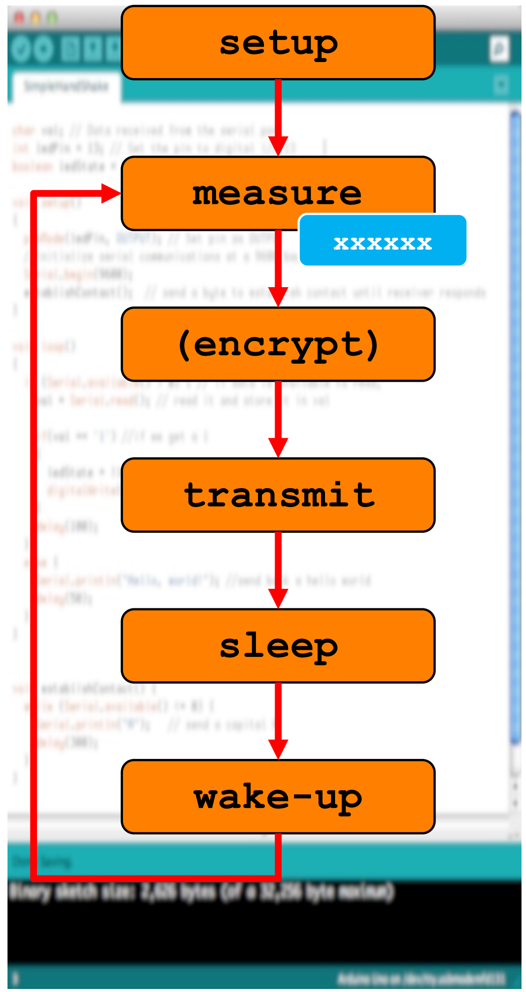
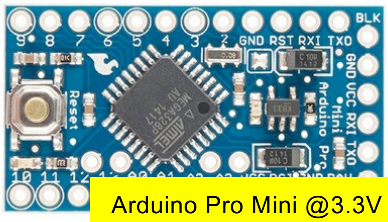
Generic & cyclic behavior

DFRobot SEN0308
capacitive soil humidity
A0 (signal), A1 (power)



Physical sensor management



Arduino Pro Mini @3.3V

Activity duty-cycle, low power

AES encryption

Long-range transmission

Logical sensor management

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Transmission to gateway

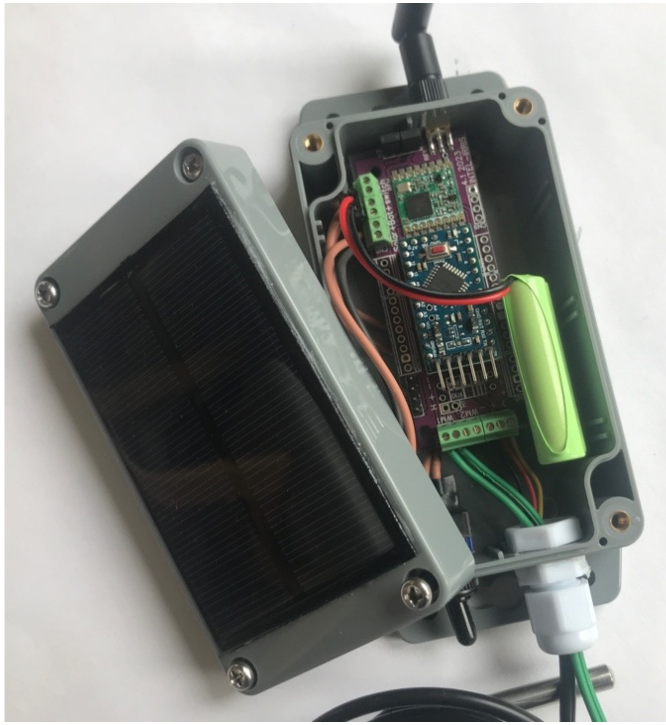

Image shown is the PCBA v4.1 from INTEL-IRRIS

LoRa

LoRa parameters

Limited LoRaWAN
SF12BW125
868.1MHz
ABP mode
Dev addr is 26011DAA
1 msg/60mins
1 sensor
XLPP data

This video from INTEL-IRRIS will show all these steps, from connecting the SEN0308 to testing transmission to the gateway
Video n°4: https://youtu.be/j-1Nk0tv0xM

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Configuration for other type of sensors

- **<u>Comment</u>** `WITH_CAPACITIVE`
- Configure the code to build the other types of sensor device
  - 1 tensiometer [+ 1 soil temp]
  - 2 tensiometer [+ 1 soil temp]
  - 1 ambient temp/hum [+ 1 soil temp]
  - 1 CO2 [+ 1 soil temp]
  - 2 soil temp or 3 soil temp


```
///////////////////////////////////////////////////////////////////
// SENSOR CONFIGURATION PART                                     //
///////////////////////////////////////////////////////////////////

///////////////////////////////////////////////////////////////////
// uncomment to have a soil humidity capacitive sensor (SEN0308 or other type of capacitive)
#define WITH_CAPACITIVE
// uncomment to use SEN0308 capacitive calibration for low voltage
#define SEN0308_CALIBRATION_LOW_VOLTAGE
// uncomment to send millivolt with SEN0308 capacitive
// #define SEN0308_TRANSMIT_MILLIVOLT
///////////////////////////////////////////////////////////////////
// uncomment to have a soil tensiometer watermark sensor
// #define WITH_WATERMARK
// only for watermark sensors, not relevant for capacitive sensors
// if a soil temperature sensor is wired, the real soil temperature can be used
#define WM_REF_TEMPERATURE 28.0
// uncomment to force the watermark to have default device address for WaziGate
// #define WM_AS_PRIMARY_SENSOR
// uncomment to have 2 tensiometer watermark sensor on the same device
// #define TWO_WATERMARK
///////////////////////////////////////////////////////////////////
// uncomment to have 1 soil temperature sensor ST using a one-wire DS18B20 sensor
// also needed for TWO_SOIL_TEMP_SENSOR and THREE_SOIL_TEMP_SENSOR
// #define SOIL_TEMP_SENSOR
// only for watermark sensors, not relevant for capacitive sensors
#define LINK_SOIL_TEMP_TO_CENTIBAR
///////////////////////////////////////////////////////////////////
// uncomment to have an additional decagon EC-5 sensor, ONLY ON IRD_PCB
// #define SOIL_EC5_SENSOR
// uncomment to have a CO2 sensor, ONLY ON IRD_PCB
// #define CO2_SCD30_SENSOR
///////////////////////////////////////////////////////////////////
// uncomment to have an ambiant air temp/hum sensor, then select which sensor model below
// #define AIR_TEMP_HUM_SENSOR
///////////////////////////////////////////////////////////////////
// uncomment to have an ambiant air temperature DHT22/AM2305 sensor, ONLY ON IRD_PCB
// #define DHT22_AM2305_TEMP_SENSOR
// uncomment to have an ambiant air humidity DHT22/AM2305 sensor, ONLY ON IRD_PCB
// #define DHT22_AM2305_HUM_SENSOR
///////////////////////////////////////////////////////////////////
// uncomment to have an ambiant air temperature SHT sensor, ONLY ON IRD_PCB
// #define SHT_TEMP_SENSOR
// uncomment to have an ambiant air humidity SHT sensor, ONLY ON IRD_PCB
// #define SHT_HUM_SENSOR
///////////////////////////////////////////////////////////////////
// uncomment to have a customized 2 soil temperature device, ONLY ON IRD_PCB
// #define TWO_SOIL_TEMP_SENSOR
// uncomment to have a customized 3 soil temperature device, ONLY ON IRD_PCB
// #define THREE_SOIL_TEMP_SENSOR
```


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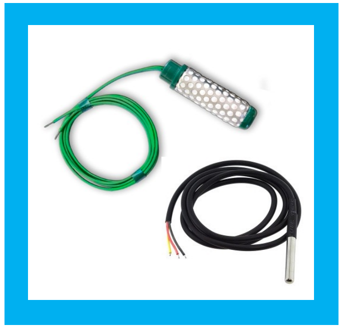
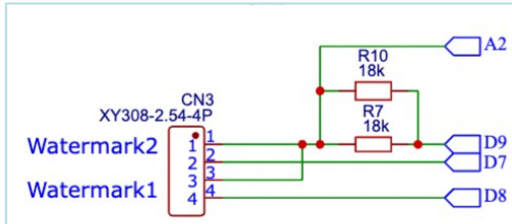
Configuration for Watermark tensiometer



- Default resistor value is 10kOhms
- Change in `watermark.h` file the `WM_RESISTOR` value to match the one you are using if needed
- The fully assembled version has a 10kOhms resistor, schematic indicates 18kOhms but it is really 10kOhms




```
#define WM_RESISTOR 10000
//we defined WM_MAX_RESISTOR=32760 because the transmitted value would be 32760/10=3276
//and currently a bug in WaziGate XLPP decoding code will limit the maximum value to 3276
#define WM_MAX_RESISTOR 32760
```

- Uncomment in main program `WITH_WATERMARK`

```
// uncomment to have a soil tensiometer watermark sensor
#define WITH_WATERMARK
// only for watermark sensors, not relevant for capacitive sensors
// if a soil temperature sensor is wired, the real soil temperature can be used
#define WM_REF_TEMPERATURE 28.0
```

Transmission with the WM sensor

- Devices with a WM sensor will have a different default address
  - 26011DB1 instead of 26011DAA for capacitive
- 2 values are transmitted
  - Centibars converted from resistance value
  - Raw resistance value measured for WM sensor, but scaled down by a factor of 10
    → 300 = 3000 Ohms

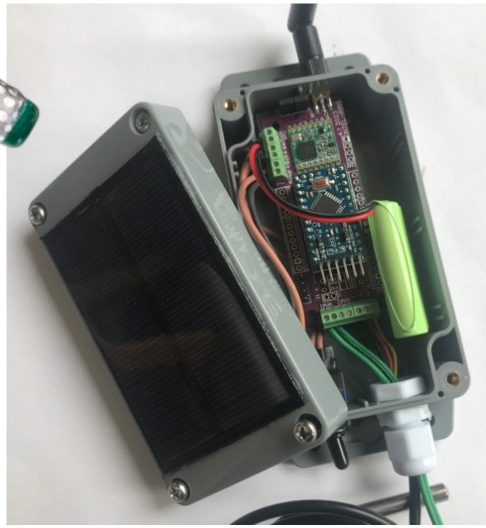
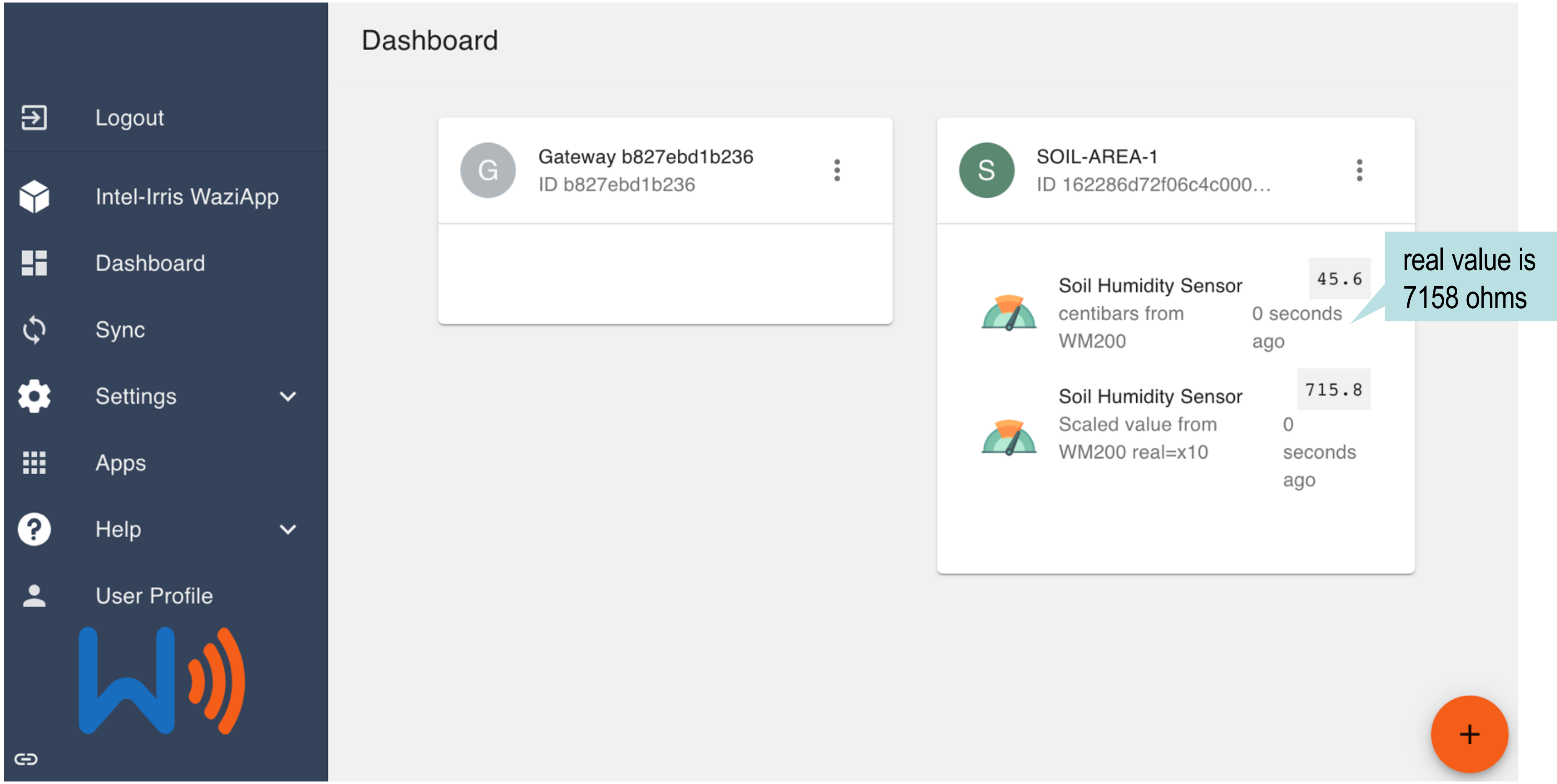


Image shown is the PCBA v4.1 from INTEL-IRRIS

LoRa parameters

Limited LoRaWAN
SF12BW125
868.1MHz
ABP mode
Dev addr is 26011DB1
1 msg/60mins
1 sensor
XLPP data

See it on the dashboard



Static soil temperature with WM sensor

- Without a real temperature sensor, the INTEL-IRRIS code uses a default soil temperature set to 28°C to convert the resistance value into centibars (see https://www.irrometer.com/200ss.html)

```
// uncomment to have a soil tensiometer watermark sensor
#define WITH_WATERMARK
// only for watermark sensors, not relevant for capacitive sensors
// if a soil temperature sensor is wired, the real soil temperature can be used
#define WM_REF_TEMPERATURE 28.0
```

- You can change this setting in the code for testing purposes
- It is better to use a real temperature sensor to dynamically get the soil temperature

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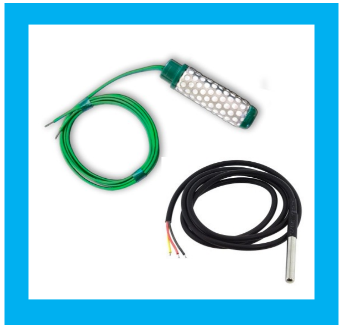
Real soil temperature with WM sensor

- A DS18B20 temperature sensor can be connected to the device



- Uncomment in main program `SOIL_TEMP_SENSOR`

```
///////////////////////////////////////////////////////////////////
// uncomment to have 1 soil temperature sensor ST using a one-wire DS18B20 sensor
// also needed for TWO_SOIL_TEMP_SENSOR and THREE_SOIL_TEMP_SENSOR
#define SOIL_TEMP_SENSOR
// only for watermark sensors, not relevant for capacitive sensors
#define LINK_SOIL_TEMP_TO_CENTIBAR
```

- Then, link the measured soil temperature to the centibar calculation performed by the device itself (set by default)

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The Capacitive and Tensiometer sensors

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SEN0308 capacitive sensor
Default address is 26011DAA

Watermark WM200
Water tension sensor
Default address is 26011DB1

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Configuration for 2 Watermark

- Uncomment in main program `TWO_WATERMARK`

```
// uncomment to have a soil tensiometer watermark sensor
#define WITH_WATERMARK
// only for watermark sensors, not relevant for capacitive sensors
// if a soil temperature sensor is wired, the real soil temperature can be used
#define WM_REF_TEMPERATURE 28.0
// uncomment to force the watermark to have default device address for WaziGate
// #define WM_AS_PRIMARY_SENSOR
// uncomment to have 2 tensiometer watermark sensor on the same device
#define TWO_WATERMARK
```

- You can still have a soil temperature sensor

```
////////////////////////////////////////////////////////////////////
// uncomment to have 1 soil temperature sensor ST using a one-wire DS18B20 sensor
// also needed for TWO_SOIL_TEMP_SENSOR and THREE_SOIL_TEMP_SENSOR
#define SOIL_TEMP_SENSOR
// only for watermark sensors, not relevant for capacitive sensors
#define LINK_SOIL_TEMP_TO_CENTIBAR
```

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Configuration for ambient air temp/hum

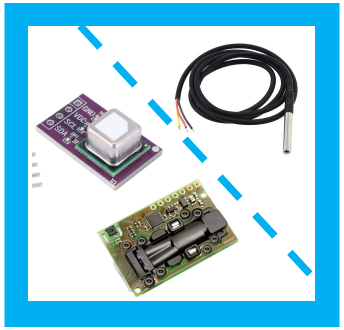
- Uncomment in main program `AIR_TEMP_HUM_SENSOR`
- …and select model: DHT22 or SHT2x
- Dev addr is 26011DC1 in ABP mode

```
////////////////////////////////////////////////////////////////////
// uncomment to have an ambiant air temp/hum sensor, then select which sensor model below
#define AIR_TEMP_HUM_SENSOR
////////////////////////////////////////////////////////////////////
// uncomment to have an ambiant air temperature DHT22/AM2305 sensor, ONLY ON IRD_PCB
// #define DHT22_AM2305_TEMP_SENSOR
// uncomment to have an ambiant air humidity DHT22/AM2305 sensor, ONLY ON IRD_PCB
// #define DHT22_AM2305_HUM_SENSOR
////////////////////////////////////////////////////////////////////
// uncomment to have an ambiant air temperature SHT sensor, ONLY ON IRD_PCB
#define SHT_TEMP_SENSOR
// uncomment to have an ambiant air humidity SHT sensor, ONLY ON IRD_PCB
#define SHT_HUM_SENSOR
```

- You can still have a soil temperature sensor

```
////////////////////////////////////////////////////////////////////
// uncomment to have 1 soil temperature sensor ST using a one-wire DS18B20 sensor
// also needed for TWO_SOIL_TEMP_SENSOR and THREE_SOIL_TEMP_SENSOR
#define SOIL_TEMP_SENSOR
// only for watermark sensors, not relevant for capacitive sensors
#define LINK_SOIL_TEMP_TO_CENTIBAR
```

Configuration for SCD30/40 CO2 sensor



- Uncomment in main program

  `CO2_SCD30_SENSOR`

  or

  `CO2_SCD40_SENSOR`

```
/////////////////////////////////////////////////////////////////////
// uncomment to have an SCD30 CO2 sensor, ONLY ON IRD_PCB
#define CO2_SCD30_SENSOR
// uncomment to have an SCD40 CO2 sensor, ONLY ON IRD_PCB
//#define CO2_SCD40_SENSOR
```

- Dev addr is 26011DE1 in ABP mode
- You can still have a soil temperature sensor

```
/////////////////////////////////////////////////////////////////////
// uncomment to have 1 soil temperature sensor ST using a one-wire DS18B20 sensor
// also needed for TWO_SOIL_TEMP_SENSOR and THREE_SOIL_TEMP_SENSOR
#define SOIL_TEMP_SENSOR
// only for watermark sensors, not relevant for capacitive sensors
#define LINK_SOIL_TEMP_TO_CENTIBAR
```

Configuration for 2 & 3 soil temp

- Uncomment in main program either
  TWO_SOIL_TEMP_SENSOR
  or
  THREE_SOIL_TEMP_SENSOR

```
////////////////////////////////////////////////////////////////////
// uncomment to have a customized 2 soil temperature device, ONLY ON IRD_PCB
#define TWO_SOIL_TEMP_SENSOR
// uncomment to have a customized 3 soil temperature device, ONLY ON IRD_PCB
// #define THREE_SOIL_TEMP_SENSOR
```

- Dev addr is 26011DD1 in ABP mode
- You MUST also have the primary soil temperature sensor

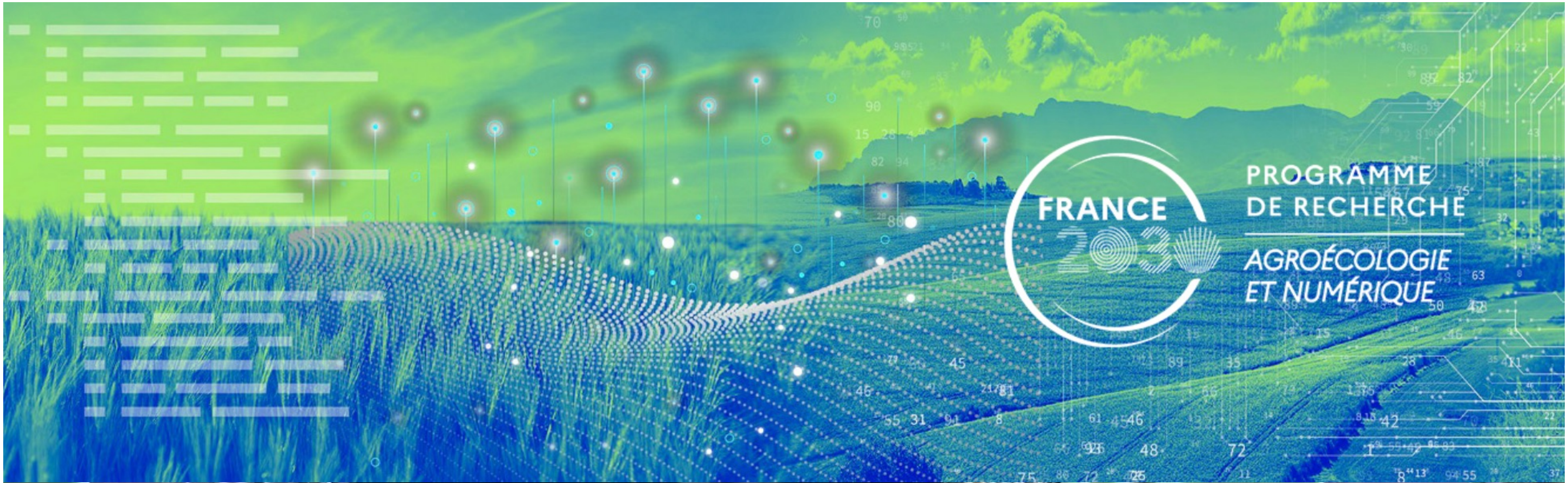
```
////////////////////////////////////////////////////////////////////
// uncomment to have 1 soil temperature sensor ST using a one-wire DS18B20 sensor
// also needed for TWO_SOIL_TEMP_SENSOR and THREE_SOIL_TEMP_SENSOR
#define SOIL_TEMP_SENSOR
// only for watermark sensors, not relevant for capacitive sensors
#define LINK_SOIL_TEMP_TO_CENTIBAR
```

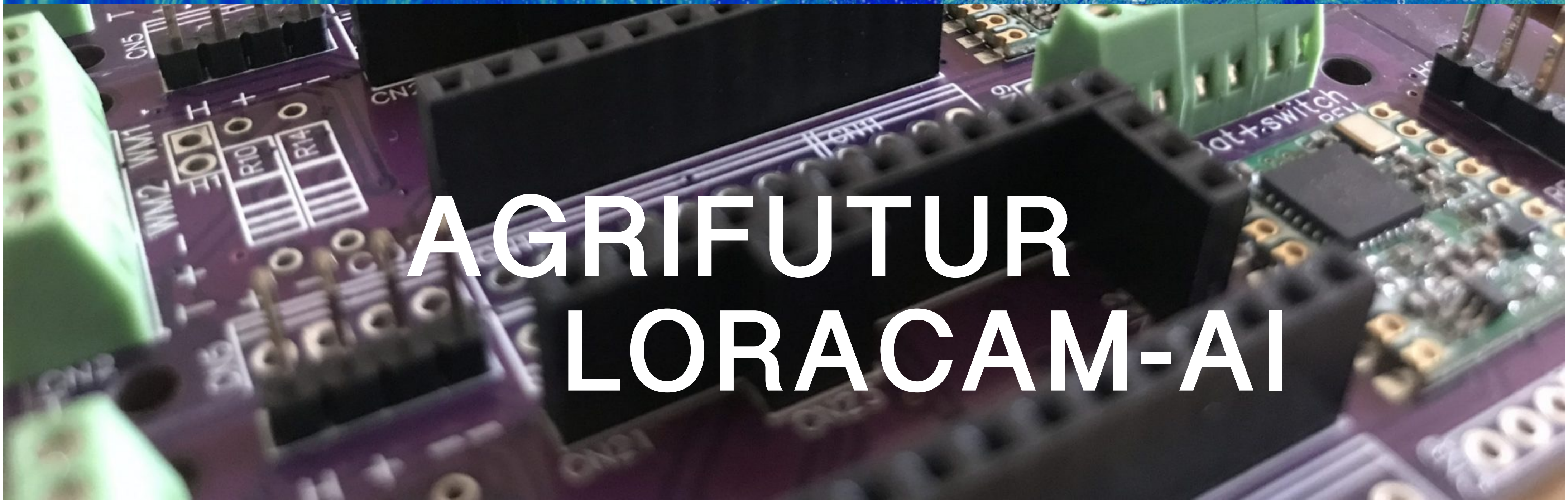
Low battery voltage indication

- The sensor device code has low battery voltage indication feature, by default, low battery voltage threshold is set to 2.85V

```
#ifdef WITH_WATERMARK
// the ATMega328P reboots at about 2.65 - 2.75
#define VCC_LOW 2.85
#else
// capacitive sensors can be impacted by low voltage, especially for very dry soil conditions
#define VCC_LOW 2.85
#endif
```

- The battery voltage will be transmitted to the gateway as well and will appear on the gateway's dashboard to warn end-user
- When low battery voltage is detected, the measure & transmission time interval will increase from 1h to 4h to preserve battery





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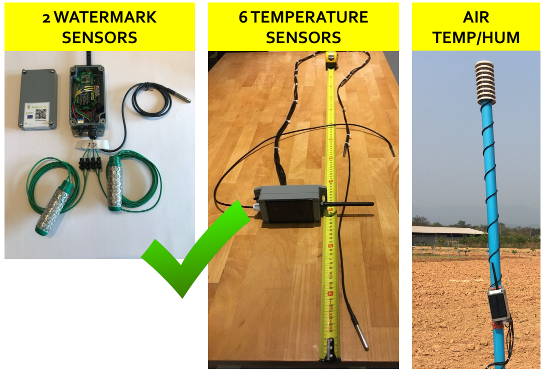
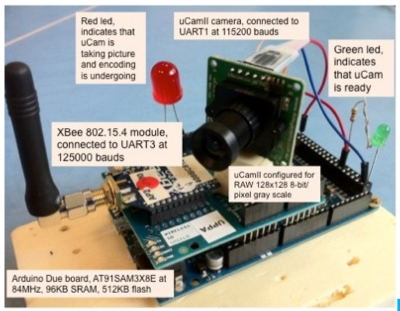
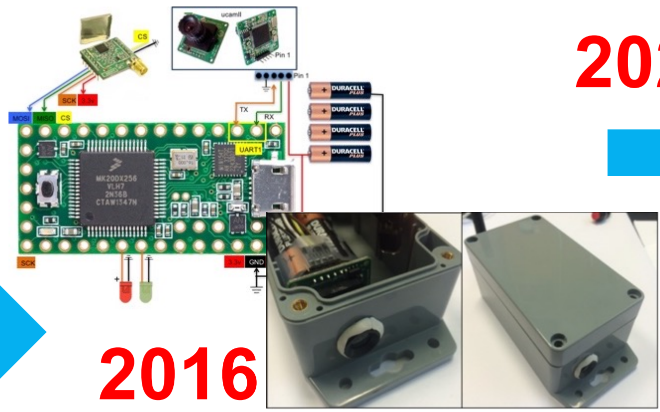
Advanced Sensing Systems



- Many applications need visual information → Image Sensing IoT



2012



2016

2024

ESP32S3 32-bit, dual-core, up to 240 MHz

XIAO ESP32S3

Prof. Congduc Pham
http://www.univ-pau.fr/~cpham

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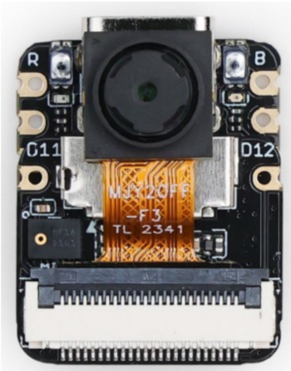
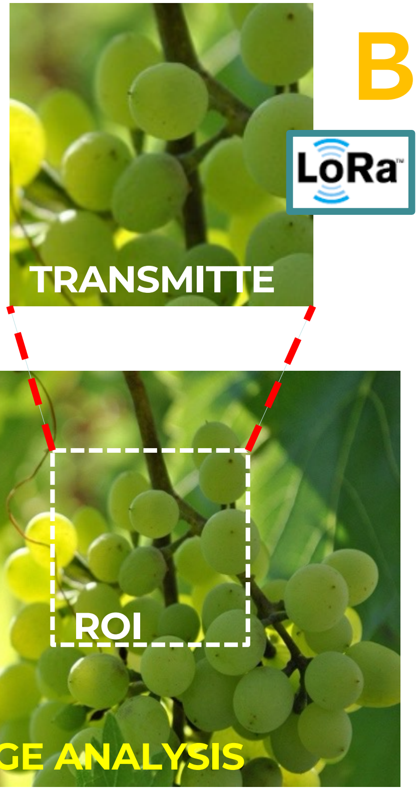
Image Sensor? Quite a challenge!





Agriculture Wildlife
Biodiversity Agroecology
Environment
Surveillance

- Low Power
- Plug-and-Sense, out-of-the-box
- Embedded AI processing
- Versatile AI learning framework
- Application-specific AI models
- Robust image encoding against losses
- Long-range radio transmissions
- Efficient usage of radio resources

Prof. Congduc Pham
http://www.univ-pau.fr/~cpham

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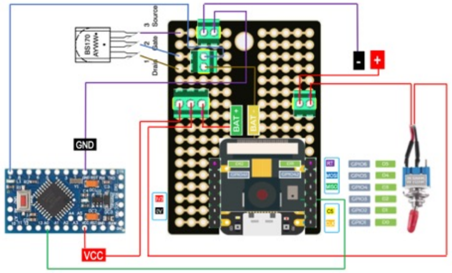
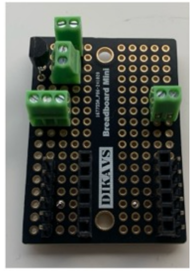
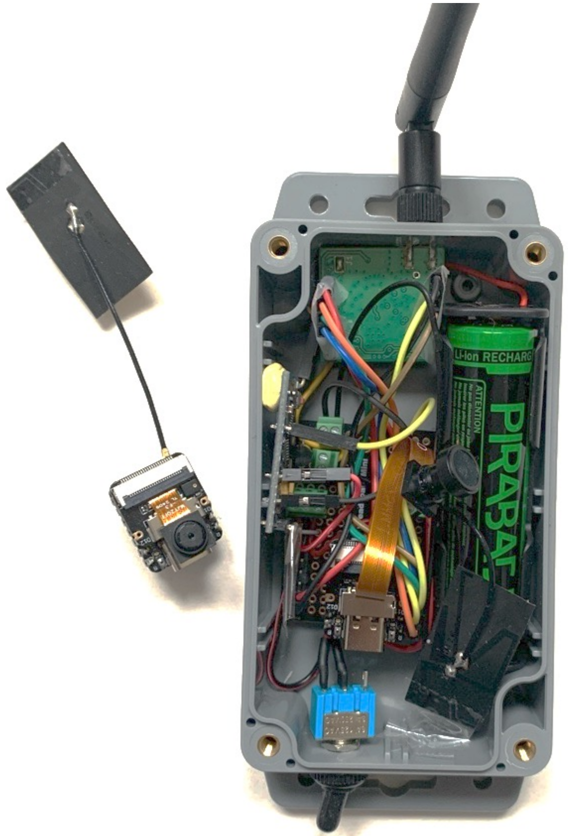
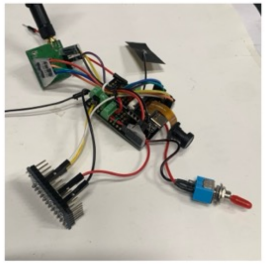
Proof-of-concept: LORACAM-AI

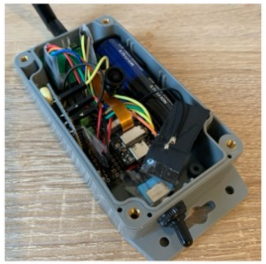
UPDATED VERSION IN 2025 FOR THE PEPR AGRIFUTUR PROJECT

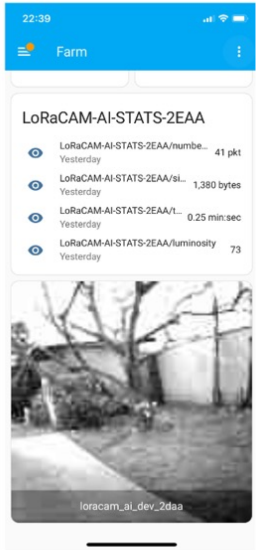












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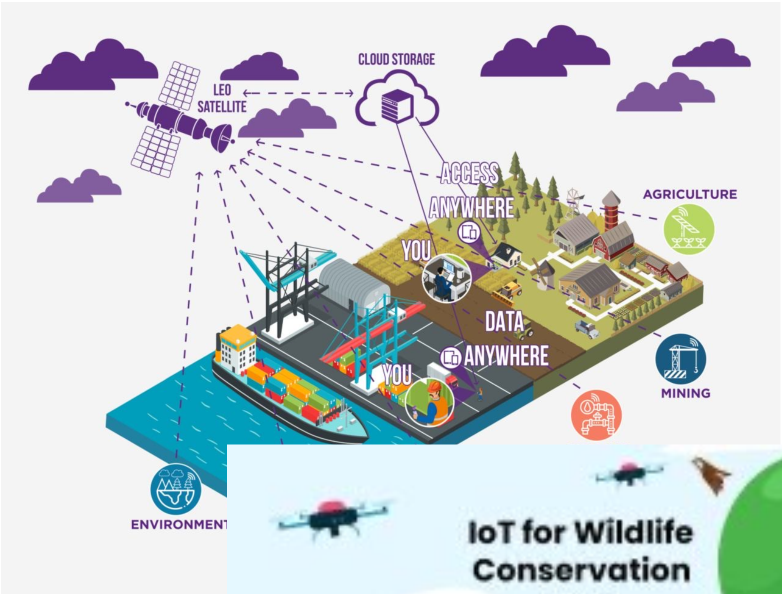
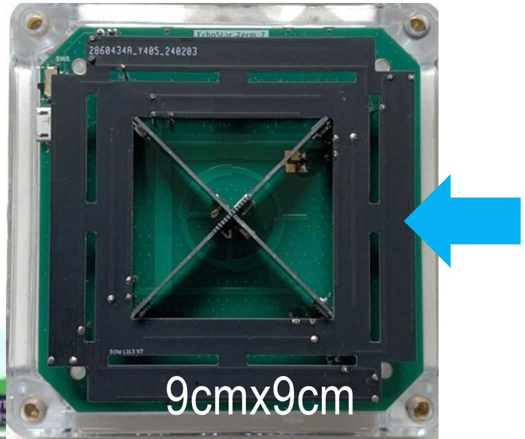
Going beyond terrestrial network!

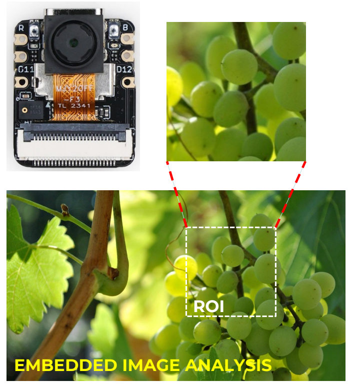
https://www.everythingrf.com/community/what-is-satellite-iot-connectivity



https://www.smartsight.in/industry-insights/iot-for-wildlife-conservation-and-environmental-monitoring/

https://github.com/nguyenmanhthao996tn/LEAT-EchoStar-Terminal-BSP





Low-cost satellite IoT

F. Ferrero & M. T. Nguyen
LEAT laboratory
University of Nice, France

Prof. Congduc Pham
http://www.univ-pau.fr/~cpham

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